

Geometric Computation: Twin Helices and the Algebra of Flow

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Abstract

This work constructs a manifold consisting of twin helices to investigate the existence of computational geometric bordisms, as part of the broader program of the cobordism hypothesis. We study its geometric, topological, algebraic, and computational properties, suggesting the emergence of a new research direction in categorical physics termed Geometric Computability Theory, situated at the intersection of geometry, gauge theory, and computability theory. We show that these “Geometrically Computable Manifolds” can encode arbitrary cyclic groups, leading to universal computational behavior in the sense of Kleene’s partial recursive functions. We further analyze generalized n -ary computational dynamics and formulate a metatheorem that describes the trade-offs inherent in these computational regimes. Ordinary differential equations are used to model abelian gauge computation, linking the construction to continuous dynamics. This perspective opens avenues for future investigation, including potential applications to Yang–Mills theory. As of writing, no algebraic systems yet exist to formalize the behavior of these manifolds; future work will develop such frameworks.

Keywords: geometric computation, dynamical systems, prequantum geometry, Yang–Mills theory, cobordism hypothesis, helices, computational gauge theory, topological quantum field theory.

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Introduction

“In the long run, we hope that some theory of language expressiveness develops into a formal theory of the programming language design space.”

— Mattias Felleisen, *On the Expressive Power of Programming Languages* (1991)

“A twisted version of four-dimensional supersymmetric gauge theory is formulated...”

— Edward Witten, *Topological Quantum Field Theory* (1988)

The following work presents a geometric model of computation.

The gap between computer science and physics has been a long-standing one. While various attempts to bridge the divide have been made over the years, a full resolution remains out of reach. On the one hand, familiar models of computation are fundamentally discrete, while much of the symmetries observed in fundamental physics are continuous. So, are the laws of physics truly computable? Penrose and others have argued that certain key physical (and cognitive) phenomena may elude algorithmic description; by contrast, mainstream practice treats numerical approximation on a continuum as acceptable. Moreover, what are the limits of how computational processes can behave? Here we bridge this divide.

Section 1 begins by constructing a deterministic geometry from manifolds composed of helices and, in doing so, lays out the coherence conditions arising from the gluing constraints imposed by chirality and torsion. From this construction arises a new class of computational objects: symbolic flow manifolds that admit non-cancellative monoidal structure. From there on, we execute a series of calculations for the group-theoretic and geometric/topological behavior of this class of manifolds.

Section 2 sets out a framework for a theory of physical computation. We lay out the definition for a “Geometrically Computable Manifold” and its resulting state vector, the “Geobit.” Here, we attempt to reconcile semantic expressiveness in programming languages with the structural stability and reversibility of geometric processes. While syntactic semantics often struggles to formalize languages with rich dynamic structure, the approach taken here replaces syntactic encoding with differential-geometric ones, specifically, moduli spaces of helices and their categorical behavior under symbolic flow. The resulting enriched category naturally supports symbolic computation with bounded state and fixed-point behavior. It exhibits expressive capacity that surpasses classical Von Neumann-like models (e.g., Turing machines and register machines) while retaining verifiability at the topological level. By analyzing this relationship, we arrive at the central result of the paper—a conservation law for computational behavior itself.

Section 3 rigorously grounds this resulting framework in the theory of Kleene’s partial recursive functions and, in doing so, provides a natural transition point from finite groups to groupoids, where computation becomes locally sensitive and context-dependent.

Section 4 describes the algebraic behavior of these manifolds as the computation evolves towards increasingly gauge-like behavior under the groupoid framework. Here, a new counterpart to the finite-state machine is introduced as an alternative to global PDEs with an extended homotopy-theoretic framework to support such a model. Formal bounds for computational behavior within such a framework are proposed along with an examination of foundational issues in moduli spaces and jet-theoretic tools for computation.

Section 5 situates our model within a larger categorical and physical context. Here, we examine the foundations of physics itself, attempting to reveal the origin of computational randomness as a quotient of geometric complexity, a coarse-graining of torsion and chirality across moduli spaces. Holonomy-based semantics are defined as we transition to non-abelian gauge-theoretical behavior. A fully constructive classification of computational behavior within TFTs is outlined. Further re-

search directions are formulated to address concrete applications in relation to both string and twistor theory, along with experimental proposals to advance the state of quantum foundations. A formal interpretation of quantum mechanics is then given in light of our results: a deterministic dynamical computational process whose semantics are so fluid they exceed any single categorical typing.

In the end, we remain hopeful that a new form of constructive computational gauge theory, grounded in the philosophy of Alexander Grothendieck, may provide a way forward in the study of HEP.

Our task here, by its nature, is a work of synthesis as much as rigor. If its scope sometimes appears unruly, or its exposition occasionally ventures beyond traditional boundaries, it is only because the questions at hand resist treatment by standard means. I have aimed for clarity without sacrificing depth. So, if arguments sometimes feel unconventional?

I hope that patient reading and engagement will reward the effort.

Ultimately, my intent is not to assert authority, but to invite the reader to explore together an unexpected landscape that may, in time, reshape our understanding of computation itself.

1 Twin Helical Manifolds

1.1 Volumetric Helices

We construct a volumetric helix by the following method. Begin with the standard parametric equation for an idealized helix. Let the general parametric form for our helix $\gamma(\theta)$ be

$$\gamma(\theta) = (R \cos(\theta), R \sin(\theta), h\theta)$$

We define the solid points of the helical object using the following equation

$$S = \{\mathbf{S}(\theta, r', \phi) \mid \theta_{\min} \leq \theta \leq \theta_{\max}, 0 \leq r' \leq \rho(\theta), 0 \leq \phi < 2\pi\}$$

Here, $\mathbf{S}(\theta, r', \phi)$ denotes the point in $\mathbb{R}^3$ obtained by starting at position $\gamma(\theta)$ on the helical spine, then moving a distance r' in the plane normal to the spine at angle ϕ . Let $\rho(\theta)$ define the cross-sectional radius of the helix for θ in some domain $[\theta_{\min}, \theta_{\max}]$.

Definition 1.1 (Helix Base and Apex). The *base* of the helix ¹ is defined as the point where $\rho(\theta)$ attains its maximal value:

$$\rho_{\text{base}} := \rho(\theta_{\min})$$

The *apex* is defined as the point where $\rho(\theta)$ attains its minimal value $\theta = \theta_{\max}$:

$$\rho_{\text{apex}} := \rho(\theta_{\max})$$

Let $\rho(\theta)$ be

$$\rho(\theta) = \rho_{\text{apex}} + (\rho_{\text{base}} - \rho_{\text{apex}}) \underbrace{\cos^2\left(\frac{\pi}{2} u\right)}_{f(u)}$$

Where

$$u = \frac{\theta - \theta_{\min}}{\theta_{\max} - \theta_{\min}}$$

and $0 < \rho_{\text{apex}} < \rho_{\text{base}}$. We select a $\cos^2$ profile, as it has vanishing derivatives at both ends. Next, to eliminate any possibility for a singular tip, we replace the terminal disk at $\theta = \theta_{\max}$ with a smooth, dome-shaped manifold patch. This patch, which we term the apical cap, is constructed as a surface of revolution that guarantees a C^1 join with the main body of the helix. Let (T, N, B) be the Frenet-Serret frame of the helical spine $\gamma(\theta)$ evaluated at the apex, $\theta = \theta_{\max}$. The construction of the cap is as follows.

¹There is not, in fact, an accepted term for this, so we adapt one from conics

Definition 1.2 (Smooth Apical Cap). The apical cap, $S_{\text{cap}}(u, v)$, is a surface parameterized for $u \in [0, \pi/2]$ and $v \in [0, 2\pi)$ by the equation:

$$S_{\text{cap}}(u, v) = \gamma(\theta_{\text{max}}) + \rho_{\text{apex}} \sin(u)T + \rho_{\text{apex}} \cos(u)(\cos(v)N + \sin(v)B)$$

where $\rho_{\text{apex}} = \rho(\theta_{\text{max}})$ is the minimal radius of the helical tube.

The boundary of this cap at $u = 0$ corresponds to the terminal circle of the main helical tube. We verify that the tangent planes of the cap and the tube are identical at this junction. The partial derivatives of the cap's parameterization at the junction ($u = 0$) are

$$\begin{aligned} \left. \frac{\partial \mathbf{S}_{\text{cap}}}{\partial u} \right|_{u=0} &= \rho_{\text{apex}} \mathbf{T} \\ \left. \frac{\partial \mathbf{S}_{\text{cap}}}{\partial v} \right|_{u=0} &= \rho_{\text{apex}} (-\sin(v)N + \cos(v)B) \end{aligned}$$

These two vectors span the tangent plane of the cap at the junction. The first vector, $\left. \frac{\partial \mathbf{S}_{\text{cap}}}{\partial u} \right|_{u=0}$, is parallel to the helical spine's tangent vector T at the junction. The second is tangent to the circular cross-section. This matches the tangent plane of the main helical body at $\theta = \theta_{\text{max}}$, where the condition $\rho'(\theta_{\text{max}}) = 0$ ensures its longitudinal tangent is also parallel to T . Therefore, the apical cap joins the main body seamlessly, and the resulting solid possesses a C^1 manifold boundary.

Remark 1.1. Let $0 < \rho_{\text{apex}} < \rho_{\text{base}}$ to avoid a cone point. We impose $\rho_{\text{base}} < R$ and $\rho_{\text{base}} < \pi h$ so that the inflated tube is embedded; no further restriction on θ_{min} is required.

1.2 Chirality and Alignment

Definition 1.3 (Chirality). A subset $O \subset \mathbb{R}^n$ is said to have *chirality* if it is not properly congruent to its mirror image. Equivalently, O is chiral if for every reflection r across a plane in $\mathbb{R}^n$, there exists no proper isometry, i.e. orientation-preserving Euclidean motion (translations or rotations), g such that $g(O) = r(O)$. We say that helices may possess either left-handed or right-handed chirality. A helix is left-handed if $h < 0$ and right-handed if $h > 0$.

Lemma 1.1 (Spiral Matching Lemma). *Two Archimedean spirals, S_1 and S_2 , with distinct centers and the same pitch, can intersect to form a C^0 curve with a matching phase if and only if they have opposite chirality.*

Proof. Let the two spirals be defined in $\mathbb{R}^2$ by the vector equations:

$$S_i(\theta_i) = C_i + a_i \theta_i \begin{pmatrix} \cos \theta_i \\ \sin \theta_i \end{pmatrix}, \quad i = 1, 2$$

We are given that the centers are distinct ($C_1 \neq C_2$) and the pitches are equal ($|a_1| = |a_2| = a > 0$). The lemma requires two conditions to be met simultaneously:

- (a) **C^0 Intersection.** The spirals meet at a common point. $S_1(\theta_1) = S_2(\theta_2)$
- (b) **Phase Matching.** The local polar angles are the same. $\theta_1 = \theta_2 =: \theta$

Substitute the phase-matching condition into the intersection equation:

$$C_1 + a_1 \theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = C_2 + a_2 \theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

Rearranging the terms to isolate the vector between the centers gives our primary relation:

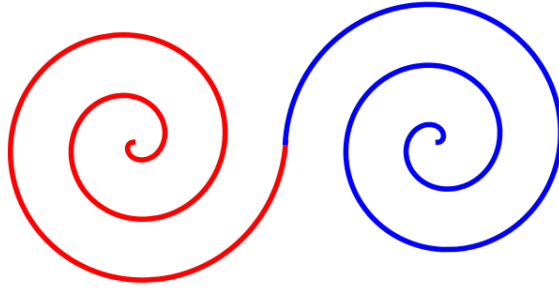
$$C_2 - C_1 = (a_1 - a_2) \theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

We analyze this relation for both chirality cases.

- (a) **Same Chirality.** ($a_1 = a_2$) If the spirals have the same chirality, the term $(a_1 - a_2)$ is zero. The relation becomes

$$C_2 - C_1 = 0 \implies C_1 = C_2$$

This contradicts the given condition that the centers are distinct. Therefore, an intersection with matching phase is impossible for spirals of the same chirality.



- (b) **Opposite Chirality.** ($a_2 = -a_1$) If the spirals have opposite chirality, the term $(a_1 - a_2)$ becomes $(a_1 - (-a_1)) = 2a_1$. The relation becomes

$$C_2 - C_1 = 2a_1\theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

Thus, an intersection with matching phase is possible if and only if the spirals have opposite chirality. $\square$

Remark 1.2. We impose the phase-matching condition $\theta_1 = \theta_2$ so that the radial directions—and hence the unit tangent directions—agree at the intersection, removing any kink, the necessary first step toward any C^1 or C^2 splice.

Corollary 1.2 (Coaxial Spiral Chirality Constraint). *Let S_1 and S_2 be Archimedean spirals centered at the same point. Then, a C^0 bounded path connecting them with matching phase at a common point can exist only if their chiralities are identical.*

Proof. This is a direct consequence of the Spiral Matching Lemma. As proven above, with $C_1 = C_2$, equation:

$$(C_2 - C_1) = (a_1 - a_2)\theta(\cos \theta, \sin \theta)$$

which gives

$$(a_1 - a_2)\theta(\cos \theta, \sin \theta) = 0$$

for $\theta \neq 0$, this forces $a_1 = a_2$, which in turn requires the spirals to have the same chirality. $\square$

Remark 1.3. This illustrates a fundamental principle in geometric design: adding a simple local constraint, such as matching phase, can enforce a relationship between the curves' global properties (chirality). In this case, the seemingly mild phase-matching condition is sufficient to rule out a connection entirely for same-chirality spirals with offset centers.

1.3 Twin Helical Manifolds

We now define a mirrored helix and perform a topological gluing operation to form a pair of helices with opposing chirality, joined at their respective bases.

Definition 1.4 (Binary Orientation of a Helix). Let H be a helix in $\mathbb{R}^3$ with axis of symmetry parallel to the z -axis. The orientation of H is defined by the monotonicity of its z -coordinate as a function of its intrinsic parameter u , increasing from base to apex:

- (a) **Upwards-oriented.** $z(u)$ is strictly increasing from base to apex.
- (b) **Downwards-oriented.** $z(u)$ is strictly decreasing from base to apex.

Definition 1.5 (Types of Twin Helices). To generate a Type 1 Twin Helix, we fix $R > 0$ and pitch $h > 0$. Let the first volumetric helix H_A be that which corresponds to the spine

$$\gamma_A(\theta) = (R \cos \theta, R \sin \theta, h\theta), \quad \theta \in \mathbb{R}$$

Let the second helix H_B be one rotated by π about the x -axis to obtain the twin:

$$\gamma_B(\theta) = (R \cos \theta, -R \sin \theta, -h\theta)$$

- γ_B is right-handed if γ_A is left-handed, and vice-versa.
- At $\theta = 0$, the spines γ_A and γ_B both pass through the point $(R, 0, 0)$. The tangent vectors at this point, $\gamma'_A(0) = (0, R, h)$ and $\gamma'_B(0) = (0, -R, -h)$, have opposite y - and z -components, ensuring the bases can be joined smoothly.

To generate the remaining types, let $\mathcal{G}$ be the specific subgroup of $E(3)$ generated by the elements as outlined in the final column of 1. Applying appropriate elements of $\mathcal{G}$ to H_B produces the remaining three configurations as displayed in 1.

To apply the topological gluing, let H_A and H_B be compact 3-manifolds with boundary embedded in $\mathbb{R}^3$, representing the helical manifolds parameterized by γ_A, γ_B respectively. Each H_i ($i = A, B$) has a single boundary component ∂H_i , which corresponds to the helix's base. Let $B_i \subset \partial H_i$ denote the base surface of H_i , corresponding to the endpoint 0 of the interval $[0, 1]$ in the product structure $D^2 \times [0, 1]$. Since we assume closed cross-sections, each B_i is homeomorphic to the closed 2-disk D^2 .

The local diffeomorphism that creates the bridge is realized by a parametric curve, $\Gamma(s)$, constructed as a Cubic Hermite spline[1]. This curve provides the central spine of the connecting manifold.

Definition 1.6 (Base-to-Base Diffeomorphic Bridge). The spine of the bridge, $\Gamma(s)$, for a parameter $s \in [0, 1]$ is given by

$$\Gamma(s) = h_{00}(s)P_A + h_{01}(s)P_B + h_{10}(s)\mathbf{m}_A + h_{11}(s)\mathbf{m}_B$$

where:

- P_A and P_B are the base positions.
- The tangent vectors $\mathbf{m}_A$ and $\mathbf{m}_B$ are defined by the unit tangents $\mathbf{t}_A, \mathbf{t}_B$, horizontal or vertical as required, and a scalar tension parameter ℓ :

$$\mathbf{m}_A = \ell \mathbf{t}_A \quad \text{and} \quad \mathbf{m}_B = \ell \mathbf{t}_B$$

- The functions $h_{ij}(s)$ are the standard cubic Hermite basis functions:

$$h_{00}(s) = 2s^3 - 3s^2 + 1$$

$$h_{01}(s) = -2s^3 + 3s^2$$

$$h_{10}(s) = s^3 - 2s^2 + s$$

$$h_{11}(s) = s^3 - s^2$$

For Types 2 and 4, we offset the neck by the bump function:

$$\tilde{\Gamma}(s) = \Gamma(s) + \alpha s^2(1-s)^2 \mathbf{n}$$

where:

- $\mathbf{n}$ is any unit vector perpendicular to both tangents (in the coaxial case, take the outward radial direction $(0, 1, 0)$)
- $\alpha > 2\rho_{\text{base}}$ guarantees the neck clears the parent tubes, which preserves C^1 boundary data while preventing the Hermite loop from intersecting the thickened tubes.

Let $\rho_{\text{bridge}}(s)$ be the radius of the bridge tube as a function of the spline parameter s . A simple choice is a constant radius $\rho_{\text{bridge}}(s) = \rho_{\text{base}}$. The points in the bridge are given by

$$P(s, r', \phi) = \Gamma(s) + r'(\cos(\phi)N(s) + \sin(\phi)B(s))$$

for $0 \leq r' \leq \rho_{\text{base}}$, where $(T(s), N(s), B(s))$ is the Frenet-Serret frame of the bridge spine $\Gamma(s)$.

For the coaxial same-orientation case (Type 2), we take the bump in a horizontal direction $\mathbf{n} \perp \mathbf{t}_A$ rather than in $\mathbf{e}_z$, moving the neck off the common axis and preventing the Hermite hair-pin from self-intersecting.

By construction, this spline and bump function rigorously satisfies the boundary conditions $\Gamma(0) = P_A$, $\Gamma(1) = P_B$, $\Gamma'(0) = \mathbf{m}_A$, and $\Gamma'(1) = \mathbf{m}_B$ that joins the two volumetric helices at their bases with a minimum of C^1 continuity, in addition to correctly realizing the global features of the Type 2, 3, and 4 configurations.

H_A and H_B are the compact 3-manifolds with boundary representing the two helices. Their bases, $B_A \subset \partial H_A$ and $B_B \subset \partial H_B$, are the surfaces to be joined. We form the disjoint union $H_A \sqcup H_B$ and define an equivalence relation $\sim$ by identifying points on the bases via a diffeomorphism, $p \sim f(p)$ for $p \in B_A$. The resulting quotient space

$$H_{AB} = (H_A \sqcup H_B) / \sim,$$

is the abstract topological manifold formed by this gluing.

To realize H_{AB} as a smooth (C^1) embedded manifold, we employ the explicit geometric bridge detailed previously. The cubic Hermite spline provides a smooth transition between the bases of H_A and H_B , ensuring that the tangent planes match precisely at the junctions and have at least C^1 (and, with higher-degree splines, potentially $C^{2,3}$) regularity.

We term this new construction, a smooth manifold embedded in $\mathbb{R}^3$, a *Twin Helical Manifold* (“THM”), or equivalently, just a *Twin Helix/Helices*.

Definition 1.7 (Twin Helical Manifold and Unit Helices). A *THM* is defined as a smooth manifold in $\mathbb{R}^n$ formed by a topological gluing of two volumetric helices and a diffeomorphic bridge that smoothly joins their respective base surfaces. The two helices composing a single *THM* are termed *unit helices*, denoted H_A and H_B .

By combining different chiralities and orientations, we define a geometric family of four canonical manifolds, denoted THM_4 . This yields a total of 4 distinct invariant configuration states.

Table 1: Essential type chart for twin helical manifolds

Type	Chirality	Orientation	Axiality	Mnemonic	Map $\Rightarrow H_B$
1	Opposite	Opposite	Coaxial	Helical Orb	$R_x(\pi)$
2	Opposite	Same	Coaxial	Twisting Pair/DNA-like	Σ_y
3	Same	Opposite	Offset	Mirrored Symmetrical Flow	$R_y(\pi), T$
4	Same	Same	Offset	Pair of Horns/Drills	$R_z(\pi), T$

However, our Type 2 twin helix has a small problem. A simple drawing shows that constructing a twin helix with the specified attributes becomes problematic in the previously defined $\mathbb{R}^3$ space. A closer investigation leads to the following conclusion.

Lemma 1.3 (Twin Helix Intersection Lemma). *No twin helical manifold whose unit helices with turns $N \geq 1/2$, sharing both axial alignment, orientation, and opposing chirality, can exist without intersection of its respective unit helices in $\mathbb{R}^3$.*

Proof. Let H_A and H_B be two coaxial helices parameterized by $t \in [0, 1]$. Both helices share the same radial function $r(t)$ and the same axial function $z(t)$. Their distinction lies in chirality and a possible phase offset.

The angular position of a point on helix H_x at parameter t is given by

$$\alpha_x(t) = 2\pi N \chi_x t + \phi_x,$$

where:

- (a) N is the number of turns in one unit helix,
- (b) χ_x is the chirality of H_x , with $\chi_A = -1$ and $\chi_B = 1$ for Type 2,
- (c) ϕ_x is the phase offset. Without loss of generality, we may set $\phi_A = 0$.

The center lines of the two helices intersect if, for some $t \in [0, 1]$, their angular positions coincide modulo 2π . That is

$$\alpha_B(t) - \alpha_A(t) = 2k\pi$$

for some integer k . Substituting the expressions for $\alpha_A(t)$ and $\alpha_B(t)$, we obtain:

$$\begin{aligned} (2\pi N \chi_B t + \phi_B) - (2\pi N \chi_A t) &= 2k\pi \\ 2\pi N t (\chi_B - \chi_A) + \phi_B &= 2k\pi \\ 4\pi N t + \phi_B &= 2k\pi. \end{aligned}$$

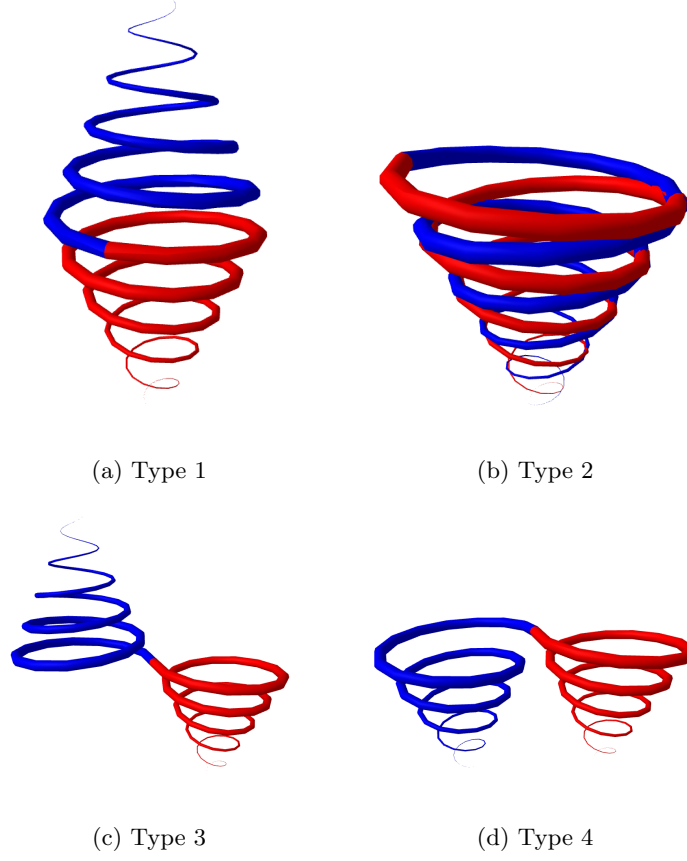


Figure 1: The THM_4 geometric family.

Define the function $f(t) = 4\pi Nt + \phi_B$. An intersection occurs if there exists a $t \in [0, 1]$ such that $f(t) = 2k\pi$ for some integer k .

The function $f(t)$ is continuous and monotonic on $[0, 1]$ (assuming $N > 0$). The values it takes range from

$$f(0) = \phi_B, \quad f(1) = 4\pi N + \phi_B.$$

This interval has length $4\pi N$. For $f(t)$ to contain at least one multiple of 2π , it suffices that

$$4\pi N \geq 2\pi \quad \Rightarrow \quad N \geq \frac{1}{2}.$$

By the Intermediate Value Theorem, since $f(t)$ is continuous on $[0, 1]$, it must attain all values in the interval $[\phi_B, \phi_B + 4\pi N]$. If this interval has length at least 2π , it must contain at least one value $2k\pi$. Therefore, for $N \geq \frac{1}{2}$, there always exists a $t \in [0, 1]$ for which $\alpha_B(t) \equiv \alpha_A(t) \pmod{2\pi}$, and thus the helices intersect.

Hence, for any $N \geq \frac{1}{2}$, it is mathematically impossible to choose a constant angular phase offset ϕ_B that avoids the intersection of the center lines of the Type 2 twin helices on $[0, 1]$. $\square$

In $\mathbb{R}^3$, a closed orientable surface has a line normal bundle; demanding a global 2-frame (to preserve both chirality and phase across the join) overconstrains codimension 1. Equivalently, any such immersion develops transverse self-intersections at the Type-2 junction, so the framed join cannot be realized without crossings. Passing to $\mathbb{R}^4$ removes this extrinsic obstruction: the normal bundle is rank 2 and can be trivialized, so one obtains a V_4 -equivariant, self-intersection-free embedding that preserves the twist data. This matches the general Whitney [2] paradigm: every smooth 2-manifold embeds in $\mathbb{R}^4$, and the extra normal direction carries the required framing.

Remark 1.4 (Chirality Rule-Reversal Between 2- and 3-Dimensional Gluing). A curious but systematic reversal occurs when one passes from planar curves to spatial curves concerning the handedness required for a C^2 splice. In the planar setting (spirals), the data to be matched at an intersection consists of

position, unit tangent, and signed curvature. Because curvature is the only first-order invariant beyond the tangent, one free phase parameter suffices to adjust its magnitude but not its sign. Consequently, two offset spirals can be C^0 -glued iff their curvature signs are opposite.

In the spatial setting (helices), a space curve adds a second scalar invariant (torsion), whose sign records right-handed versus left-handed screw motion. The gluing constraints now depend on the relative placement of the helical axes:

Proposition 1.4 (Rules for Smooth Gluing of Helices). *Let*

$$\gamma_i(t) = c_i + R_i(\cos t \mathbf{u}_i + \sin t \mathbf{v}_i) + p_i t \mathbf{e} \quad (i = 1, 2)$$

be two circular helices in $\mathbb{R}^3$ whose axes are parallel to the same unit vector $\mathbf{e}$.

Define curvature and torsion in the usual way:

$$\kappa_i = \frac{R_i}{R_i^2 + p_i^2}, \quad \tau_i = \frac{p_i}{R_i^2 + p_i^2},$$

so $\text{sign}(\tau_i)$ records right- or left-handedness.

- (a) **Coaxial case.** *If the centres c_1, c_2 differ only along the common axis $(c_2 - c_1) \parallel \mathbf{e}$, then one can splice γ_1 to γ_2 by a C^2 curve through an intersection point P iff*

$$\tau_1 + \tau_2 = 0, \quad \text{equivalently} \quad \text{sign}(\tau_1) = -\text{sign}(\tau_2).$$

- (b) **Offset-axis case.** *If the axis directions coincide but the centres are displaced sideways, $(c_2 - c_1) \perp \mathbf{e}$, then a C^2 splice exists iff*

$$\tau_1 = \tau_2, \quad \text{equivalently} \quad \text{sign}(\tau_1) = \text{sign}(\tau_2).$$

Proof sketch. At the chosen meeting point P we must match three items:

- (a) Position P itself.
- (b) Unit tangent directions.
- (c) Signed curvature vector, i.e., curvature magnitude and the normal direction determined by torsion.

For coaxial helices, only one phase angle (say t_1) can be changed independently; this lets us match the curvature magnitude but forces the torsion signs to cancel. With offset axes, we have two independent phase angles (t_1, t_2) , giving enough freedom to align the normal planes provided the torsion signs agree. A short trigonometric computation makes these statements precise. $\square$

Remark 1.5. Passing from $\mathbb{R}^2$ to $\mathbb{R}^3$ promotes the matching problem from the 1-jet to the 2-jet bundle: curvature sign is now unsigned (since $\kappa > 0$), while torsion sign becomes the relevant orientation datum. Whether this new constraint binds or is absorbed by an extra phase degree of freedom hinges on the axial configuration. In effect, the added dimension introduces a framing (normal-binormal plane) whose sign can obstruct or permit the splice depending on how many phase parameters survive the geometric symmetry.

Proposition 1.5 ($THM_4 \cong V_4$). *The group of transformations generated by the Chirality Swap (X) and Orientation Swap (Y) operators, which acts on the set of four canonical THM types, is isomorphic to the Klein four-group, V_4 .*

Proof. Let $H_{\text{types}} = \{\text{Type 1, Type 2, Type 3, Type 4}\}$, representing the four configurations of unit helices determined by combinations of chirality (Same vs Opposite) and orientation (Same vs Opposite).

Define the following transformations acting on H_{types} :

- Identity I : Trivial transformation. $I(T_i) = T_i$ for all $T_i \in H_{\text{types}}$.
- Chirality Swap X : Swap the chirality relationship (Same $\leftrightarrow$ Opposite), while preserving orientation.
- Orientation Swap Y : Swap the orientation relationship (Same $\leftrightarrow$ Opposite), while preserving chirality.

- Composite $Z = X \circ Y$: Acts by applying both swaps.

The action on the four manifold types proceeds as follows:

Type	(Chirality, Orientation)	X	Y
H_1	(O, O)	H_3	H_2
H_2	(O, S)	H_4	H_1
H_3	(S, O)	H_1	H_4
H_4	(S, S)	H_2	H_3

Composition rules:

$$\begin{aligned}
Z &= X \circ Y = Y \circ X \\
Z(H_1) &= X(Y(H_1)) = X(H_2) = H_4 \\
Z(H_2) &= X(Y(H_2)) = X(H_1) = H_3 \\
Z(H_3) &= X(Y(H_3)) = X(H_4) = H_2 \\
Z(H_4) &= X(Y(H_4)) = X(H_3) = H_1
\end{aligned}$$

All non-identity elements have order 2:

$$X^2 = I, \quad Y^2 = I, \quad Z^2 = (X \circ Y)^2 = X^2 \circ Y^2 = I$$

Pairwise compositions:

$$\begin{aligned}
X \circ Y &= Z, \quad Y \circ X = Z \quad \Rightarrow \quad X \circ Y = Y \circ X \\
X \circ Z &= X \circ (X \circ Y) = (X \circ X) \circ Y = I \circ Y = Y \\
Y \circ Z &= Y \circ (X \circ Y) = (Y \circ Y) \circ X = I \circ X = X
\end{aligned}$$

The set of transformations $G = \{I, X, Y, Z\}$ forms a group under composition. All elements have order 1 or 2, and the group is abelian. Thus, $G \cong V_4$. $\square$

Table 2 of the calculated FSG progressions via the GAP has been provided for ease of reference.

Table 2: Group progression of the THM_4

Group Structure	Order	Defined by...	Type
$V_4 \cong C_2 \times C_2$	4	Chirality and orientation combinations of the 4 canonical THM_4 types	abelian
$D_4 \cong (C_4) \rtimes C_2$	8	Chirality flips + swapping identity of the two helices	non-abelian
C_2^4	16	Independent flips of (c_A, c_B, d_A, d_B) for each unit helix	abelian
$C_2^4 \rtimes C_2$	32	Above + helix identity swapping symmetry	non-abelian
$(C_2^4 \rtimes C_2) \times C_4$	128	Adds 90° global rotation symmetry around z -axis	non-abelian
$(C_2^4 \rtimes C_2) \times C_{4h} \cong (C_2^4 \rtimes C_2) \times (C_4 \times C_2)$	256	Full group including xy -plane reflection symmetry (σ_h)	non-abelian

In addition, the generators of THM_{256} and lower orders are shown in Table 3. These symmetries exhibit a progression toward non-abelian behavior as more generators and state permutations are introduced. Preliminary computations suggest a rapidly growing group order (originally estimated at 256, but more accurately in the range of 1028 or higher when accounting for all independent orientation and chirality swaps between the two helices).

We note that this group structure may not be unique to the present construction. Similar generator sets arise in various contexts of symmetry and braid groups. The explicit relations and whole structure

Table 3: Generators of the THM_{256} symmetry group

Generator	Action
c_1	Flip chirality of Helix A
c_2	Flip chirality of Helix B
d_1	Flip orientation of Helix A (up/down)
d_2	Flip orientation of Helix B (up/down)
S	Swap identity of Helix A and Helix B
R_{90}	Global 90° rotation of the twin helix around the z -axis
σ_h	Reflect the entire twin helix across the xy -plane

of THM_4 , including its precise order, are left open for further investigation, ideally by those with more expertise in computational group theory.

We will now explore the topology of the THM . A single solid helix is a compact 3-manifold whose boundary, ∂H , is a smooth 2-manifold homeomorphic to a sphere. This boundary can be decomposed into the helical tube, the apical cap, and the base disk. The circular edge where the base disk meets the tube is a feature of this geometry. When we glue together the bases of two such unit helices in a proper manner, we form a larger compact 3-dimensional manifold with a single boundary. The original bases become an internal surface, and the boundary of the new THM consists of the original helical side surfaces and the tapered tips. THM is homeomorphic to D^3 in topological terms, giving genus 0.

Singularities at the apex are indeed a threat if the volume of the helix were to taper suddenly at the end to a point. But note that we have eliminated them earlier in the paper by way of a non-linear $\rho(\theta)$ volume decay and diffeomorphisms that produce smooth apex to THM 's unit helices.

Definition 1.8 (Singularity–Smoothness Principle). In geometric computation, every computational state is realized by a smooth manifold together with a prescribed flow. A necessary condition for logical consistency and well-posed evolution is the absence of geometric singularities. Concretely, the manifold and its flow must possess at least C^1 differentiability so that position and tangent fields vary continuously, guaranteeing the existence and uniqueness of solutions to the associated ordinary differential equations.

Definition 1.9 (Minimal Smoothness by Computational Regime). The following differentiability thresholds are established:

- **Abelian Regimes** (Symbolic and Quasi-Fluidic.) For stable logic and deterministic integration, it suffices that each twin helix manifold be C^1 ; continuous tangents ensure well-defined local ODEs. Higher regularity (C^2 or smoother) is preferred but not strictly required.
- **Non-Abelian Regimes** (Fully Fluidic and Gauge.) Dynamics now depend explicitly on curvature and higher-order invariants. Hence, a minimum of C^2 smoothness is mandated, with higher regularity (C^k , $k \geq 3$, or analytic) desirable for robust geometric evolution and topological stability.

Remark 1.6. The Singularity–Smoothness Principle establishes that geometric smoothness is equivalent to logical consistency. The minimal smoothness requirements specify the differentiability needed in each computational regime, ensuring that computation proceeds without singular breakdown. For readers who are more comfortable with the subject matter at hand and wish to see the full landscape ahead of time, one may skip ahead to Section 5.6 to obtain an overview of the classification.

2 Geometrically Computable Manifolds

We now begin the construction of a Turing Machine (“TM”) from a lattice of THM . The feasibility of this approach rests on the fact that THM , unlike most manifolds, provides a rich but controlled symmetry space. It is precisely because most well-known shapes in this metric (e.g., circles, tori, spheres) possess too much symmetry—not too little—that they fail to offer the structural asymmetry needed for computational behavior. The THM construction breaks this impasse.

2.1 Geometrically Computable Manifolds

Definition 2.1 (Geometrically Computable Manifold). A **GCM** on a space is:

$$\mathcal{G} = (\mathcal{M}, \{R_i\}_{i \in I}, \pi : E \rightarrow \mathcal{M}, \mathcal{I}, \text{Ev})$$

consisting of:

- (a) **Smooth Manifold** $\mathcal{M}$ — a smooth manifold (often a finite working region, possibly with boundary). A finite family of compact submanifolds $\{R_i\}_{i \in I}$ (cells) covers the working region and provides locality (finite overlap pattern).
- (b) **State bundle** $\pi : E \rightarrow \mathcal{M}$ — a smooth fiber bundle with finite dimensional fibers. A *global state* is a section $\sigma \in \Gamma(E)$. Discrete/symbolic, toroidal/phase, and integer-valued components may all live in the fiber.
- (c) **Interfaces** $\mathcal{I}$ — for each nonempty overlap $R_i \cap R_j$, an interface map:

$$I_{ij} : (\sigma|_{R_i}, \sigma|_{R_j}, \text{local data}) \mapsto (\sigma'|_{R_i}, \sigma'|_{R_j}),$$

attached to a portion of $\partial R_i \cap \partial R_j$. Interfaces couple neighboring cells and enforce locality: updates at R_i depend only on σ over R_i and finitely many overlaps $R_i \cap R_j$.

- (d) **Evolution** Ev — either:
 - discrete time: an update operator $U : \Gamma(E) \rightarrow \Gamma(E)$ that is local with respect to $\{R_i\}$, or
 - continuous time: a flow of sections $(\sigma_t)_{t \in \mathbb{R}}$ solving $\dot{\sigma}_t = F(\sigma_t)$ for a smooth, fiber-wise vector field F over $\mathcal{M}$;

possibly accompanied by a base diffeomorphism $\Phi_t : \mathcal{M} \rightarrow \mathcal{M}$. In either case, evolution is non-singular; it exists for all steps/times on admissible initial states and remains in a fixed finite-energy subset of E .

So what is a GCM, really? A GCM is a manifold that can encode its computable properties. A single *THM* can qualify as a GCM. A lattice of *THM* is also a GCM—just with more computing capacity. We call $\mathcal{G}$ a GCM when its evolution is local, interface-driven, and closed under composition: finite products of such systems and gluing along matched interfaces yield new systems of the same kind.

2.2 Geobits—Bits, Geometric Bits, and State Space

Definition 2.2 (Geometric Bits — Symbolic and Enriched). Let $\mathbf{S}_{\text{GCM}}$ be the category of state space embedding that all GCMs will use. Fix an ambient manifold region $\mathcal{M} \subset \mathbb{R}^n$ and a twin helix template H (a pair of embedded tubular helices) together with a framing: a preferred axial direction $\hat{z}$ and a reference scale r_0 . Let $\text{Emb}^{\text{fr}}(H, \mathcal{M})$ denote the set of framed embeddings $e : H \hookrightarrow \mathcal{M}$ modulo framed ambient isotopy that preserves the axial framing and the scale r_0 so that radius and orientation remain part of the state.

Let $H_{\text{ext}} \leq O(n)$ be a chosen ambient symmetry group (allowing reflections so that chirality changes are representable) and let H_{int} be a finite internal symmetry group acting on the labels of the twin (e.g. the swap of the two helices). Set $G := H_{\text{ext}} \times H_{\text{int}}$, acting on $\text{Emb}^{\text{fr}}(H, \mathcal{M})$ by pre/post-composition.

- (a) **Symbolic geobit.** Define the symbolic geobit state set:

$$\mathbf{S}_{\text{sym}} := \left\{ [e; \chi, \omega] \mid e \in \text{Emb}^{\text{fr}}(H, \mathcal{M}), \chi, \omega \right\} / G,$$

where $[e; \chi, \omega]$ denotes a framed embedding decorated by chirality (χ) and axial orientation (ω) relative to $\hat{z}$, and the quotient is by the G -action, so states are defined up to global symmetry. A *symbolic geobit* is an element $s \in \mathbf{S}_{\text{sym}}$. The finite subgroup generated by “flip chirality” C , “flip orientation” D , and (optionally) “swap” $S \in H_{\text{int}}$ acts on $\mathbf{S}_{\text{sym}}$; in the canonical twin helix unit this induces a $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ action on the labels (χ, ω) .

- (b) **Enriched (quasi-fluidic) geobit.** To allow compact phase and quantized charges, extend the label by a finite-type fiber:

$$P \cong (S^1)^m \times \mathbb{Z}^p \times F,$$

where the S^1 -coordinates encode bounded phases (e.g. axial angle), the $\mathbb{Z}$ -coordinates encode twist/winding/linking numbers, and F is a finite set of discrete tags (e.g. mode, parity). Set

$$\mathbf{S}_{\text{enr}} := \left\{ [e; \chi, \omega, p] \mid e \in \text{Emb}^{\text{fr}}(H, \mathcal{M}), \chi, \omega, p \in P \right\} / G,$$

with G acting equivariantly on the S^1 -factors (by rotation) and preserving the $\mathbb{Z}$ -charges. An *enriched geobit* is an element of $\mathbf{S}_{\text{enr}}$. The G -action need not be transitive; its orbits in $\mathbf{S}_{\text{enr}}$ are the *geobit types*. The symbolic case is recovered by taking $m = p = 0$ and F trivial.

In both cases, the geobit is a G -set (not necessarily a group): the finite operations generated by C, D (and S when present) act on states, yielding the canonical V_4 action in the symbolic unit. The enrichment by P provides the quasi-fluidic degrees of freedom (compact phases and integer counters) used elsewhere in the text.

For the time, we work only with S_{sym} . Now, let:

$$(H_1, H_2, H_3, H_4) \mapsto (00, 01, 10, 11)$$

where each H_n is a configuration from the THM_4 symmetry schema defined in Table 1, preserving the order of the original type chart.

We assign bit values based on symmetry: if the chirality and orientation of the unit helices are the same, assign bit value 1; if they differ, assign 0. This yields a total of 2 bits per THM_4 configuration.

Definition 2.3 (Geobit Capacity). Let L be the finite set of *internal labels* for a GCM cell (e.g. chirality/orientation choices before identifying symmetries), and let the *encoding symmetry* O act on L by intrinsic transformations (excluding extrinsic rigid motions). The set of *observable* states is the orbit set L/O , and the *geobit capacity* is

$$\text{Cap}(L, O) := \log_2 |L/O|.$$

When $L = O$ with the left-regular action, this reduces to $\text{Cap} = \log_2 |O|$.

Example 2.1 (Embedding Relationships for Helices). Let G act on a configuration space and let $O \subseteq G$ be the intrinsic encoding subgroup; put L for the raw label set and compute observable states via L/O .

- (a) THM_{16} . $O \cong C_2^4$, $L = O$ with the left-regular action. Then $|L/O| = |O| = 16$, so

$$\text{Cap} = \log_2 16 = 4 \text{ bits} \quad (\text{“2 bits from } \chi, 2 \text{ from } \omega\text{”}).$$

- (b) THM_4 . Impose identifications (e.g., identity-swap/pairing symmetries) modeled by a normal subgroup $H \trianglelefteq O$ of order 4; the effective encoding collapses to $O_{\text{eff}} \cong O/H \cong C_2^2$. With $L = O_{\text{eff}}$ (regular action),

$$\text{Cap} = \log_2 |O_{\text{eff}}| = \log_2 4 = 2 \text{ bits}.$$

- (c) THM_2 . If only orientation is swappable, the effective group is $O_{\text{eff}} \cong C_2$, hence

$$\text{Cap} = \log_2 2 = 1 \text{ bit}.$$

- (d) THM_8 . Let the raw labels be $L = C_2 \times C_4$ with elements (χ, ω) (chirality $\chi \in \{\pm 1\}$, orientation $\omega \in \mathbb{Z}/4$). Let $O = D_4 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$ act by

$$r \cdot (\chi, \omega) = (\chi, \omega + 1), \quad s \cdot (\chi, \omega) = (-\chi, -\omega).$$

Then $|L| = 8$, but the action has nontrivial stabilizers (reflections identify (χ, ω) with $(-\chi, -\omega)$), yielding $|L/O| = 4$ orbits. Hence

$$\text{Cap} = \log_2 4 = 2 \text{ bits} \quad \text{while} \quad \log_2 |O| = 3,$$

illustrating the reduction caused by non-commuting symmetries (orbit collisions).

This leads to a key structural pattern. Let O act on the finite label set L :

Abelian Case. If $O \cong (C_2)^n$ acts by independent flips on L with a free action (e.g. the regular action on $L = O$), then $|L/O| = |O| = 2^n$ and the capacity is n bits.

Non-abelian Case. When O is non-abelian, typical elements introduce larger stabilizers and merge labels across conjugate directions; consequently, $|L/O| < |L|$ and often $\text{Cap} < \log_2 |O|$ (as in the D_4 example).

Remark 2.1. Why the focus on abelian vs non-abelian groups?

In an abelian group, every state can be constructed as the direct sum of independent generators. That is, every geobit state is uniquely addressable by an n -tuple over $\mathbb{C}_2^n$. There are no collisions. The total number of distinguishable geobit states is exactly 2^n , and the encoding is non-redundant.

In contrast, a non-abelian group acts less cleanly. Its action “mixes” states: some group elements act non-trivially, creating identifications among configurations. Orbits form under conjugation or subgroup stabilizers, and the full state space becomes quotient-like. Redundancy becomes intrinsic—some geobit states are indistinguishable under the group action. As a result, the number of distinct encodable states in the orbit space may be strictly less than the order of the group itself.

This distinction becomes crucial when interpreting a GCM as a computational substrate:

abelian structure maximizes symbolic resolution, while non-abelian structure imposes constraints that must be managed or exploited.

Table 4: Relationship between geobits and their groups

Group	Order	Dimension	# Bits	Comments
C_2	2	2	1	Classical bit
$(C_2)^2 = V_4$	4	4	2	Base level 2-geobit
$(C_2)^4$	16	4	4	Canonical abelian 4-geobit
D_4	8	3	≤ 3	Non-abelian, simple global swap
$D_4 \ltimes (C_2)^4$	128	≥ 5	< 8	Non-abelian, global rotations

Remark 2.2. The embedding dimension for a THM symmetry group is the minimal integer n such that all distinct geometric configurations corresponding to the elements of the group can be realized as non-self-intersecting embedded submanifolds in Euclidean space. It is essential to note that global reflections or entire rotational symmetry groups do not contribute to the intrinsic geobit capacity in $\mathbb{R}^4$. These operations require action on the global geometry and do not alter the intrinsic state of the unit helices. Thus, group orders beyond 16 that arise from external symmetry do not increase state space in a computable sense. Table 5 compares geobits with their close cousin.

Definition 2.4 (Geometric Kets). We refer to an individual geobit ket array as “ n -geobit,” directly paralleling the usage of qubit arrays with ket-style notation. Following the same notation, we employ a modified bra-ket notation.

Let the microstate of a twin helix cell be

$$\alpha = (C_A, O_A, C_B, O_B) \in C_2^4.$$

Where C is chirality and O is orientation. We write the *state ket* $|\alpha\rangle := |C_A O_A C_B O_B\rangle$. Intrinsic flips act by the Klein-four group

$$V_4 = \{e, s_A, s_B, s_A s_B\},$$

where s_A flips (C_A, O_A) and s_B flips (C_B, O_B) coordinatewise. A 1-bit readout is any fixed function $r : C_2^4 \rightarrow C_2$ (e.g. parity $r = o_A \oplus o_B$ or equality $r = [o_A = o_B]$).

Example 2.2 (Update Step with V_4). Take parity readout $r = o_A \oplus o_B$ and the microstate

$$|\alpha\rangle = |1101\rangle.$$

Then $b = r(x) = 1 \oplus 1 = 0$. Suppose the next bit should be $b' = 1$. Apply any toggler in V_4 , e.g. s_A :

$$s_A \cdot |\alpha\rangle = |1001\rangle, \quad r(s_A \cdot \alpha) = 0 \oplus 1 = 1 = b'.$$

Table 5: Qubit vs. Geobit

Property		Quantum qubit	Geometric geobit
Pure state space		$\mathbb{C}P^1 \cong S^2$ (Bloch sphere)	Finite labels $\times$ continuous geometric parameters (e.g. chirality/orientation, phase)
Dynamics		Unitaries (e.g. $SU(2)$ rotations)	Geometric transformations (flips, rotations, deformations)
Measurement/Readout		Born rule (probabilistic)	Geometric probe/partition (deterministic given microstate)
Accessible info	classical	≤ 1 bit per use ²	Depends on chosen encoding and symmetries
Elementary operations	opera-	Single-/two-qubit gates; entanglement via tensor product	Local geometric updates; composition by connecting/gluing components
Key feature		Coherence and entanglement central	Robustness tied to analog stability

(If instead $b' = b$, use a parity-preserver e or $s_A s_B$.) Optionally, a coarse projection $\pi : C_2^4 \rightarrow C_2^2$ such as

$$\pi(C_A, O_A, C_B, O_B) = (C_A \oplus C_B, O_A \oplus O_B)$$

can be displayed as $|\pi(\alpha)\rangle$; note that s_A or s_B need not descend to operators on the quotient, so actions are applied on C_2^4 before projecting. The resulting ket would then become $|11\rangle$.

2.3 The GCM Lattice

Definition 2.5 (GCM Lattice). A *GCM Lattice* or *GL* is a finite or locally finite assembly of GCMs arranged over a graph. Formally, it is the data:

$$GL = (\mathcal{L}, G = (I, E), \{\mathcal{G}_i\}_{i \in I}, \{\iota_i\}, \mathcal{I}^{\text{lat}}, \text{Ev}, \mu)$$

with:

- (a) **Ambient region** $\mathcal{L}$ — smooth manifold (often a bounded working region) into which cells will be embedded.
- (b) **Lattice graph** $G = (I, E)$. A finite or locally-finite graph (e.g. a box of $\mathbb{Z}^d$) with uniformly bounded degree; vertices index the cells, edges encode adjacency.
- (c) **Cells** $\{\mathcal{G}_i\}$ — For each $i \in I$, a GCM $\mathcal{G}_i = (\mathcal{L}_i, \{R_\alpha^i\}, \pi_i : E_i \rightarrow \mathcal{L}_i, \mathcal{I}_i, \text{Ev}_i)$ as in Def. 2.1.
- (d) **Placements** $\{\iota_i\}$ — Smooth embeddings $\iota_i : \mathcal{L}_i \hookrightarrow \mathcal{L}$ with pairwise disjoint interiors. The support of cell i is $\text{supp}(i) := \iota_i(\mathcal{L}_i)$.
- (e) **Lattice interfaces** $\mathcal{I}^{\text{lat}}$ — For each edge $(i, j) \in E$, a family of interface maps attached to portions of $\partial \text{supp}(i) \cap \partial \text{supp}(j)$,

$$I_{ij} : (\sigma|_{\text{supp}(i)}, \sigma|_{\text{supp}(j)}, \text{local flow data}) \mapsto (\sigma'|_{\text{supp}(i)}, \sigma'|_{\text{supp}(j)}),$$

which are local (depend only on a finite stencil around i, j) and equivariant with respect to the chosen cell symmetries.

- (f) **Evolution** Ev — Either (i) a discrete, synchronous update $U : \Gamma(E) \rightarrow \Gamma(E)$ on the global state bundle $E := \bigsqcup_{i \in I} \iota_i E_i$, or (ii) a continuous flow of sections $(\sigma_t)_{t \in \mathbb{R}}$ solving $\dot{\sigma}_t = F(\sigma_t)$ and compatible with all interfaces. Evolution is local and non-singular on admissible initial states.
- (g) **Symbolic observation** μ — A readout map $\mu : \Gamma(E) \rightarrow \Sigma^N$ that, at each cell i , selects a coding $q_i : E_i \rightarrow \Sigma^{n_i}$ of its geobits; the total capacity is $N = \sum_{i \in I} n_i$.

We say a *GL* is *computationally coherent* if the symbolic dynamics induced by the geometric evolution commutes with readout, i.e., there exists $\Phi_s : \Sigma^N \rightarrow \Sigma^N$ with:

$$\mu(U(\sigma)) = \Phi_s(\mu(\sigma)) \quad (\text{discrete time}), \quad \text{or} \quad \frac{d}{dt} \mu(\sigma_t) = F_s(\mu(\sigma_t)) \quad (\text{continuous time}).$$

Remark 2.3. The architecture of geometric computability is decidedly a non-Von Neumann one. A GL is necessary to compute programs that are not trivial, since one can't discretely read out results on a single GCM with symbolic or quasi-fluidic computation, not without a projection layer or a symbolic "slice." In a GCM there is no separation of program and data, no instruction pointer, or fetch-decode-execute cycle, the geometry *is* the computation. Computation is intrinsic, continuous, and spatially distributed. While one can clearly see the similarity to cellular automata in the discrete case, parallels can be drawn to neural fields or reaction-diffusion systems if an outside example is required.

2.4 Achieving Turing Completeness in R4

Finite state machines (FSMs) are the building blocks of classical computation. A FSM consists of a quintuple $(S, \Sigma, \delta, s_0, F)$: a finite set of states S , a finite input alphabet Σ , a transition function $\delta : S \times \Sigma \rightarrow S$, an initial state $s_0 \in S$, and a designated accepting set $F \subseteq S$ (or an output map). Given an input word over Σ , the machine updates its state by iterating δ ; acceptance (or output) is determined by the terminal state.

Proposition 2.1 (FSM Embedding in Symbolic Mode). *In the symbolic regime, any finite-state machine $(S, \Sigma, \delta, s_0, F)$ embeds into a GCM tile with finite internal labels and symbolic interfaces.*

Sketch. Let a cell carry a finite internal label set $L \cong S$ (e.g. $L = C_2^k$ with a fixed encoding of S). Inputs are a finite set of discrete interface signals Σ (boundary toggles). Specify a local update table $\Delta : L \times \Sigma \rightarrow L$ implementing δ . Fix an initial label $\ell_0 \in L$ encoding s_0 . Choose a readout predicate $R : L \rightarrow \{0, 1\}$ and define the accepting set $F = \{\ell \in L : R(\ell) = 1\}$.

A single tile thus realizes the FSM; a single-dimensional lattice with nearest-neighbor symbolic communication realizes synchronous products. $\square$

Table 6 compares classical computational constructions with symbolic-regime geometric logic.

Table 6: Classical vs Geometric Symbolic Computation

Feature	Classical Computability	Geometric Computability (Symbolic Mode)
Objects	Strings, tapes, finite alphabets	Finite label sets on embedded units
Primitive state	Symbol / tape cell	Local geometric label
Operations	Rewrite rules; TM head updates	Finite group actions on local labels
Update law	Local transition function δ	Local transition table compiled to label updates (nearest-neighbor)
Readout	Symbol on tape / register bit	Fixed readout of label (e.g. parity/equality of components)
Constraints	Consistency of δ	Topological smoothness
Symmetry	Automata over discrete alphabets	Discrete subgroups acting on label space
Composition	Product of machines / CA	Lattice of tiles with symbolic interfaces (1D/2D CA)

Proposition 2.2. *A one-dimensional GL whose cells are THM_4 tiles with symbolic nearest-neighbor communication can simulate Rule 110.*

Proof. Each cell contains a twin helix with two binary orientation labels

$$(o_A, o_B) \in C_2 \times C_2,$$

and intrinsic orientation flips act by the Klein four group

$$V_4 \cong C_2 \times C_2 = \{e, s_A, s_B, s_A s_B\},$$

where s_A flips o_A , s_B flips o_B , and $s_A s_B$ flips both.

Readout/encoding. Fix the bit readout as the parity (XOR)

$$r(o_A, o_B) := o_A \oplus o_B \in \{0, 1\}.$$

Any fixed local encoding is acceptable; parity is chosen because s_A or s_B toggles r , while $s_A s_B$ preserves r .

Cellular update. Let $b_i^t := r(o_{A,i}^t, o_{B,i}^t)$ be the bit at site i and time t . Compute the Rule 110 local rule

$$b_i^{t+1} = \delta(b_{i-1}^t, b_i^t, b_{i+1}^t), \quad \delta = 110 \rightarrow 1, 101 \rightarrow 1, 011 \rightarrow 1, 010 \rightarrow 1, 001 \rightarrow 1, \text{ else } 0.$$

Now choose a group element $g_i^t \in V_4$ to enforce the new readout:

$$g_i^t = \begin{cases} e \text{ or } s_A s_B, & \text{if } b_i^{t+1} = b_i^t \text{ (preserve parity);} \\ s_A \text{ or } s_B, & \text{if } b_i^{t+1} \neq b_i^t \text{ (toggle parity).} \end{cases}$$

Apply the intrinsic update

$$(o_{A,i}^{t+1}, o_{B,i}^{t+1}) = g_i^t \cdot (o_{A,i}^t, o_{B,i}^t).$$

This uses only nearest-neighbor bits to compute b_i^{t+1} and a local V_4 action to realize it; hence, the induced dynamics on the readout $\{b_i^t\}_{i \in \mathbb{Z}}$ is exactly Rule 110.

By Cook's [3] theorem, such a lattice simulates a cyclic tag system and therefore a Turing machine. Consequently, the GL with THM_4 cells and the above encoding is Turing complete. $\square$

2.5 The Shape of Computation

A regular bit naturally lives on a Boolean number line, existing as 0 or 1. A qubit, as a state vector, naturally lives on the Bloch Sphere. Where does a geobit live? The answer is tentative. As we established, the group operations determine the number of bits used by a quantized GCM.

We now turn to a geometric source to construct a state-space visualization of the geobits of THM_4 . Polytopes are used to represent the state space of geobits. In this case, the vertices of the polytope are the THM states, and the edges are the group transformations.

Starting with V_4, D_4 , they can fit easily into T, O . Although they encode the same number of bits, O represents an additional flip bit and requires additional edges. Now, however, we need a subgroup for $(C_2)^4$. Consider the binary octahedral group $2O$. This group, together with its brothers $2T$ and $2I$, is a double-cover subgroup of the group of 3D rotations $SO(3)$. Given that we are primarily interested in representing a computational group of order 16, a group of order 48 is the best choice for clean subgroup embedding by Lagrange's Theorem.

Or, so it would seem. But a not often mentioned theorem leads to a paradoxical result that prevents a clean embedding of the THM computational group model into a state space of dimensionality congruent to the dimensions of its natural geometry.

Proposition 2.3 (Minimal Embedding Dimension for THM_{16} Symmetry Group). *A faithful embedding of the group THM_{16} (with symmetry group $(C_2)^4$) as a symmetry subgroup requires at least $SO(5)$.*

Proof. The Sylow 2-subgroups of the binary octahedral group $2O$ (a double cover of the octahedral rotation group in $SO(3)$) are isomorphic to the generalized quaternion group Q_{16} (the binary dihedral group $2D_4$). Q_{16} is non-abelian and contains elements of order 8, while $(C_2)^4$ is abelian and all non-identity elements have order 2. Moreover, $2O$ only has one central element of order 2 (corresponding to $-I$ in $SU(2)$), whereas $(C_2)^4$ requires 15 distinct elements of order 2. Therefore, $(C_2)^4$ cannot embed into $2O$ as a subgroup.

One might expect $SO(4)$ to be sufficient, since the geometry of THM_4 naturally lives in 4D. However, a direct analysis of the group's structure shows that this is impossible. It is a standard result in the theory of Lie groups [4] that the largest possible elementary abelian 2-subgroup of $SO(4)$ has rank 2, being isomorphic to the Klein four-group.

Hence, embedding $(C_2)^4$ as a symmetry group requires $SO(5)$ at minimum. Embedding $(C_2)^5$ would require $SO(6)$, but such a group does not arise in the THM progression, which becomes non-abelian above order 16 due to the use of semi-direct products.

Therefore, the state space embedding of THM_{16} necessitates a “dimensional jump”—manifesting emergent 5D algebraic behavior, despite the geometric construction living in 4D.³ $\square$

To embed all three original computational groups, we consider the full symmetry group of the 5-dimensional hypercube (the penteract), which is the Coxeter group B_5 . This group has order $2^5 \cdot 5! = 32 \cdot 120 = 3840$. The relevant rotation subgroup, called the *demipenteract*, is $D_5 \cong (C_2^4) \rtimes S_5$, of order 1920. This group contains the diagonal $(C_2)^4$ symmetry and subgroups isomorphic to D_4 and V_4 . The demipenteract has 16 vertices and provides a rigorous setting for realizing our algebraic structures on an actual Lie group.

Since it is not possible to faithfully visualize a 5-dimensional shape, we settle for visualizations of V_4 and the regular tetrahedral group T (of order 12, embedded in $SO(3)$). In these visualizations, the polyhedral points correspond to the allowable states of the GCM and serve as the vertices of a 3D Cayley graph. The edges of the polyhedron represent transitions between these states. Figure 2a displays such a rendering, with colorings chosen to highlight these symmetries.

Remark 2.4. This “discretized” Bloch sphere can be constructed by representing the state space T as a spherical tessellation. In this approach, each vertex corresponds to one of the four fundamental geobit states. At the same time, edges represent single group-symmetry operations, and faces correspond to minimal closed loops of these actions. Notably, only four colors are required, one for each geobit state, making the discrete structure explicit. Figure 2b illustrates this method. This discrete tessellation highlights the analogy between the continuous state space of quantum systems and the finite, symmetry-driven state space in symbolic geometric computation.

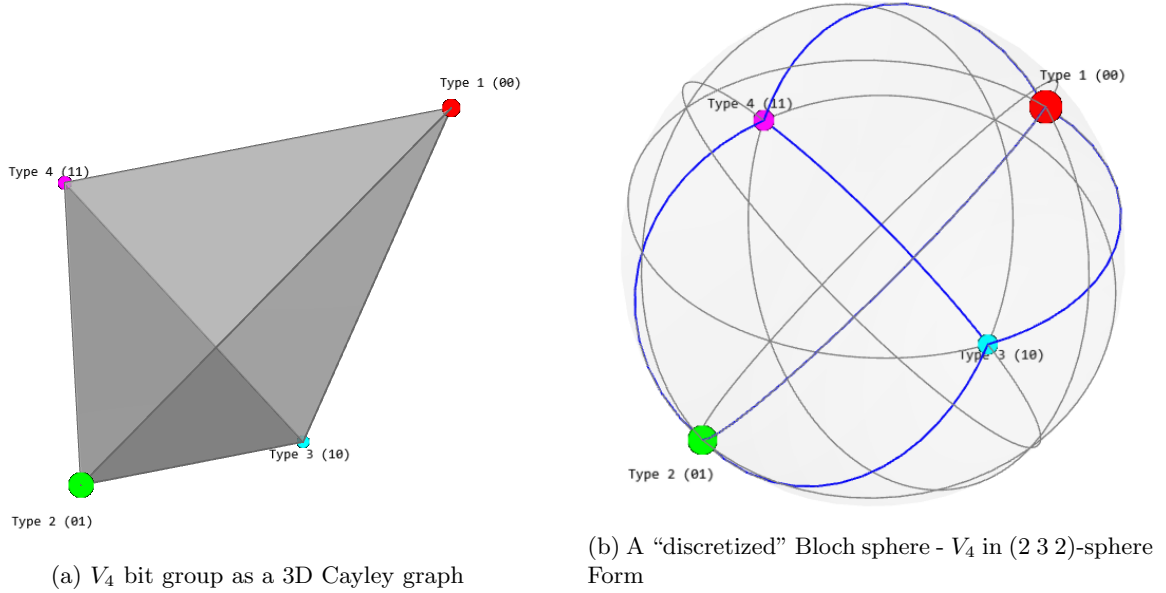


Figure 2: Different representations of the V_4 group.

Remark 2.5. The tetrahedral structure of the V_4 GCM state space is highly notable in light of results by Baez [6], who observed similar collapses in the quantum geometry of 3D vs 4D tetrahedra. However, in our case, this structure arises entirely from combinatorial flow logic, with no reference to spin-network quantization. This suggests a deeper relationship between geometry, computational state space constraints, and the topology of state polyhedra—one that may precede quantization itself. See 5.3.

2.6 The Logic of Geometry

GCMs can encode not only classical binary logic, but also three-valued logic and fuzzy logic. This is accomplished by introducing a third value to represent uncertainty, which is implemented by modifying the join angle between unit helices. This geometric logic can take two main forms:

³See Feynman’s *Lectures on Computation* [5, p. 111].

- (a) **Ternary/ n -ary logic** — where bits take values in $\{0, U, 1\}$ (with U an “undecided” value not computed, but defined on $[0, 1]$), or more generally in $\{0, 1, \dots, n-1\}$ for n -ary logic. For example, pentary logic uses values 0 through 4.
- (b) **Fuzzy logic** — where bits may take values in $\{0, P, 1\}$, where P is a real number in $[0, 1]$, computed by a function such as $f(\theta) = \cos^2(\theta)$. The specific choice of function f determines the logic’s behavior and can be tailored to computational requirements.

If the tangent surface between the two unit helices at the glue joint remains non-singular, then helices may be joined at arbitrary angles—not just 0° or 180° . This flexibility enables both n -ary and fuzzy logic encodings. For instance, simple three-valued logic is realized by a “trinary join” determined by a function:

$$f : (n \cdot 120^\circ) \mapsto \{0, U, 1\}$$

where n indexes the n -ary digit system, running from 0 to $n-1$. All angle measurements are taken from the positive z -axis, proceeding clockwise.

So far, we have deliberately excluded infinite groups from our analysis, for two distinct reasons:

- (a) Allowing arbitrary join angles breaks the natural algebraic group structure of THM_4 . Closure under group operations is lost when joins are not restricted to a finite set.
- (b) The cross section of unit helices would be forced into a single conic (the circle), limiting geometric expressiveness and precluding the use of ovals or more general curves.

Both forms of logic, however, entail distinct geometric, algebraic, and computational tradeoffs. Three- or four-valued logic preserves finite group structure, since 120° or 90° rotations allow for finite algebraic closure. Fuzzy logic, by contrast, requires infinitesimal join rotations and destroys all vestiges of finite group structure, in addition to enforcing geometric isotropy.

Given that there exists a fine-grained tradeoff between the capacity for geometric expressiveness, algebraic group closure, and logical computational power, **this suggests the existence of a conservation law**. Rather than drawing potentially dangerous analogies to Noether’s Theorem [7], it should be noted that instead of dealing with the process of how symmetries give conserved quantities, this is a conservation law for symmetry *itself*. That is, it measures how one gives up potential possibilities in geometries to gain symmetry.

Lemma 2.4 (Geometric Logic Conservation Lemma). *For any GCM G , there exists a conservation law between its algebraic and geometric computational capacities. Specifically, there is a constant K such that*

$$\mathcal{C}(G) + \mathcal{E}(G) \leq K$$

where $\mathcal{C}(G)$ denotes the algebraic closure of the logic gate set induced by G (i.e., which logical compositions remain valid under repeated joins or symmetries), and $\mathcal{E}(G)$ denotes a measure of geometric expressiveness (e.g., the number of distinct cross-sectional geometries supported by G). The constant K is determined by the topological and group-theoretic invariants of the system.

Here $\mathcal{C}(G)$ counts the $K(G)$ -invariant gate clone and $\mathcal{E}(G)$ counts $K(G)$ -orbit-distinguishable geometries at fixed readout; K depends only on topological and group-theoretic invariants of G . One path becomes a clean, algebraic infinite-state computation model. The other path devolves into an increasingly purely geometric, analog, continuous processor. A GL made of THM_4 sits at the cusp of this tradeoff. Its geometric expressiveness is relatively high—as we will see. In classical physics, symmetry preserves energy and momentum, but Lemma 2.4 in GCT preserves computability and algebraic closure within a tightly constrained topological economy.

Remark 2.6. The essential point is that the symmetry of the system’s configuration space directly induces an invariant quantity in the system’s dynamics. Every symmetry break in topology introduces a logical ambiguity, and vice versa. Although it is difficult to visualize, given that we are only in the initial regime of computation, one can imagine higher-order manifolds in which the joint can be deformed outright into asymmetric 2-manifolds. These higher-order manifolds would devolve into pure fluidic computability, leaving discrete logic altogether and entering increasingly abstract and chaotic forms of analog computation. See 4.1 for direct visual explanation.

Definition 2.6 (Canonically Oriented Submanifold). Let M be a smooth n -dimensional manifold and $\Sigma \subset M$ a smooth, connected k -dimensional submanifold. We say that Σ is canonically oriented if:

- (a) **Orientable.** The tangent bundle $T(\Sigma)$ is orientable in the usual sense (there exists a nowhere-vanishing k -form on Σ).
- (b) **Canonical choice of orientation.** There exists a smooth, nowhere-vanishing k -vector field $X \in \Gamma(\Lambda^k T\Sigma)$ that is distinguished by the geometry of Σ within M , in the sense that it is invariant under the whole group of geometric symmetries preserving Σ as an embedded object.

The distinction between orientable and non-orientable surfaces is well known. However, in our setting, we now require a slightly stronger definition of the helix, namely a “constantly-oriented manifold.”

Example 2.3 (Rules for Geometric Quantization). Any finite symmetry group of rotations of the circle, $C_n \subset U(1)$, enables n -ary non-Boolean logic in geometric computability. For instance, one can define a 5-valued (pentary) logic using C_5 —the cyclic group of order 5. More generally, we encode n -ary logic as follows.

Let $x \in \{0, 1, \dots, n-1\}$ be the index of the rotation step, and m_x the associated logical value.

$$f : x \mapsto m_x, \quad m_x \in \mathcal{L}_n, \quad x \in \{0, 1, \dots, n-1\}$$

where

$$\mathcal{L}_n = \{\ell_0, \ell_1, \dots, \ell_{n-1}\}$$

and the geometric correspondence is given by

$$\theta_x = x \cdot \frac{2\pi}{n}$$

so the angle θ_x represents the logical value m_x .

Thus, a rotation of the helix by θ_x encodes the logical value ℓ_x , allowing n -ary logic representation via angular encoding. This mapping may be compactly written:

$$f(x) = m_x = \ell_x, \quad \theta_x = x \cdot \frac{2\pi}{n}$$

Definition 2.7 (Orientation Bit Assignment for n -ary Unit Helices and ω -logic). Let the axis of symmetry of helix i be represented by a vector $\mathbf{v}_i = (v_x, v_y, v_z)$ in $\mathbb{R}^3$. We define a reference frame where the primary axis is $\hat{\mathbf{z}} = (0, 0, 1)$ and the reference direction for angle zero is $\hat{\mathbf{x}} = (1, 0, 0)$.

The n -ary logical state, or orientation bit $b_i \in \{0, 1, \dots, n-1\}$, is determined by the angle of the vector’s projection onto the xy -plane. This angle, θ_i , is calculated using the two-argument arctangent function:

$$\theta_i = \text{atan}^2(v_y, v_x)$$

The arctan^2 function returns a value in the range $(-\pi, \pi]$. We map this to the range $[0, 2\pi)$ to ensure positive angles:

$$\theta_{\text{norm}} = \begin{cases} \theta_i & \text{if } \theta_i \geq 0 \\ \theta_i + 2\pi & \text{if } \theta_i < 0 \end{cases}$$

The continuous state (or fuzzy value) s_i of the geobit is defined as the normalized angle itself, representing a value in the real interval $[0, 1)$:

$$s_i := \frac{\theta_{\text{norm}}}{2\pi} \in [0, 1)$$

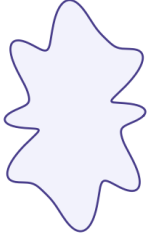
For n -ary discrete logic, the state b_i is derived by partitioning the continuous state space into n discrete bins:

$$b_i := \lfloor n \cdot s_i \rfloor = \left\lfloor n \cdot \frac{\theta_{\text{norm}}}{2\pi} \right\rfloor \in 0, 1, \dots, n-1$$

This is termed “ ω -logic.”

Table 7: The chart of n -logics

Geometric Expressiveness	Join Symmetry	Algebraic Closure Group	Result
Extreme	—	Diff(M) (Maximal)	Computational gauge field
High	1-fold (360°)	C_1	Geometric chaotic analog logic flow
Medium	2-fold (180°)	C_2	Classic boolean logic gate
Low	16-fold (22.5°)	C_{16}	Hexadecimal logic gate
None	n-fold (Infinite)	$U(1)$ (Lie)	Algebraic fuzzy analog logic flow



(a) Expressiveness in binary vs hexadecimal

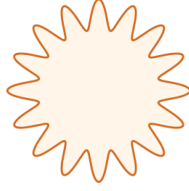
(b) n -logical gluing maps for—3,4,5-ary

Table 7 organizes several models, illustrating the relationship between join symmetry and the resulting computational system.

This pattern reveals an astonishing phenomenon: at the extremes, logical computational flow manifests in two analog forms—one chaotically geometric, the other excruciatingly algebraic. This suggests that low-value n -ary logic, such as Boolean logic, constitutes a kind of *logical habitable zone* where real machines can function in physical space, much like how our own planet is the hospitable zone in our solar system.

It is not merely an engineering quirk that noise and error rates make n -ary processors difficult to realize in practice. Even in an idealized, noiseless world, n -ary logical computers are “sandwiched” between two analog extremes. The physically stable and computationally tractable zone is inherently narrow. Those regimes that are C_2 and above will be termed *algebraic*, and those that are below this threshold will be termed *geometric* computational regimes.⁴

For n -ary logic, the cross-section of the *THM* can be any smooth closed manifold with at least n distinguished points on its boundary that realize the requisite “logical angularities.” There are, of course, infinitely many such shapes. The primary family consists of sinusoidal perturbations (“clovers”), whose cross sections can be globally smooth and are easily visualized in polar coordinates—see Figure 3b.

However, one can readily imagine more expressive cross sections that realize the required angular structure but offer even richer forms of expressiveness. Figure 3a illustrates such alternatives and compares expressiveness across different logical regimes.

Definition 2.8 (Geometric Expressiveness). Let $\mathcal{F}_1$ denote the space of all smooth, closed, simple curves (or hypersurfaces) in $\mathbb{R}^n$, parameterized up to some fixed regularity (e.g., finite-degree trigonometric polynomials, or a suitable Sobolev space). Let G_n be a symmetry group (e.g., the cyclic group C_n of order n), and let $\mathcal{F}_n \subseteq \mathcal{F}_1$ be the subspace of shapes invariant under the action of G_n .

We define the *geometric expressiveness* at symmetry level n as the normalized dimension:

$$E(n) = \frac{\dim(\mathcal{F}_n)}{\dim(\mathcal{F}_1)},$$

where $\dim(\cdot)$ denotes the functional dimension (e.g., the number of independent Fourier modes up

⁴Please note a slight conflict regarding nomenclature: although C_1 is an abelian group, its inability to define, at minimum, binary discreteness means it falls under a non-abelian computational regime.

to a fixed cutoff, or the dimension of a relevant function space). Thus $E(n) \in [0, 1]$ quantifies the proportion of the whole shape space available under the symmetry constraint.

Let $D(n, K)$ denote the dimension of the space of trigonometric polynomials of degree $\leq K$ and n -fold symmetry:

$$D(n, K) = 2 \left\lfloor \frac{K}{n} \right\rfloor + 1,$$

where the formula counts both cosine and sine harmonics, plus 1 for the constant. As n increases for fixed K , $D(n, K)$ decreases in a stepwise fashion.

For finite K ,

$$E(n) \approx \frac{2 \lfloor K/n \rfloor + 1}{2K + 1}.$$

The set of all 2π -periodic, smooth functions is infinite-dimensional. The subset of n -fold symmetric, smooth functions is also countably infinite-dimensional, but it becomes increasingly sparse as n increases. In the limit as $n \rightarrow \infty$, $E(n) \rightarrow 0$; the relative dimension shrinks to zero.

Furthermore, the total L^2 variation of $r(\theta)$ from a constant also decreases as n increases, since the only possible deviations are high-frequency harmonics, and these are increasingly penalized by the requirement of smoothness.

Definition 2.9 (Algebraic Verifiability). Let $\mathcal{G}$ be a GCM, and let $O = \text{Sym}(\mathcal{G})$ be the maximal group of algebraic symmetries acting faithfully on $\mathcal{G}$. The algebraic verifiability $\mathcal{V}(\mathcal{G})$ is a normalizable measure of the size and structure of O , representing the ability to distinguish and verify discrete computational states within $\mathcal{G}$. For a finite group,

$$\mathcal{V}(\mathcal{G}) = \log_2 |O|$$

A high $\mathcal{V}(\mathcal{G})$ corresponds to many, clearly distinguishable algebraic states (e.g., quaternary, hexadecimal), while a lower $\mathcal{V}(\mathcal{G})$ reflects fewer verifiable states or continuous/chaotic flows. For infinite or Lie groups, $\mathcal{V}(\mathcal{G})$ can be defined via the rank of a maximal discrete subgroup or another appropriate algebraic invariant (e.g., dimension of a maximal abelian subgroup).

Proposition 2.5. *The set of all closed, sufficiently smooth 1-manifolds in $\mathbb{R}^2$ reduces to the circle when subjected to arbitrary rotational symmetry constraints.*

Proof. A general sufficiently smooth closed curve in polar coordinates can be expressed as a Fourier series:

$$r(\theta) = a_0 + \sum_{k=1}^{\infty} [a_k \cos(k\theta) + b_k \sin(k\theta)].$$

Suppose the curve is required to be invariant under rotation by $2\pi/n$ for some integer $n > 1$:

$$r(\theta + 2\pi/n) = r(\theta) \quad \forall \theta.$$

This constraint eliminates all terms except those where k is a multiple of n , so only terms $\cos(kn\theta)$ and $\sin(kn\theta)$ can remain for integer k .

As $n \rightarrow \infty$, the only nonzero term that can survive is the constant a_0 . Therefore, in this limit, the curve reduces to:

$$r(\theta) = a_0,$$

so, under arbitrarily high-order rotational symmetry, the only possible closed 1-manifold in $\mathbb{R}^2$ is the circle. This proves Lemma 2.4. $\square$

This is the essential phenomenon as the size of the rotational symmetry group grows. Once the symmetry group becomes continuous, the system's logic becomes identical to its geometry. We stress that this is *not* merely a clever representation:

The computation and the manifold are outright one and the same.

This is the core idea of geometric computation, and it is why such systems can, in principle, exceed the computational power of a Turing machine—first by leveraging infinite algebra, then by accessing transcendental geometry. The inevitable result is that “fuzzy logic” in this setting naturally leads to a computational state space governed by a compact, simple Lie group—the ability to execute hypercomputation, not on an idealized machine but on a topologically stable medium.

Remark 2.7. It is natural to question what precisely the point of “geometric expressiveness” is. Why not make all cross-sections a circle? The shape of the actual helices does not matter for classical computation insofar as the manifold remains smooth. This is a valid point, but it illustrates that in the symbolic mode of operation, the true geometric versus algebraic tradeoffs cannot be fully realized. It is only as we move into the next steps of computation that the implications become clear. For now, it is taken at face value.

We now approach the bit space governed by chirality. While orientation in GCMs supports arbitrary n -valued logic through the structure of rotation groups of $U(1)$, chirality remains fundamentally locked to ternary. The only available values are left, right, and a *degenerate unknown state*, in which torsion itself vanishes. Imagine if you will, a solid but smooth-ended toroidal elbow, a paradoxical helix with no twist. It extends outward for a short distance from the glue joint, then upward at a smooth angle. It is an inflection point between chiralities.

This explicit geometric sacrifice once again embodies Lemma 2.4. Greater logical expressiveness at the expense of geometric sharpness. The resulting manifold supports ternary logic in the chirality subspace. This is termed “ χ -logic.”

Even in higher dimensions, no known geometry allows for a richer chirality logic within a single connected manifold. Any attempt to construct such a system merely produces additional degenerate regions, not genuinely new logical values. The “no twist” position in the chiral space is a sort of “smooth singularity”⁵ an inflection point between winding and unwinding. This asymmetry reflects the deeper topological distinction between angular (periodic) and chiral (binary with singularity) symmetries.

2.7 Syntax, Semantics, and Computation

We now synthesize three convergent lines of thought, alongside our geometric model.

First, following Felleisen’s account of expressiveness [8], we distinguish *micro-* from *macro-expressiveness*: adding new local states or notations versus adding genuinely new global, compositional power.

Second, in the compositional semantics of processes of Categorical Quantum Mechanics: Abramsky–Coecke, with graphical calculi surveyed by Selinger, the meaning of a composite is determined by the wiring of its parts; contextuality and entanglement arise as *compositional* phenomena rather than add-ons [9, 10]. This gives us a precise, syntax-agnostic way to talk about how meanings compose.

Third, our geometric/groupoid models instantiate these ideas concretely: GCMs and decorated paths make composition, context, and macrostate emergence *geometric*. Macrostates are equivalence classes of geometric/computational microstates; macro-expression corresponds to traversing or constructing new geometric cobordisms.

Finally, in the spirit of Baez–Stay’s “Rosetta Stone” perspective on physics, topology, logic, and computation [11], we regard computation itself as a process theory: objects are interfaces, morphisms are computations, and composition is gluing. In this language, the familiar tradeoffs become geometric/categorical phenomena rather than informal slogans. We do not merely interpret computation using geometry; we *construct the geometry of computation itself*. The following definitions guide our landscape.

Definition 2.10 (Micro-Expression). A language extension is *micro-expressive* w.r.t. a base language L if every program P in the extended language L' can be mechanically translated (compiled/macro-expanded) to a program P_0 in L without changing its semantic structure. Formally: for all $P \in L'$, there exists a derivable $P_0 \in L$ with equivalent semantics; the extension is syntactic sugar, not added power.

Definition 2.11 (Macro-Expression). A language extension is *macro-expressive* if it enables programs or control/data abstractions that cannot be obtained from the base language by any mechanical or structure-preserving translation. Formally: there exists $P \in L'$ such that no equivalent $P_0 \in L$ can be constructed by a uniform, semantics-preserving translation.

⁵I know this is imprecise, it is intended here to suggest a topologically regular but functionally critical transition point.

Felleisen’s framework makes this distinction precise and operational [8], and we adopt it as the semantic backbone for what follows. They define precise criteria for when a language extension is merely syntactic (micro-expressive, or “eliminable by mechanical translation”) versus when it adds genuine new expressive power (macro-expressive, or “semantically irreducible”). Just as in programming, where some features create behaviors not reducible to simpler forms—gluing submanifolds can create new topological invariants only if the gluing is smooth (singularity-free).⁶ As systems grow more complex, obstructions to smooth composition proliferate, paralleling the limits of macro-expression in computation.

Definition 2.12 (Verifiability). Let $\mathcal{S}$ be a computational/semantic system with predicates $\mathcal{P}$ over its states/behaviors. The *verifiability* $\mathcal{V}(\mathcal{S})$ is a normalized measure of how many *nontrivial* properties in $\mathcal{P}$ admit effective, pre-execution decision procedures. Equivalently: $\mathcal{V}(\mathcal{S})$ measures the size/structure of the subset of $\mathcal{P}$ consisting of properties decidable by algorithmic, structural, or analytic means (e.g., checkable invariants or geometric signatures), without brute-force evaluation.

Verifiability captures the capacity to certify properties *by construction*. In algebraic regimes, it correlates with rigid “ledgers” (e.g., line bundles/cohomology classes that distinguish states). Still, it is not identical to micro-expression: the former is semantic/algorithmic, whereas the latter is syntactic/generative. High verifiability corresponds to systems where many global or local properties can be decided by construction; low verifiability signals the presence of undecidable, chaotic, or non-algorithmic behavior.

In the process semantics of Abramsky–Coecke, contextual/entangled behavior is a *compositional* effect of how processes wire together. Our geometric program inverts the usual narrative: start from deterministic geometric computation, then see probabilistic/quantum-like structure *emerge* as the statistical/compositional shadow of complex gluing and flow.

Definition 2.13 (Tractability). Let $\mathcal{S}$ be a formal system (e.g., a programming language, a computational model, or a geometric computation system). Tractability is the property that there exists a procedure (algorithmic or semantic) to:

- For machines: Execute or evaluate a statement/program/configuration in $\mathcal{S}$ with feasible computational resources (e.g., polynomial time or some agreed practical bound).
- For humans: Understand, reason about, or verify the intent and behavior of statements in $\mathcal{S}$ without overwhelming cognitive complexity.

More precisely: A property or decision problem in $\mathcal{S}$ is tractable if it can be decided or realized with computational resources (time, space, etc.) bounded by a polynomial (or similarly ‘practical’) function of the input size. A system is tractable if most of its intended computations and the process of programming/understanding them fall within humanly or machine-manageable complexity.

Tractability, in our framework, refers to the ease with which both a machine can execute and a human can understand or reason about a computational statement in a formal system. For a programming language or system of symbols, tractability is both a computational and a cognitive phenomenon. The degree of tractability depends not only on algorithmic complexity but also on the alignment between the system’s formal structure and the way meaning is constructed and communicated among its users.

With these parts in place, we can now *macro-express* a new theorem.

Conjecture 2.6 (CORE THESIS: Expressiveness/Verifiability/Tractability (“EVT”) Hypothesis). *The conjecture formally comprises two parts:*

- (A) **Unverifiability of Macro-expressive Systems.** *Any system of computation supporting macro-expression is globally algorithmically unverifiable.*
- (B) **Expressiveness/Verifiability/Tractability Trilemma.** *Within such a system, every deterministic computational model is fundamentally constrained by a three-way tradeoff among expressiveness (geometric or semantic generality), verifiability (algebraic checkability or formal runtime*

⁶The geometric mismatches noted earlier explain why, in the language of Feynman diagrams, certain gluings fail: the logic of the bordisms must agree.

assurance), and tractability (algorithmic or computational efficiency). These attributes satisfy the following trilemma:

No system can simultaneously maximize all three properties. For any given model, at most two of the three can be optimized beyond a critical threshold, while the third must necessarily be sacrificed.

Formally, the achievable combinations of these properties are bounded by a 2-dimensional simplex in the space of attributes. The precise tradeoff is determined by the system’s local symmetries and structural constraints.

This is the central motivating result of this work.⁷ We take the phenomenon isolated in Lemma 2.4 and generalize it to deterministic computation in full generality. We claim that there is a principled tradeoff between *verifiable* (“algebraic”) and *expressive* (“geometric”) computational structure that governs *all* deterministic substrates once their categorical primitives (e.g. finite products, natural numbers objects, cartesian closedness or suitable enrichment) and their operational semantics (e.g. what counts as “tractable”) are made explicit. Conceptually, the result should be read as a computation-wide analogue of Lawvere’s fixed-point principle: fixed-point/diagonal phenomena are not an accident of syntax but a structural shadow of sufficient expressiveness.

The key to understanding lies in Rice’s theorem. In our framework, Rice’s conclusion is the natural consequence of the first half of the EVT Hypothesis: once a substrate is expressive enough to support self-interpretation and free composition, fixed points are endemic; extensional semantic predicates collapse to halting-hard instances. There is a deep parallel here with Gödel’s incompleteness. Rice says that every non-trivial extensional semantic property of partial computable functions is undecidable. Gödel says that for any consistent, sufficiently strong, recursively axiomatized theory, there are true arithmetical statements unprovable in that theory. These are two faces of the same diagonal/fixed-point mechanism—logic-side and computation-side—unified categorically by Lawvere-type fixed-point arguments (and, operationally, by Kleene’s Recursion Theorem together with the *s-m-n* theorem). What has been missing on the computational side is an explicit meta-theorem that, a priori, explains why “Rice-like” undecidability is generic once macro-expressiveness is admitted.

In any deterministic computing category that encodes programs and data within the same ambient semantics, admits the Recursion Theorem in operational form (programs can feed their own descriptions to program transformers), and evaluates properties extensionally on computed partial functions, every non-trivial semantic property is undecidable.

It is fair to say that theoretical computer science has long known Rice’s theorem and has many device-specific explanations, but lacks a single structural *meta-explanation* in the spirit of Lawvere’s logical fixed-point account. Here, we provide the missing layer: the EVT packages the conditions under which Rice is generic, and it quantifies the corresponding loss of verifiability unless expressiveness is constrained.

Mathematics contains undecidability by axiomatic constraints: many working theories are deliberately complete/decidable. Computation, by default, collapses program/data and admits unbounded self-reference; without constraints, one is always at the knife’s edge of diagonalization. In EVT language: classical machine models maximize expressiveness at the expense of global verifiability; categorical/typing disciplines trade some expressiveness for proof power and tractability.

The EVT generalizes Lemma 2.4 from our geometric case to arbitrary deterministic computation by identifying the structural preconditions under which Lawvere/Kleene fixed points become unavoidable. Under those preconditions, Rice’s theorem ceases to be a collection of ad hoc reductions and becomes the generic case; conversely, regimes that raise verifiability do so by withdrawing precisely the structural fuel that would otherwise reintroduce fixed points.

Recall the spectrum of *n*-logics generated from join symmetry 7. It is tempting to attribute the pervasiveness of undecidability to the use of binary logic: a two-symbol alphabet lies near a “chaotic edge,” offering high compositional freedom while providing few built-in algebraic constraints. However, the alphabet’s arity is not entirely to blame for the phenomenon that underpins Rice’s theorem. What matters for diagonalization is not *micro-expressiveness*, but *macro-expressiveness*. Any finite *n*-ary digital substrate with these macro features exhibits the same Rice-style barriers.

In principle, one could seek to purchase additional verifiability in hardware by enforcing richer symbolic alphabets. In practice, device physics, noise margins, and error-correction overwhelmingly

⁷Yes, I understand Yang-Mills and QG are also important. Please bear with me here. It all ties together in the end.

favor binary encodings, and more importantly, even ternary or quaternary devices recover Turing universality as soon as conditionals, memory, and unbounded composition are present. Thus, changing n does not neutralize diagonalization. To mitigate Rice in a principled way, one withdraws macro-expressiveness: forbid general recursion, stratify effects, or impose linear/affine and resource-sensitive typing. This is precisely how proof assistants based on Coquand’s Calculus of Constructions [12] control semantics: they trade macro-expressiveness for verifiability.⁸

Lawvere’s diagonal/fixed-point insight [13] isolates the structural engine behind self-reference: in a setting with sufficient cartesian closure and a weak surjectivity condition, endomaps acquire fixed points. Operationally, Kleene’s Recursion Theorem and the $s-m-n$ theorem implement the same mechanism for programs. In the EVT Hypothesis, we adopt the following viewpoint: the “fixed point” is not an accident of syntax but the *computation itself*—more precisely, the availability of an evaluator that can be fed its own description. Once a deterministic substrate is macro-expressive enough to encode and compose its own interpreters, self-reference becomes endemic; most additional macro constructs preserve or amplify this capacity. Turing’s halting proof is then a canonical *instance* of the general meta-principle later made explicit by Lawvere/Kleene, not a disconnected trick.

While Rice’s theorem is often paraphrased as “all non-trivial semantic properties are undecidable,” its force is conditional on macro-expressiveness and extensionality. When those preconditions are *removed* (e.g. in total, strongly normalizing calculi), large classes of semantic questions become decidable, even though some equivalence and extensional properties may remain hard. The EVT packages this calibration: increasing macro-expressiveness drives the system toward fixed points and pervasive undecidability; engineering for verifiability withdraws precisely the structural fuel that diagonalization consumes.

The second half of the EVT Hypothesis cleanly explains the odd “dual nature” observed in functional programming languages. On the one hand, languages with strong static typing offer compile-time guarantees (type safety, parametricity principles), at the cost of meta-linguistic restrictions during construction (Haskell). On the other hand, dynamically typed languages maximize meta-linguistic expressiveness at construction time (macros, reflection), while giving up static guarantees at runtime and shifting assurance to testing and contracts (Scheme).

Bridging attempts proceed along multiple axes: *gradual typing* (Typed Racket) inserts types with higher-order contracts at typed/untyped boundaries, and *staging* (MetaOCaml) separates compile-time code generation from runtime execution. From the proof-theoretic direction (Coquand), total dependent type theories recover strong guarantees by forbidding general recursion. From the pragmatic language research side, (Felleisen) shows how large untyped codebases can be migrated toward sound types via occurrence typing and higher-order contracts—at the cost of explicit blame and potential runtime overhead—an archetypal EVT tradeoff between meta-linguistic freedom and verifiability.

The EVT explains why these bridges are inherently delicate, as when many powerful features co-exist, three generic pressure points appear:

- (a) **Static complexity** — creates extremely rich type features (GADTs, type families, dependent types) that can render inference undecidable, forcing annotations; combining dependency with effects or staging threatens phase separation unless carefully constrained.
- (b) **Compile-time liveness** — can cause macros or multi-stage generation to diverge at compile time; expressive compile-time evaluators reintroduce the very fixed-point phenomena the EVT warns about.
- (c) **Runtime boundaries** — means gradual typing preserves type safety via contracts, but introduces blame and potentially unbounded overhead; interoperation across typed/untyped modules trades static guarantees for dynamic checks.

These are not accidents of particular implementations; they are manifestations of the EVT tradeoff: increased *macro-expressiveness* (reflection, self-interpretation, unconstrained composition) strains *global verifiability*, unless restrictions restore structure. Hence, an explicitly stated conservation law for computational behavior explains these difficulties.

The reason this meta-theorem has gone unformulated is that its essential tradeoff has long been hidden in plain sight—embedded in the very architecture of classical computation we all use in our

⁸One may picture an n -ary geometric computer as a “hardware proof assistant” in a heuristic sense: local states may be rich (*micro*-expressive), but the key control of undecidability comes from the global discipline that rules out unconstrained self-interpretation and general recursion.

day-to-day lives. The Von Neumann architecture, by treating program and data uniformly and allowing maximal flexibility (any memory can store code or data, and programs can modify themselves), enables *extraordinary* expressiveness. However, this very flexibility comes at the cost of global verifiability: it becomes almost impossible to reason about or prove properties of the system as a whole. When the hardware is instead fixed algebraically—restricting the set of possible states, operations, or transitions—the system gains predictability and becomes globally checkable, but loses expressive power.

The tradeoff does not assert a single conservation equality. There exist regimes with both low expressiveness and low verifiability—the infamous “Turing tarpit,” (e.g., bash, Malbolge) where everything is theoretically possible but practically inexpressible. The EVT is better read as a *no-free-lunch inequality* with phases: beyond a threshold of macro-expressiveness, fixed points become endemic, and Rice-type barriers are pervasive; to regain verifiability, one must withdraw some of that structure. Between these extremes lie transitional corridors in which small feature additions can cause qualitative shifts in what can be verified.

Because these tradeoffs depend on ambient structure, the EVT treats *local* symmetries (of a model or architecture) rather than postulating a global invariant across all substrates. In a Von Neumann setting, a surprisingly crucial local symmetry is cognitive: the language’s conventions and composition patterns induce de facto invariants that transiently “abelianize” parts of the system (routines, schemas), while reflective features reintroduce non-commuting structure. Hence, on a classical machine, one of the most consequential local symmetries is inside that of the skull of the programmer.

Remark 2.8. Regarding tractability, hardware is less of a concern for geometric computation. If one had a stacked array of GLs, they could form a hyperaware, self-modeling FPGA capable of self-synthesis and logical introspection, switching between discrete and analog computational modes at will. In this regime, even the concept of a universal compiler or processor breaks down: every “program” defines its own hardware. In real life, there exist physical corollaries to Rice’s Theorem and Lawvere’s Fixed Point, natural limits on the engineering of computation itself in the form of finite, well-defined instruction set architectures, clear separation between compilation and execution, and static boundaries between code and metacode. As a result, although the entire trilemma must be stated as-is, we will not explore tractability as extensively in this paper, as it is more of a physical constraint for humans and the machines we build than for the universe.

But in a GCM, modeling the expressiveness/verifiability relationship becomes much more tractable, since there is *no separation between logic and memory*, and tractability is built in, thereby forcing tradeoffs between expressiveness and verifiability to the forefront. A GL is an idealized version of a *non-Von Neumann* architecture existing on a substrate defined purely by math—nothing but raw group theory, geometry, and topology. By fusing logic and language into an analog form, here one can *see* how the operational semantics forcibly deforms the computational substrate, and model how the algebraic closure group eats away at the possible expressions in geometry with near infinite amounts of n -ary logic with every increasing bit!

As Baez writes in *Getting to the Bottom of Noether’s Theorem* [7]

“Atiyah [14] warned us of the dangers of algebra: “...*algebra is to the geometer what you might call the Faustian offer. As you know, Faust in Goethe’s story was offered whatever he wanted (in his case the love of a beautiful woman), by the devil, in return for selling his soul. Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine.*”

Baez notes that “While this is sometimes true, algebra is more than a computational tool: it lets us express concepts in a very clear and distilled way.” While we do agree that the best of mathematics occurs when the two are brought together to work harmoniously, our contention here is that computability is, unfortunately, one particular case of the bargain where the devil shall not be cheated. Under the watchful eye of the ZFC, algebra and geometry can play nicely together, but the act of computation (in our case—the “gluing”) *enforces constraints on how* the flow can be structured.

But why physics? It is reasonable for readers to ask why a metatheorem about computation is developed in a physics paper. Why do this? Two pragmatic reasons guide this choice. First, differential

geometry and geometric analysis provide a language in which *flows, invariants, and gluing* are native: variational principles, conservation laws, and well-posed evolution already live there. Second, many of the sharpest constraints on computation are *physical*—finite precision, energy/entropy budgets, and measurement back-action—so a formulation that is testable against laboratory and numerical practice is preferable to one that remains purely syntactic (e.g. in terms of abstract syntax trees and Galois connections).

This does not diminish the interest of the EVT Hypothesis for computer science, education, or the philosophy of natural laws. Versions of the underlying tension have appeared in many domains, though rarely isolated as a single, explicit selection principle. A few recurring patterns (illustrative, not exhaustive) are:

- **Logic and programming.** As previously stated, total/strongly normalizing calculi gain verification power at the price of general recursion; reflective/metaprogramming gains flexibility at the cost of global guarantees.
- **Numerics and PDEs.** Stability constraints (e.g. CFL-type conditions) and structure-preserving schemes (e.g. symplectic integrators) enforce invariants but restrict admissible updates; aggressive damping improves robustness while destroying time-reversibility and fine invariants.
- **Many-body/field theory.** Integrable corners admit exact charges and factorization but are measure-zero in parameter space; generic regimes are expressive yet analytically resistant. Undecidability phenomena (e.g. spectral-gap questions) underscore that some semantic properties of dynamics cannot be decided uniformly.

But the most pressing applications of a *computational* selection principle are presently in *quantum gravity* and *lattice gauge theory*, where one must extract dynamics from large, ongoing physical computations (experiments or simulations) under resource and precision constraints. The EVT trilemma is intended to bound *how* such processes can behave and what can be recovered from them, independently of any particular quantization scheme.

Mathematics has Gödel–Tarski style incompleteness, and computer science has the halting problem and Rice-type no-go results. Physics, despite its rich family of conservation laws, lacks a comparably explicit bound on the *simultaneous* attainability of global verifiability and unrestricted expressive power in *physical* information-processing. Various desiderata for such a bound have been voiced (e.g. calls for a “Gödelization” of fundamental theory in public remarks), and adjacent programs (such as Constructor Theory [15]) have sought axioms phrased in terms of counterfactual capabilities; our starting point diverges in ontology but shares the motivation.

The EVT Hypothesis is proposed as a *selection principle* for the behavior and modeling of computational processes: physical computation—discrete or analog—is not substrate-independent magic; it is a *gauge-constrained flow* carried by a geometric substrate, with invariant quantities and resource monotones that cannot be simultaneously optimized without limit. From this viewpoint, the structures that make computation possible (*how* we compute) are as consequential as the extensional outcomes (*what* we compute). Any physical theory of computation should, at a minimum, respect this geometric constraint.

3 Kleene Computation and Geometry

Stephen Kleene’s work in the 1930s–40s established the modern theory of computability via the partial recursive functions, providing a machine-independent account that undergirds Turing machines and the Church–Turing thesis. In this section, we re-express GCT within Kleene’s formalism. The point is twofold: (a) partiality, products, and groupoid symmetries appear naturally in our setting; (b) the presentation is platform-independent and does not commit us to a specific syntactic calculus (e.g., λ -calculus).

Our strategy is constructive. Concretely, we give semantics of primitive recursion inside our geometric category and then extend to partial maps (Kleene’s μ -operator) in a way compatible with our groupoid symmetries. The categorical summary (via a Lawvere–theory view of primitive recursion and a partial-maps completion) will follow the operational definitions and examples below.

3.1 Fibered Kets and Product States

To track both the observable and internal degrees of freedom of a geobit, we use a notation reminiscent of 2.4 but extended. This is purely organizational: it packages a pair of states, one observable and one internal. When needed, we functorially linearize to a genuine Hilbert space; until then, the symbol $\otimes$ denotes the cartesian product of state spaces.

Definition 3.1 (Fibered Ket for Geobits). Let A be the (finite) observable state space encoding chirality and orientation, and let B be the internal state space (fluidic variables, or other hidden degrees of freedom). The *state object* is

$$S := A \times B.$$

A *fibered ket* is a notational pairing:

$$|\alpha\rangle \otimes |\beta\rangle \hat{=} (\alpha, \beta) \in S,$$

where:

- (a) $|\alpha\rangle \in A$ encodes the observable component (chirality/orientation).
- (b) $|\beta\rangle \in B$ encodes the internal component (e.g. fluidic or topological data).
- (c) the ambient state space S is an object of our geometric category $\mathcal{G}$, which is enriched by a groupoid of symmetries (see Section 4.6).

An elementary update (or local move) is a morphism $T : S \rightarrow S$, written on kets as

$$(|\alpha\rangle \otimes |\beta\rangle) \mapsto |\alpha'\rangle \otimes |\beta'\rangle.$$

We write $G \triangleright T$ for the right action of an update T on a state $G \in S$ (so $G \triangleright T := T(G)$). When T factors by components, we write $T = T^\alpha \otimes T^\beta$ with $T^\alpha : A \rightarrow A$ and $T^\beta : B \rightarrow B$; otherwise T may couple the factors.

Remark 3.1. Throughout this section, the symbol $\otimes$ in the fibered ket notation is purely mnemonic and denotes the cartesian product of state spaces in our geometric category $\mathcal{G}$. No linear or inner-product structure is assumed, and no topological tensor product is used. If genuine tensor products are ever needed, they should be introduced explicitly; they play no role in the development of primitive recursion or partiality here.

Operationally, we will regard a GCM as an object subjected to computational flows: endomorphisms of S generated by finite vector fields or discrete moves. The recursor we define below iterates such updates, matching Kleene's scheme of composition and primitive recursion. In the categorical explanation that follows the examples, this will materialize as preservation of products and the natural-numbers-object (NNO) recursor in $\mathcal{G}$, with partiality handled by the partial-maps completion.

3.2 Partial Recursive Functions—Primitives and Composition

Classically, the partial recursive functions over $\mathbb{N}$ are generated from three primitives:

Zero $Z(\vec{x}) = 0$.

Successor $S(x) = x + 1$.

Projections $P_k^n(x_1, \dots, x_n) = x_k$.

and three closures:

Composition given $f : \mathbb{N}^k \rightarrow \mathbb{N}$ and $g_i : \mathbb{N}^n \rightarrow \mathbb{N}$, define $h(\vec{x}) = f(g_1(\vec{x}), \dots, g_k(\vec{x}))$.

Primitive recursion given $f : \mathbb{N}^m \rightarrow \mathbb{N}$ and $g : \mathbb{N}^{m+2} \rightarrow \mathbb{N}$, define $h : \mathbb{N}^{m+1} \rightarrow \mathbb{N}$ by

$$h(0, \vec{x}) = f(\vec{x}), \quad h(n+1, \vec{x}) = g(n, h(n, \vec{x}), \vec{x}).$$

Unbounded μ -recursion given $g : \mathbb{N}^{n+1} \rightarrow \mathbb{N}$, define the partial map $f : \mathbb{N}^n \rightarrow \mathbb{N}$ by

$$f(\vec{x}) = \mu y [g(\vec{x}, y) = 0],$$

i.e. the least y (if any) with $g(\vec{x}, y) = 0$.

We begin with the three primitive constructors of partial recursive functions. Let $(N_G, 0_G, S_G)$ be an NNO in $\mathcal{G}$; write N_G^n for products.

(a) **Zero.** For each arity n ,

$$Z_n := !; 0_G : N_G^n \longrightarrow 1 \longrightarrow N_G,$$

the constant arrow to 0_G .

(b) **Successor.**

$$S_G : N_G \longrightarrow N_G.$$

(c) **Projections.**

$$\pi_k^n : N_G^n \longrightarrow N_G, \quad \pi_k^n(x_1, \dots, x_n) = x_k.$$

Example 3.1 (Zero as a Reset). Fix a readout $\text{tw} : B \rightarrow N_G$ that extracts a discrete orientation count (or any integer-coded internal variable), and a setter $\text{set0} : B \rightarrow B$ that overwrites this count with 0. Define the local move $T_0^\beta : B \rightarrow B$ by $T_0^\beta(\beta) = \text{set0}(\beta)$ and act trivially on A . Then

$$(|\alpha\rangle \otimes |\beta\rangle) \triangleright (\text{id}_A \otimes T_0^\beta) = |\alpha\rangle \otimes |\text{set0}(\beta)\rangle.$$

At the controller level, the associated numeric output is $Z_n(\vec{x}) = 0_G$, independent of inputs, matching the logical role of the zero function.

Example 3.2 (Successor as a Single Rotation). Let $\text{tw} : B \rightarrow N_G$ be as above and let $\text{inc} : B \rightarrow B$ implement $\omega(\text{inc}(\beta)) = \omega(\beta) + 1$. Define $T_{+1}^\beta : B \rightarrow B$ by $T_{+1}^\beta = \text{inc}$ and act trivially on A :

$$(|\alpha\rangle \otimes |\beta\rangle) \triangleright (\text{id}_A \otimes T_{+1}^\beta) = |\alpha\rangle \otimes |\text{inc}(\beta)\rangle.$$

On the numeric side, this is governed by $S_G : N_G \rightarrow N_G$, i.e. the successor on the orientation counter.

Example 3.3 (Projection as Readout/Selection). Given an input tuple $(x_1, \dots, x_n) \in N_G^n$ serving as control parameters (e.g. planned numbers of twists, swaps, or steps), the projection π_k^n selects the k th control

$$\pi_k^n(x_1, \dots, x_n) = x_k.$$

Operationally, one can route $\pi_k^n(\vec{x})$ to an update family $\{T_y\}_{y \in N_G}$ to choose which local move to apply next; no change to the current geobit state is implied by projection itself.

Primitive Composition. Write $\pi_A : S \rightarrow A$ and $\pi_B : S \rightarrow B$ for projections. Fix a total map $v : B \rightarrow B$ (e.g. a one-step flow update).

$$\beta' := v(\beta).$$

Define a discrete controller: $\sigma : B \rightarrow \{+1, -1\}$ by the sign of (some readout of) β' :

$$\text{sgn}(\beta') := \begin{cases} +1, & \text{if the chosen readout of } \beta' \text{ is } > 0, \\ -1, & \text{otherwise.} \end{cases}$$

Any two-valued partition works; sign is a convenient choice. Observable update gated by the controller. Define $f : A \times \{+1, -1\} \rightarrow A$ by

$$f(\alpha, +1) = \alpha, \quad f(\alpha, -1) = -\alpha.$$

Then the observable component updates as

$$\alpha' := f(\pi_A(\alpha, \beta), \text{sgn}(v(\pi_B(\alpha, \beta)))).$$

The whole composite step is the composite arrow:

$$T := \langle f \circ \langle \pi_A, \text{sgn} \circ v \circ \pi_B \rangle, \quad v \circ \pi_B \rangle : S \rightarrow S,$$

which is of the classical PRF composition form $f \circ \langle g_1, \dots, g_k \rangle$.

Example 3.4 (Flow-controlled chirality flip). Take initial state $|\alpha\rangle \otimes |\beta\rangle = | +1 \rangle \otimes | 0.7 \rangle$. Let $v(\beta) = \beta - 1$ (one-step slowdown). Then

$$\beta' = v(0.7) = -0.3, \quad \text{sgn}(\beta') = -1,$$

followed by:

$$\alpha' = f(+1, -1) = -1.$$

Thus the updated state is $| -1 \rangle \otimes | -0.3 \rangle$.

Basic linear encoding. Choose a two-point basis for A : $| +1 \rangle = (1, 0)$, $| -1 \rangle = (0, 1)$, and a finite code for flow levels in B , e.g. $| \text{low} \rangle = (1, 0, 0)$, $| \text{med} \rangle = (0, 1, 0)$, $| \text{high} \rangle = (0, 0, 1)$, together with a coarse-graining $q : B \rightarrow \{\text{low}, \text{med}, \text{high}\}$ (so $q(0.7) = \text{med}$, $q(-0.3) = \text{low}$). Then the pair (α, β) is represented (after linearization) by the Kronecker product:

$$| +1 \rangle \otimes | \text{med} \rangle = (1, 0) \otimes (0, 1, 0) = (0, 1, 0, 0, 0, 0),$$

and updates act by block-permutation matrices implementing f on A and v (then q) on B .

Recursive composition. Fix an initializer $i : X \rightarrow S$ (the geometric constructor from inputs $\vec{x} \in X$), and a family of local moves $T_n : S \rightarrow S$ indexed by $n \in N_G$. Each T_n is an endomorphism of S (and in reversible regimes, an automorphism). We write right action as $G \triangleright T := T(G)$.

Define the state sequence by

$$G_0(\vec{x}) = i(\vec{x}), \quad G_{n+1}(\vec{x}) = G_n(\vec{x}) \triangleright T_n.$$

If T_n factorizes by components we write $T_n = T_n^\alpha \otimes T_n^\beta$; in general T_n may be a coupled update on $A \times B$. In fibered-ket notation

$$|G_n\rangle = |\alpha_n\rangle \otimes |\beta_n\rangle, \quad |G_{n+1}\rangle = (T_n^\alpha |\alpha_n\rangle) \otimes (T_n^\beta |\beta_n\rangle),$$

with the obvious coupled variant. After k steps

$$|G_k\rangle = (\cdots ((|G_0\rangle \triangleright T_0) \triangleright T_1) \cdots) \triangleright T_{k-1}.$$

Equivalence with the PRF recursor. Let $u : S \times N_G \rightarrow S$ be the step map $u(s, n) = T_n(s)$. The natural-numbers-object (NNO) recursor yields a unique

$$h = \text{rec}(i, u) : X \times N_G \rightarrow S$$

such that $h(\vec{x}, 0) = i(\vec{x})$ and $h(\vec{x}, n+1) = u(h(\vec{x}, n), n)$. Then for all k , $G_k(\vec{x}) = h(\vec{x}, k)$.

Example 3.5 (Commuting Flips on Separate Factors). Let $|\Psi_0\rangle = | \text{left} \rangle \otimes | \text{up} \rangle$. Let $C : A \rightarrow A$ swap $\text{left} \leftrightarrow \text{right}$ and $O : B \rightarrow B$ flip $\text{up} \leftrightarrow \text{down}$. Define $T_0 = C \otimes I$ and $T_1 = I \otimes O$. Then

$$|\Psi_1\rangle = (C \otimes I) |\Psi_0\rangle, \quad |\Psi_2\rangle = (I \otimes O) |\Psi_1\rangle.$$

Hence:

$$|\Psi_2\rangle = (I \otimes O)(C \otimes I) |\Psi_0\rangle = (C \otimes O) |\Psi_0\rangle,$$

since C and O act on different factors and therefore commute.

Unbounded μ -recursion is interpreted as geometric search and partiality. We interpret Kleene's unbounded search inside the partial-maps completion $\text{Par}(\mathcal{G})$. Fix a total $g : N_G^{n+1} \rightarrow N_G$ in $\mathcal{G}$ (a decidable numeric predicate with 0_G as “true”). The *partial* map:

$$\mu(g) : N_G^n \rightharpoonup N_G$$

is defined on $\vec{x}$ iff there exists y with $g(\vec{x}, y) = 0_G$, and then

$$\mu(g)(\vec{x}) = \min\{y \in N_G \mid g(\vec{x}, y) = 0_G\}.$$

Constructively, $\mu(g)$ is realized as the increasing union of the finite-search partial maps $w_k : N_G^n \rightarrow N_G$ defined by primitive recursion:

$$w_0(\vec{x}) = \begin{cases} 0 & \text{if } g(\vec{x}, 0) = 0_G, \\ \text{undefined} & \text{otherwise,} \end{cases}$$

$$w_{k+1}(\vec{x}) = \begin{cases} w_k(\vec{x}) & \text{if } w_k(\vec{x}) \text{ is defined,} \\ k+1 & \text{if } g(\vec{x}, k+1) = 0_G, \\ \text{undefined} & \text{otherwise.} \end{cases}$$

Then $\mu(g)$ is the partial map whose graph is $\bigcup_k \text{graph}(w_k)$.

Geometric reading. For inputs $\vec{x} \in N_G^n$, an initializer $i : X \rightarrow S$ chooses a starting state, and a family of local moves $U_y : S \rightarrow S$ (or a step map $u : S \times N_G \rightarrow S$) generates candidates

$$|G_y\rangle = U_y(i(\vec{x})).$$

A geometric halting predicate $H : S \rightarrow N_G$ (e.g. “symmetry satisfied”, “entered special region”) defines $g(\vec{x}, y) := H(|G_y\rangle)$. Then $\mu(g)(\vec{x})$ is the least y with $H(|G_y\rangle) = 0_G$ if such a y exists; otherwise the search is undefined at $\vec{x}$ (diverges).

Example 3.6 (Divergence via Mutual Triggering in a Minimal GL). Consider two neighboring GCMs A and B with state factors $|\alpha_\bullet\rangle \in A$ (orientation: up/down) and $|\beta_\bullet\rangle \in B$ (continuous flow). The joint state is

$$|\Psi\rangle = (|\alpha_A\rangle \otimes |\beta_A\rangle) \otimes (|\alpha_B\rangle \otimes |\beta_B\rangle) \in (A \times B) \times (A \times B).$$

Let $T : (A \times B) \times (A \times B) \rightarrow (A \times B) \times (A \times B)$ be the local transition that, at each step

- increases β_A ; if it crosses the threshold 1, flips α_A and triggers an increase of β_B ;
- symmetrically for B (threshold 1, flip α_B , trigger A).

Set $|\Psi_{n+1}\rangle := T|\Psi_n\rangle$ and let the halting predicate be $H(|\Psi\rangle) = 0_G$ iff a designated stop condition holds (e.g. both flows = 0, or a marked symmetry cell reached). If T perpetually re-triggers the threshold crossings, then for all n , $H(|\Psi_n\rangle) \neq 0_G$ and the search never halts

$$|\Psi_0\rangle \xrightarrow{T} |\Psi_1\rangle \xrightarrow{T} |\Psi_2\rangle \xrightarrow{T} \dots$$

Hence for the corresponding $g(\vec{x}, y) := H(|\Psi_y\rangle)$ one has $\neg \exists y (g(\vec{x}, y) = 0_G)$, so $\mu(g)(\vec{x})$ is “undefined” as a partial arrow in $\text{Par}(\mathcal{G})$. This realizes unbounded μ -recursion geometrically as an endless sequence of fibered-ket transitions with no fixed point.

3.3 Kleene Computability and Computational Groupoids

For a compact categorical definition to encapsulate our PRF correspondence, we turn to a Lawvere category.

Definition 3.2 (Lawvere Theory for Primitive Recursion). Let LPR be the small cartesian category freely generated by an object N with arrows $0 : 1 \rightarrow N$, $S : N \rightarrow N$, finite products, and the primitive recursion operator characterized by the usual universal property.

Definition 3.3 (Geometric NNO). In our semantic category $\mathcal{G}$ of geometric computations, a natural numbers object is an object N_G with maps $0_G : 1 \rightarrow N_G$, $S_G : N_G \rightarrow N_G$ such that for all X , $f : X \rightarrow N_G$, $g : X \times N_G \rightarrow N_G$ there exists a unique $\text{rec}(f, g) : X \rightarrow N_G$ satisfying the recursion equations.

Theorem 3.1 (Representation of PRF). *There exists a strict cartesian functor $F : \text{LPR} \rightarrow \mathcal{G}$ with $F(N) = N_G$, $F(0) = 0_G$, $F(S) = S_G$, preserving products and primitive recursion. Consequently, every primitive recursive function $f : \mathbb{N}^k \rightarrow \mathbb{N}$ is realized by a unique morphism $F(\ulcorner f \urcorner) : N_G^k \rightarrow N_G$.*

Corollary 3.2. *Zero, successor, projections, composition, and primitive recursion are realized in $\mathcal{G}$.*

Let $\text{Par}(\mathcal{G})$ denote the partial maps category over $\mathcal{G}$. It is cartesian, inherits N_G , and admits a (partial) unbounded search operator μ for decidable PR predicates. Thus:

Theorem 3.3 (Kleene Normal Form in $\text{Par}(\mathcal{G})$). *There exist primitive recursive T, U in $\mathcal{G}$ such that every partial recursive ϕ_e is represented as $\phi_e(\vec{x}) = U(\mu y. T(e, \vec{x}, y) = 0)$ in $\text{Par}(\mathcal{G})$.*

Since all partial recursive functions are defined, Kleene’s Normal Form is now in effect. Following this, reversible, state-dependent micro-moves are modeled by a groupoid: $\mathcal{H} \rightrightarrows S$ acting on the geobit state object $S = A \times B$: an arrow exists only when the move is admissible for that source state, and all arrows are invertible (local symmetries, chart changes, reversible twists).

Irreversible evolution (lossy updates, non-existence of a next state, or unbounded search) is not a groupoid phenomenon. We model it in the partial-maps completion $\text{Par}(\mathcal{G})$: objects are those of $\mathcal{G}$, arrows $X \rightarrow Y$ are partial morphisms (e.g. represented by spans or via the partiality monad $1 + -$). The μ -operator lives here.

Remark 3.2. Groupoids capture where reversible moves are allowed, no arrow means forbidden; $\text{Par}(\mathcal{G})$ captures when computation can legitimately be undefined (divergence, crash, or absence of a solution).

We formalize “programs that can act on their own code” using only the Cartesian structure, the NNO, and partial maps, without any additional analytic structure. Fix a state object S in $\mathcal{G}$. Let C be an object of “codes for realizable partial endomaps of S ”, equipped with an *evaluator*:

$$\text{eval} : C \times S \rightarrow S \text{ in } \text{Par}(\mathcal{G}).$$

Assume there is a surjective coding $\langle - \rangle : N_G \rightarrow C$ and a *universal* partial evaluator:

$$U : N_G \times S \rightarrow S \text{ with } \text{eval}(\langle e \rangle, s) = U(e, s).$$

Finally, assume that total computable transformers on codes are represented by total arrows $\psi_C : C \rightarrow C$ induced from total $\psi_N : N_G \rightarrow N_G$ through $\langle - \rangle$. These hypotheses are standard in realizability/PRF semantics: the NNO supplies internal arithmetic and Gödel coding; the μ -operator in $\text{Par}(\mathcal{G})$ supplies unbounded search; U plays the role of a universal machine.

Theorem 3.4 (Internal Kleene Recursion). *For every total computable $\psi_C : C \rightarrow C$, there exists a code $c^* \in C$ such that*

$$\text{eval}(c^*, -) = \text{eval}(\psi_C(c^*), -) \quad \text{as partial endomaps } S \rightarrow S.$$

Proof sketch. Work internally over N_G . By the usual s - m - n construction for partial recursive functions (available from $\text{PR} + \mu$), there is a total $\mathbf{s} : N_G \rightarrow N_G$ such that: $U(\mathbf{s}(e), s) = U(\psi_N(e), s)$ for all e, s . Take $e^* = \mathbf{s}(e^*)$ given by Kleene’s classical diagonalization (internalized via the NNO), and set $c^* = \langle e^* \rangle$. Transporting along $\langle - \rangle$ yields the displayed equality for eval . $\square$

Corollary 3.5 (Geometric Quine). *Let $W : S \rightarrow S$ be the partial update that writes a presented code into a designated register of S (and then proceeds). Define the code transformer ψ_C to “prepend W ” to any code’s behavior. By 3.4, there exists c^* with $\text{eval}(c^*, -) = \text{eval}(\psi_C(c^*), -)$, so running c^* causes the system to write (an encoding of) c^* into its own state before continuing a canonical geometric quine.*

The following section develops the analytic layer, including groupoid flows and ODE-driven updates that act as total or partial arrows, with existence/uniqueness assumptions made explicit. The discrete PRF/ μ semantics remain the controlling interface; the continuum dynamics provide realizations of those controllers in geometric time.

4 Lie Theory and Abelian Gauge Computation

Recent work on the computability and complexity of fluid flows (e.g. Cardona et al. [16]), together with Tao’s conjectural program on universality phenomena for the Euler equations [17], highlights a recurring theme: *continuous media can support computation*, but often only through dynamics whose analytic behavior is difficult to control.

In this section, we introduce the *quasi-fluidic band* as an *abelian gauge* mode of computation. Informally: a GL is a lattice of geometric cells whose internal degrees of freedom carry abelian phase

data (typically $U(1) \simeq S^1$, or a finite cyclic approximation C_n), together with discrete labels and quantized counters.

As established earlier, a geometric computer is neither strictly a Turing machine nor a quantum computer. Quantum computation is organized around a global state vector and coherence constraints; locality and scaling are mediated by entanglement. By contrast, a GL scales by composition: cells are wired together by interfaces and boundary conditions, and the operative degrees of freedom are abelian phases and discrete readouts that compose functorially across the lattice. There are no complex amplitudes and no Born-rule collapse; instead, one reads gauge-invariant features of geometry (e.g. phase differences and winding).

As a motivating example, consider the classic experiment in which a hose filled with running water is placed across a vibrating speaker. The resulting jet forms a visible oscillating pattern: the driving frequency controls coil tightness, amplitude controls the spiral radius, and phase controls the initial offset. This is an instructive caricature of the quasi-fluidic regime: periodic forcing induces an effective $U(1)$ -phase along a geometric substrate, and the observable shape provides a coarse readout of that phase. In the language adopted later (4.2), the phase offset belongs to the T -layer (S^1 data), while winding/twist counts correspond to quantized Z -invariants; discrete readouts arise from coarse partitions of these quantities. Helical and twisted structures appear here not as an aesthetic choice, but as energetically natural carriers of phase and twist under periodic driving. A visual demonstration by brusspup [18] makes this phenomenon concrete.⁹

If one could reliably cross or couple streams of such geometric bordisms (as in THM -style constructions), one would obtain a physical intuition for a lattice of GCM modules coupled by local interfaces and driven by global pulses.

Remark 4.1 (Computational Geometric Bordisms). The vibrating-hose experiment may be viewed as an analog of a *computational geometric bordism*: a smooth evolution mediating between boundary conditions (fixed endpoints, imposed phase/frequency, initial and final configurations). Here, the bordism is “internal” to the geometry of the evolving flow, and the computable content is not the PDE solution itself but the stable, gauge-invariant features one can read from it. See Section A.4 for cobordism formalisms.

Proposition 4.1. *The geometric topology of a GCM constrains its information-bearing degrees of freedom, but the effective information content is relative to a chosen readout (coarse-graining). In particular, geometric computation is deterministic at the microstate level, while information is defined relative to an encoding.*

Unlike in classical computation—where the bit is treated as an absolute, context-free primitive—in geometric computation, the notion of a “geobit” is always specified with respect to a readout map (or partition) on the underlying state space. Concretely, one fixes an observable or coarse-graining

$$\rho : \Sigma \longrightarrow B,$$

where Σ is the (possibly structured) microstate space of the GCM and B is a finite label set used for symbolic computation. Once ρ is fixed, information becomes an objective property of the pair (Σ, ρ) : the dynamics may be fully deterministic on Σ , yet the induced evolution on B depends on which features are chosen as readable and which are treated as hidden or irrelevant.

This is directly analogous to physics, where measurements and reference choices enter through specified observables rather than through subjectivity. The boundary between symbolic and quasi-fluidic computation is therefore not a matter of taste. Still, of modeling choice: symbolic regimes arise when the operative degrees of freedom are purely discrete (a finite B with static invariants), while the quasi-fluidic regime appears when one enlarges the local fiber to include compact abelian phases and quantized topological counters (the $B/T/Z$ structure of Definition 4.2). Geometric deformation then permits interpolation between discrete and continuous logic *without* introducing probabilistic amplitudes.

This does not mean the quasi-fluidic regime is vague. In many ways, it has already appeared implicitly: even in earlier symbolic constructions, evolution was driven by an explicit action stream (global pulses and local update schedules) acting on finite geometric labels. The quasi-fluidic mode

⁹Regarding an unconventional reference; the point is purely pedagogical: it is a compact way to visualize phase, winding, and coarse readout in a physical system.

makes this action stream geometric by allowing abelian phase transport and thresholded topological events, while retaining the core principle of GCT: computability is realized by local, deterministic transition structure rather than by solving global differential equations.

4.1 A New Kind of Computation

A central motivation of GCT is to avoid the classical reliance on global PDE evolution as the *definition* of computation over continuous media (fluids, fields, elastic bodies). As Tao has repeatedly emphasized, even well-posed PDE models may develop loss of regularity, extreme sensitivity to initial conditions, or finite-time singular behavior; these analytic pathologies are not merely technicalities, but can become *computationally relevant* in unfavorable ways.

Historically, analog and fluidic devices exploited continuous physics to compute. Mechanical differential analyzers and wartime fire-control machines, as well as later fluidic and hydraulic integrators, demonstrate that a physical medium can implement nontrivial transforms. Yet this tradition also exposes a basic paradox of analog logic: the computational substrate is not insulated from the mathematics it is tasked with realizing. If the governing model admits blow-up, stiffness, or chaotic amplification, then the physical device inherits the same fragility. In contrast to digital machines—which often fail noisily via discrete faults—*analog computation can fail by gradual loss of identifiability: the state remains well-defined physically while becoming unreadable with respect to the intended encoding.*

GCT responds by shifting the locus of computability from *solving equations to enforcing structure*. A GCM is designed so that the fundamental computational step is a local, deterministic update at an interface: an equivariant transformation of a structured state fiber (in the quasi-fluidic band, the $B/T/Z$ decomposition of Definition 4.2), driven by controlled inputs (pulses, boundary perturbations, or local couplings). The ambient dynamics may be smooth, but the computation is carried by the admissible morphisms of the system—what changes, what cannot change, and what invariants must persist. In this sense, GCMs are naturally piecewise smooth: not because we approximate a PDE, but because computation is organized around local transitions and thresholded events rather than global analytic integration.

Accordingly, we do not attempt to compute Navier–Stokes or Euler solutions by “bit-flips” internal to the manifold. Instead, we encode logic into the topology, phase structure, and allowable deformations of embedded geometry, and we read outputs through fixed observables (coarse-grainings) of that geometry. In a GL, the computer is the problem, and the problem is the solution.

This structural viewpoint is not without precedent. Goriely and collaborators, whose earlier work on nonlinear geometry and helices we will cite later, have recently advocated a related shift in applied modeling: replacing monolithic PDE descriptions by networked, local interaction rules in neuroscience and biophysics. Their work illustrates how complex global behavior can arise from local coupling laws on a discrete architecture, providing a concrete example of how one may trade PDE-centric descriptions for robust, compositional dynamics (see, e.g., Thompson et al. [19]).

4.2 Fluidic State Systems

We now introduce the notion of a *Fluidic State System* (FSS). The purpose of this definition is not to abandon symbolic computation, but to formalize the fact that GCM dynamics are not purely discrete: even in the quasi-fluidic band, evolution can depend on continuous parameters (phases), boundary conditions, and geometric deformations. Classical finite-state machines (FSMs) are therefore an insufficient primitive if one wishes to treat flow and geometry as part of the computational mechanism rather than as an external implementation detail.

At scale, computation in a GL emerges from the coordinated action of many such units coupled by local interfaces. In this setting, μ -recursion should be understood as a *structured search* over geometric configurations and their induced readouts, driven by local update laws and global pulses, and terminating when a designated halting condition is detected at the readout layer.¹⁰ In the same way that FSMs serve as a basic building block for discrete computation, an FSS serves as a basic building block for quasi-fluidic and fluidic computation in GCT.

Definition 4.1 (Fluidic State System). An FSS is a dynamical system consisting of a smooth ambient manifold L equipped with a finite or countable family of distinguished regions $\{\mathcal{R}_i\}$, a family of

¹⁰In later work, the fully non-abelian (gauge) regime may require a further generalization (e.g. a “Topological State System”) in which state is tracked up to gauge equivalence.

continuous flows $\{\Phi_t\}$ governing evolution on L , and a collection of transition interfaces $\mathcal{I}$ implementing local update laws. Inputs and outputs may be realized as boundary conditions, applied deformations, or perturbations of the driving flow field. Formally, an FSS is a 4-tuple

$$(L, \{\mathcal{R}_i\}, \{\Phi_t\}, \mathcal{I}).$$

Elaborated:

- (a) **State.** L is the ambient lattice/manifold supporting the computation. A subregion $L_x \subseteq L$ may be interpreted as a connected collection of cells coupled by interfaces and transported by flows. Each cell carries a structured local state (in the quasi-fluidic regime, a $B/T/Z$ -type fiber), and in the concrete twin helix model one may take the basic geometric coordinates to include chirality (χ), orientation (ω), and optional geometric invariants such as twist/winding data (τ) and scale parameters (e.g. a global radius r). The global state is the assignment of these local states across the lattice.
- (b) **Input.** External influence enters through the driving family $\{\Phi_t\}$, interpreted as a smooth vector field (or controlled pulse) applied to the ambient geometry. In practice this could be mechanical torque, acoustic forcing, optical pulses, or other physically realizable perturbations. The role of the input is to drive local evolution; it is not itself a stored register.
- (c) **Transition.** As input acts, cells deform and local states update. Mathematically, this is modeled by smooth evolution on L together with local update maps at interfaces. Each interface $I_{ij} \in \mathcal{I}$ is supported on a non-empty portion of the common boundary $\partial\mathcal{R}_i \cap \partial\mathcal{R}_j$ and is specified by a smooth map

$$I_{ij} : \Sigma_i \times \Sigma_j \times \mathbb{R}^d \longrightarrow \Sigma_i \times \Sigma_j,$$

where Σ_i and Σ_j are the state fibers carried by $\mathcal{R}_i$ and $\mathcal{R}_j$, and $\mathbb{R}^d$ packages any local data extracted from the ambient drive (e.g. flux, shear, curvature, or phase increment). Operationally, I_{ij} is a local rule that:

- reads the current state on both sides of the interface,
- samples the instantaneous drive along the interface,
- outputs the updated states after an infinitesimal time step.

Thus, $\mathcal{I}$ provides the compositional “logic gates” that couple neighboring cells.

- (d) **Output.** A result may be signaled either by convergence to a stable geometric configuration (a recognizable fixed point in the readout) or by the emission of an observable propagating feature (e.g. a traveling pulse) that can be read downstream. In both cases, the output is a property of the final geometry together with the chosen readout.

The conceptual map in Table 8 is provided for the ease of the reader.

Table 8: Classical vs. Quasi-Fluidic computation

Classical Computation	Geometric Computation — Quasi-Fluidic Mode
Finite State Machine (FSM)	Fluidic State System (FSS)
Turing Machine (TM)	GCM Lattice (GL)
Bit	Geobit
State space S	Local fiber $\Sigma_i \cong B_i \times T_i \times Z_i$
Transition function δ	Interface update maps I_{ij} (equivariant local rules)
Clock/step semantics	Driving pulses / flow family $\{\Phi_t\}$
Tape/wires	Interfaces and coupling geometry (adjacency/contiguity)
Input alphabet Σ	Boundary conditions/perturbation channel
Output	Readout $\rho : \Sigma \rightarrow B$ (observable manifold projection)

Definition 4.2 (Quasi-Fluidic Computation and Regime Phases). Given an FSS, we say a GCM operates in the *quasi-fluidic regime* if each region R_i carries a finite-dimensional state fiber:

$$\Sigma_i \cong B_i \times T_i \times Z_i,$$

with the following structure:

- (QF1) **Discrete core (symbolic).** B_i is finite, typically $B_i \cong (C_2)^{k_i}$; in the twin helix unit one has $k_i = 4$ for chirality/orientation bits.
- (QF2) **Phase layer.** T_i is a continuous but compact abelian Lie group (a torus), $T_i \cong (S^1)^{m_i} \simeq U(1)^{m_i}$, encoding axial phase(s), orientation angles, or other bounded “analog” degrees of freedom.
- (QF3) **Quantized invariants.** Z_i is a finitely generated abelian group (e.g. $\mathbb{Z}^{P_i}$), recording twist/winding/linking numbers or other topological charges.
- (QF4) **Abelian symmetry and equivariance.** The local symmetry acting on Σ_i is

$$K_i \cong (C_2)^{k_i} \times (S^1)^{m_i},$$

and every interface map $I_{ij} \in \mathcal{I}$ and the induced flow on Σ_i are K -equivariant, that is, they depend on phase differences and commute with flips/rotations.

- (QF5) **Flow coupling.** The ambient flow $\{\Phi_t\}$ transports the T -coordinates continuously (phase evolution) and can modify Z only by isolated, energy-threshold events (phase slips) prescribed by the interface rules; B updates by the same local rules. No non-abelian monodromy is present.
- (QF6) **Non-singularity and bounded energy.** Trajectories have no finite-time blow-ups; curvature/torsion remain bounded; state updates at interfaces are well defined for all t .

A system is symbolic when $m_i = 0$ and Z_i is static; it is purely fluidic when B_i is trivial and non-abelian holonomy appears. The quasi-fluidic band sits between these: it combines a finite discrete core with $U(1)$ -type phases and integer topological counters, all governed by abelian symmetry.

4.3 Composition of Quasi-Fluidic GCMs

In the symbolic model, a GL is composed of *discrete interconnect*: neighboring GCMs exchange a finite readout and update by a local rule, precisely as in a cellular automaton with richer internal state. In the quasi-fluidic regime, this picture remains valid. Still, each cell now carries an additional compact phase and a quantized invariant structure (the $B/T/Z$ fiber of Definition 4.2), so the internal dynamics of a single unit are no longer adequately described by a purely discrete transition table.

There is also a second, more geometric notion of composition which becomes natural in this band: *flow-circuit composition*. Rather than identifying adjacency inside a single ambient lattice, one regards a GCM as a module with an *input interface* and an *output interface*, together with a local update law. The module, therefore, induces a (generally state-dependent) transfer map on the relevant readouts and phase variables, and connecting two modules corresponds to wiring output interfaces into input interfaces. At the level of these transfer maps, composition is ordinary function composition.

Operationally, “wiring” is encoded by the choice of interfaces $\mathcal{I}$ and their coupling strength: one may couple two neighboring regions by a nontrivial interface map I_{ij} , or effectively *disconnect* them by selecting a trivial (e.g. identity/no-op) interface, or by imposing a boundary condition that forces the coupling term to vanish. In this way, reconfiguration is expressed within the same functorial framework: one modifies the *interface structure* (the allowed local morphisms) rather than introducing an additional intrinsic “undefined” state of the geobit. This viewpoint aligns with the guiding principle of the quasi-fluidic regime: computation is carried by the admissible local moves and their compositions, while the ambient drive (pulses, boundary perturbations) supplies controlled input to those moves.

4.4 Abelian Lattice Gauge Dynamics in the Quasi-Fluidic Mode

In the QF band, the appropriate “computation” physics of we should work with is not a discrete logic model as the symbolic band but an *abelian lattice gauge dynamics*: compact phases transported and coupled locally, with evolution governed by a bounded local energy and (optionally) a driven-damped relaxation law. This section is *interpretive*, it provides a stable dynamical model for analyzing QF behavior, without elevating differential equations to the definition of geometric computability.

Lattice variables (QF state). Let $\Gamma = (V, E)$ be the interaction graph of a GL (cells/modules are vertices, interfaces are edges). For each cell $i \in V$, we take an internal phase (orientation layer)

$$\theta_i \in U(1) \cong \mathbb{R}/2\pi\mathbb{Z},$$

and for each interface $(i, j) \in E$ we take a *link phase* (a discrete connection 1-form)

$$A_{ij} \in \mathbb{R}/2\pi\mathbb{Z}, \quad A_{ji} = -A_{ij},$$

equivalently $U_{ij} := \exp(iA_{ij}) \in U(1)$. (If one wishes to remain in the finite cyclic subband, replace $U(1)$ by C_N and interpret all equations below as a continuous approximation to a stroboscopically driven C_N update.)

Gauge redundancy. A $U(1)$ gauge transformation is a choice of angles $\alpha_i \in \mathbb{R}/2\pi\mathbb{Z}$ at each cell,

$$\theta_i \mapsto \theta_i + \alpha_i, \quad A_{ij} \mapsto A_{ij} + \alpha_i - \alpha_j,$$

under which the *gauge-invariant phase difference*

$$\Delta_{ij} := \theta_i - \theta_j - A_{ij} \pmod{2\pi}$$

is unchanged. In this sense, “independence of local modules” is a choice of gauge/frame, while couplings are encoded invariantly by Δ_{ij} .

Local energy (interface functional). A minimal gauge-invariant energy on Γ is

$$\mathcal{E}(\theta, A) = \sum_{(i,j) \in E} J_{ij} (1 - \cos(\Delta_{ij})) + \sum_{p \in \mathbf{P}} \kappa_p (1 - \cos((dA)_p)), \quad (1)$$

where $J_{ij} \geq 0$ are interface couplings and $\mathbf{P}$ is a chosen set of plaquettes (2-cells) with discrete curvature

$$(dA)_p := \sum_{(i,j) \in \partial p} s_{p,ij} A_{ij} \pmod{2\pi}, \quad s_{p,ij} \in \{\pm 1\}$$

the oriented sum around ∂p . The first term penalizes a mismatch between neighboring phases modulo the link variable; the second penalizes nonzero curvature (“flux”).

Gradient-flow dynamics. A simplest well-behaved evolution law is driven by gradient flow on the compact state space:

$$\dot{\theta}_i = -\gamma_\theta \frac{\partial \mathcal{E}}{\partial \theta_i} + \eta_i(t), \quad \dot{A}_{ij} = -\gamma_A \frac{\partial \mathcal{E}}{\partial A_{ij}} + \eta_{ij}(t), \quad (2)$$

with mobilities $\gamma_\theta, \gamma_A > 0$ and external drives $\eta_i(t), \eta_{ij}(t)$ (global pulses, boundary forcing, or localized injections). Concretely, the Euler-Lagrange components are

$$\frac{\partial \mathcal{E}}{\partial \theta_i} = \sum_{j \sim i} J_{ij} \sin(\Delta_{ij}), \quad (3)$$

$$\frac{\partial \mathcal{E}}{\partial A_{ij}} = -J_{ij} \sin(\Delta_{ij}) + \sum_{p \ni (i,j)} \kappa_p s_{p,ij} \sin((dA)_p), \quad (4)$$

so (2) relaxes toward gauge-invariant consistency ($\Delta_{ij} \rightarrow 0$) and low curvature ($(dA)_p \rightarrow 0$) unless forced otherwise. Because the variables are compact and $\mathcal{E}$ is bounded below, this regime is dynamically stable and aligns with the QF “no blow-ups” philosophy.

Remark 4.2 (Scope and Interpretation). Equations (1)–(2) are not proposed as fundamental laws of GCT; they are a *normal form* for the abelian corridor: local couplings, compact phases, gauge redundancy, and dissipative relaxation. In particular, continuous flows of this type do *not* implement jump-style updates by themselves; any discrete “computational” layer is induced later by readout/partitioning (return maps).

Auxiliary geometric ODE variables. To retain an explicit connection to helix geometry, one may attach slowly varying, bounded geometric parameters to each cell,

$$r_i(t) \in \mathbb{R}_{>0} \quad (\text{effective radius/scale}), \quad (5)$$

$$\tau_i(t) \in \mathbb{R} \quad (\text{effective twist rate}), \quad (6)$$

$$\mathcal{L}_i(t) \in \mathbb{R}_{>0} \quad (\text{effective length}) \quad (7)$$

and couple them to *local energy density* and/or *local phase current*. A simple choice is:

$$\dot{r}_i = -\lambda_r (r_i - r_0) + \alpha_r \sum_{j \sim i} (1 - \cos(\Delta_{ij})), \quad (8)$$

$$\dot{\tau}_i = -\lambda_\tau (\tau_i - \tau_0) + \alpha_\tau \sum_{j \sim i} \sin(\Delta_{ij}), \quad (9)$$

$$\dot{\mathcal{L}}_i = -\lambda_L (\mathcal{L}_i - \mathcal{L}_0) + \alpha_L r_i |\tau_i|, \quad (10)$$

with $\lambda_\bullet > 0$ ensuring relaxation to baseline values $(r_0, \tau_0, \mathcal{L}_0)$ and $\alpha_\bullet$ setting coupling strength. These equations encode the informal idea that mismatch/coupling stress can thicken the effective geometry (8), phase transport can bias twist (9), and geometry co-scales (10). One may also let the mobilities in (2) depend on τ_i (twist as a local “clock” factor), e.g. $\gamma_\theta = \gamma_\theta(\tau_i)$, without changing the gauge structure.

Remark 4.3 (Event Exclusions in the Pure QF band). In the strict QF corridor, one typically *excludes* threshold events (phase slips/defects) so that integer topological counters remain frozen and the dynamics stays smooth and energy-bounded. Allowing such events corresponds to leaving the pure abelian corridor toward more expressive regimes. The return-map/readout layer (introduced next) is the mechanism by which discrete computation is recovered without invoking singular PDE behavior.

4.5 Lie Theory and Vectorial Computation

The QF dynamics of Section 4.4 are continuous (compact phases with local energy and gradient flow). As such, they do *not* natively implement jump-style updates. The bridge back to classical symbolic computation is therefore *stroboscopic*: we sample the system once per pulse and apply a *discretized readout* (a quantizer) to obtain an induced update on a finite alphabet. In this sense, discrete computation in the QF band is a *return-map phenomenon*, not an ODE phenomenon.

Definition 4.3 (Pulse Times and Stroboscopic Map). Fix a pulse period $\Delta t > 0$ and define $t_n := n\Delta t$. Let $P : L \rightarrow L$ denote the time- Δt map on the QF state space L induced by one pulse:

$$P := \Phi_{\Delta t}.$$

We write $x^n \in L$ for the post-pulse state at time t_n , so $x^{n+1} = P(x^n)$.

Because the abelian corridor carries gauge redundancy, any robust readout should be phrased in terms of *gauge-invariant local observables*. In the lattice-gauge normal form, the basic invariants are:

$$\Delta_{ij} = \theta_i - \theta_j - A_{ij} \pmod{2\pi} \quad \text{and} \quad (dA)_p = \sum_{(i,j) \in \partial p} s_{p,ij} A_{ij} \pmod{2\pi}.$$

We therefore define a cellwise observable $y_i = y_i(x)$ as a finite list of such invariants in a bounded neighborhood (e.g. the Δ_{ij} on links incident to i , and optionally nearby plaquette fluxes $(dA)_p$).

Definition 4.4 (Discretized Readout/Quantizer). Fix a finite alphabet B and a (piecewise constant) quantizer

$$Q : \mathcal{Y} \longrightarrow B,$$

where $\mathcal{Y}$ is the range of local invariants y_i . The induced discrete configuration for a state $x \in L$ is

$$b = \text{Read}(x) \in B^V, \quad b_i := Q(y_i(x)).$$

For example, one may partition S^1 into arcs and threshold a chosen invariant angle, or partition the flux $(dA)_p$ into bins. The specific partition is a modeling choice.

Definition 4.5 (Vector Cellular Automata on an Induced Symbolic Return Map). Write x_i^n for the local continuous state in cell i at pulse n (phases θ , links A , and any auxiliary variables such as $r, \tau, \mathcal{L}$). Because the QF energy and interfaces are finite-range, there exists a finite neighborhood $\mathcal{N}(i)$ such that the stroboscopic update is local:

$$x_i^{n+1} = \mathcal{U}\left(\{x_j^n\}_{j \in \mathcal{N}(i)}\right), \quad x_i^n \in \mathbf{X},$$

for some compact state manifold X (a product of circles and bounded geometric variables). This is precisely a *vector cellular automaton* (continuous alphabet, local rule, synchronous clock).

Composing with the readout gives an induced discrete return map on B^V :

$$\mathcal{F}_B := \text{Read} \circ P \circ \text{Lift} : B^V \longrightarrow B^V,$$

where Lift selects a representative microstate $x \in L$ consistent with a discrete labeling (e.g. by choosing canonical phases inside each bin).

Remark 4.4 (What is “Computed”?). In the QF corridor, the *continuous* dynamics are the physical substrate, and the *symbolic* computation appears only after (a) stroboscopic sampling and (b) a discretized readout. This is the same conceptual move used in electronics: continuous circuit laws are real, while digital logic is the stable quotient obtained by thresholding and clocking.

The $U(1)$ Corridor and an E/V Paradox. It is at this point necessary to issue some disclaimers regarding the use of the full S^1 state spaces, that is, the $U(1)$ corridor. As established, within the quasi-fluidic band, the ω -logic layer may be taken either as a finite cyclic alphabet C_N or as a continuous phase $U(1)$. Passing to the full S^1 limit is mathematically tidy (abelian, compact), but it also oddly backfires, creating a regime with atypical E/V trade-off behavior. The following remarks record the current hypothesis of this paper:

- **Low geometric expressiveness.** Pure $U(1)$ phase composition is commutative. As a result, sequencing power collapses: one cannot exploit non-commuting compositions to build richer path-dependent behavior, and the phase layer alone tends to support only limited families of smooth, energetically bounded dynamics. In this corridor, most “computational complexity” must therefore be carried either by the discrete core B (via readout and interface thresholds) or by leaving the abelian setting (e.g. via threshold events/topological defects or non-abelian holonomy).
- **Low algebraic verifiability.** Any *finite* readout must discretize phase (explicitly or implicitly) by thresholds. In the $U(1)$ corridor, near-resonant phase evolution makes this discretization brittle: small phase perturbations can move trajectories across readout boundaries, producing significant changes in the inferred discrete state. Moreover, phase-locking reduces redundancy (many degrees of freedom share a common “clock”), weakening independent cross-checks.¹¹
- **Transition, not destination.** For these reasons, the $U(1)$ phase limit is best regarded as a *critical corridor*: visually orderly and analytically tractable, yet brittle under discretized readout and shallow in purely geometric expressiveness unless supplemented by discrete structure. It serves as a staging layer for robust computation on B and as a boundary marker before the onset of genuinely non-abelian (fluidic/gauge) behavior.

4.6 Application of Lie Groupoids to the Quasi-Fluidic Regime

As the state space of transformations in large abelian (C_n) geobit groups begins to resemble compact simple Lie groups in the “fuzzy” logic limit, and since Kleene’s theory of partial recursion suggests groupoids as the most natural structure for partial transformations, it is natural to ask: Can these perspectives be unified? The answer is yes. The concept of a *Lie groupoid*—first introduced by Charles Ehresmann in the 1950s—provides precisely this synthesis.

A Lie groupoid is, informally, a groupoid in which the space of objects and the space of morphisms are smooth manifolds, and all structural maps (source, target, identity, inversion, and composition) are differentiable. Like Lie groups, they are built on differentiable manifolds. Still, as groupoids, they admit a more flexible local symmetry structure: rather than a single global group action, we have objects and invertible morphisms between them. The classification of orbits and stabilizers in a Lie groupoid encodes a richer computational structure, offering a spectrum of intermediate computational classes bridging the digital/symbolic and analog/fluidic extremes.

A definition is formulated below that grounds the underlying fundamentals of a GCM, a category whose morphisms are manipulations of homotopies “decorated” by the underlying geometry.

¹¹Ironically, $U(1)$ has *too many* microstates, and this points toward a concrete mathematical reason why large-scale analog computers were mostly abandoned by the late 1950s.

Definition 4.6 (S–decorated Groupoid). Let $(S, *)$ be a monoid with unit e , and write $\mathbf{BS}$ for the one-object category whose endomorphism monoid is S with composition given by $*$. An *S–decorated groupoid* is a pair $(\mathcal{G}, d)$ consisting of a small groupoid $\mathcal{G}$ and a functor

$$d : \mathcal{G} \longrightarrow \mathbf{BS}.$$

Equivalently, each morphism $f \in \text{Mor}(\mathcal{G})$ is assigned a label $d(f) \in S$ such that $d(g \circ f) = d(g) * d(f)$ and $d(\text{id}_x) = e$ for all composable f, g and objects x . Since functors preserve isomorphisms, $d(f)$ lies in the group of units $S^\times \subseteq S$ for all f .

Definition 4.7 (Morphisms of S–decorated Groupoids). A morphism $(\mathcal{G}, d) \rightarrow (\mathcal{H}, \delta)$ of *S–decorated groupoids* is a functor $F : \mathcal{G} \rightarrow \mathcal{H}$ such that $\delta \circ F = d$. This defines the category DecGrpd_S of *S–decorated groupoids*.

Remark 4.5. The use of the term “decorated” in this work is inspired by its widespread adoption in modern geometry and category theory (see, e.g., Penner [20], and Fong [21]). In particular, Baez’s [22] construction of “decorated cospans” provides a categorical framework for attaching auxiliary data (the decorations) to the objects or morphisms of a base category, preserving compositionality and enabling a rich interplay between structure and process. The notion of a decorated groupoid used here is entirely in this spirit: we enrich the fundamental groupoid with additional geometric or computational data, in a way that is compatible with groupoid composition and equivalence.

When dealing with a groupoid, we should ask about its fundamental invariants. Let $THM \cong S^1_\theta \times D^2_{(r, \varphi)}$ be the solid–torus twin helix with core circle $S^1 \times \{0\}$. The projection $\text{pr}_{S^1} : S^1 \times D^2 \rightarrow S^1$ induces a winding-number map on paths: for a path $\gamma : [0, 1] \rightarrow THM$ with endpoints $x, y \in THM$, the integer:

$$\omega([\gamma]) := \deg(\text{pr}_{S^1} \circ \gamma) \in \mathbb{Z}$$

depends only on the homotopy class $[\gamma] \in \Pi_1(THM)(x, y)$. Consequently, for any x, y the hom-set $\Pi_1(THM)(x, y)$ is a $\mathbb{Z}$ –torsor, and each isotropy group $\Pi_1(THM)(x, x)$ is canonically isomorphic to $\mathbb{Z}$. In particular, the (undecorated) fundamental groupoid is determined by winding about the core circle.

To encode the helical geometry used in computation, we equip THM with orientation/chirality structure and an $U(1)$ –phase, yielding a decoration target: S (e.g. $S = \mathbb{Z} \times C_2$ or $S = S^1 \times C_2$). This gives a functor:

$$d : \Pi_1^{\text{thin}}(THM) \longrightarrow \mathbf{BS}, \quad [\gamma] \longmapsto (\text{winding/phase along } \gamma, \text{ chirality along } \gamma),$$

which is constant on thin homotopies. Thus, the homotopical data relevant for GCMs is not merely the abstract $\mathbb{Z}$ –valued winding from Π_1 , but the *decorated* classes $([\gamma], d([\gamma]))$ that package the explicitly parametric helical geometry.

In particular, if $p_A, p_B \in THM$ are the distinguished apex points of the two unit helices, then $\Pi_1(THM)(p_A, p_B)$ is a $\mathbb{Z}$ –torsor measured by net turns about the core; the decoration d refines this by recording, for each representative path, the (chirality–resolved) twist data prescribed by the helical geometry. Any deformation through thin homotopies with endpoints fixed leaves d unchanged; changes in d arise only from altering the geometric structure (e.g. switching chirality or modifying the phase law), not from the bare homotopy class alone.

Definition 4.8 (Decorated Fundamental Groupoid). Let M be a smooth (or stratified) manifold, and let $\mathcal{S}$ denote a specified set of geometric, combinatorial, or dynamical structures on M —such as winding numbers, twists, chiralities, orientation data, flows, or other local parameters.

The *decorated fundamental groupoid* $\Pi_1^{\mathcal{S}}(M)$ is defined as follows:

- (a) **Objects.** Points $x \in M$ (as in the classical fundamental groupoid).
- (b) **Morphisms.** Homotopy classes of continuous paths $\gamma : [0, 1] \rightarrow M$ from x to y , together with an assignment of “decorations” from $\mathcal{S}$ to each path, compatible with the composition law. That is, a morphism from x to y is an equivalence class $[\gamma, d]$, where d records the geometric or combinatorial data along γ .
- (c) **Composition.** Given $[\gamma_1, d_1] : x \rightarrow y$ and $[\gamma_2, d_2] : y \rightarrow z$, their composite is $[\gamma_2 * \gamma_1, d_2 * d_1]$, with decorations composed via a specified rule (e.g., addition of winding numbers, concatenation of twists, etc.).

Two decorated manifolds $(M, \mathcal{S})$ and $(N, \mathcal{S}')$ have equivalent decorated groupoids if there exists a functor between their groupoids preserving both the path composition and the decoration data.

In practice, the decoration $\mathcal{S}$ can be:

- A winding number function assigning an integer to each loop (e.g., in $\pi_1(M)$ or its universal cover).
- A combinatorial label (such as twist, chirality, or a physical field value) defined locally or globally along paths.
- A program, automaton, or any computable process attached to each morphism, as in synthetic homotopy type theory.

The classical fundamental groupoid is recovered by taking all decorations to be trivial (i.e., admit weak homotopy equivalence again).

In a GCM, the fundamental groupoid is a combinatorial-topological hybrid: it records both geometric (number of twists) and logical (chirality) data.

Example 4.1 (Fundamental Groupoid and Covering Groups). Fix $S = \mathbb{Z} \times C_2$ with operation:

$$(\tau_2, \chi_2) * (\tau_1, \chi_1) = (\tau_1 + \tau_2, \chi_1 \chi_2),$$

where $\tau \in \mathbb{Z}$ encodes net twist count (in units of 2π) and $\chi \in C_2 = \{\pm 1\}$ encodes chirality. For a smooth manifold M , one may refine to the thin fundamental groupoid $\Pi_1^{\text{thin}}(M)$ and define a geometric decoration:

$$d : \Pi_1^{\text{thin}}(M) \longrightarrow \mathbf{B}(\mathbb{Z} \times C_2), \quad d([\gamma]) = (\tau(\gamma), \chi(\gamma)),$$

with path reversal sending $(\tau, \chi) \mapsto (-\tau, \chi^{-1})$. In general, d does not factor through the ordinary $\Pi_1(M)$, since $\tau(\gamma)$ can vary under thin homotopy while remaining constant under ordinary homotopy.

So, for a given path $[\gamma] \in \pi_1(M)$ in a twin helical manifold, the decoration is explicitly computable

$$[\gamma] \mapsto (\tau_A \cdot \chi_A) + (\tau_B \cdot \chi_B)$$

where A and B index the two unit helices. To study the ensemble or statistical behavior across a collection of such helices (or geobit types), we may define the average decorated invariant:

$$\langle [\gamma] \rangle = \frac{1}{n} \sum_{i=1}^n \tau_i \cdot \chi_i$$

where n is the total number of possible configurations in the groupoid space.

In the fully symmetric regime (V_4), the average winding number over all geobit types is $(0 \times 2 + 2 \times 2)/4 = 1$. In more general or dynamical regimes, where twist and chirality are imbalanced or can vary more freely (e.g., ternary undefined space in chirality eliminates winding numbers on half the manifold), this average may take arbitrary values, and the distribution of invariants can be studied using tools from measure theory and ergodic theory.

Definition 4.9 (Homotopy Computation, Informal). Let $(M, \text{str}) \in \mathcal{ESM}^+$ with an S -decoration $d : \Pi_1^*(M) \rightarrow \mathbf{BS}$ (or $d : G \ltimes M \rightarrow \mathbf{BS}$). A *homotopy computation* is a decorated path/state evolution $\gamma : [0, 1] \rightarrow \mathcal{SGCM}$ whose updates are given by the arrows of the decorated groupoid, and whose outcome is the composite label $d(\gamma) \in S$. Two computations are considered equivalent if there exists a 2-cell (thin homotopy rel. endpoints) $H\gamma \Rightarrow \gamma'$ along which d is constant, so they have the same outcome in S .

Remark 4.6. This definition is by intent not fully formalized, but it should be noted that the homotopy is such that:

- The path encodes evolution under flows/diffeomorphisms/group actions.
- Homotopies of computations correspond to deformations preserving the S -invariants.

So, even when the underlying space is topologically trivial (e.g. $THM \simeq B^3$), a choice of geometric structure induces a decoration,

$$d : \Pi_1^{\text{thin}}(M) \longrightarrow \mathbf{BS},$$

so the S -decorated groupoid $\Pi_1^S(M, \text{str})$ is typically nontrivial. Hence, the computational content lives in the decoration d , not in the bare homotopy type of M . Composition must respect both topology and S -labels, yielding a stable, compositional invariant that records the history of transformations.

In lattice models, one naturally obtains a diagram of interacting decorated groupoids $\{\Pi_1^S(H_i, \text{str}_i)\}$ coupled by flow functors; this forms the state-transition layer of the computation. Much of the theory is faithfully captured at the 1-categorical level; higher categorical structure is added only when coherence or field-theoretic extensions (decorated bordisms, evaluation functors Z) are required.

Heuristically, it seems that transcendental geometry encodes computation, whereas algebraic topology organizes proofs. This is the practical duality guiding the framework.

Conjecture 4.2 (Extended Grothendieck Homotopy Hypothesis). *Let M be a smooth or stratified manifold equipped with additional geometric or combinatorial structure (such as twist, chirality, or orientation), and let $\mathcal{G}$ denote its decorated fundamental ∞ -groupoid: the structure encoding homotopy classes of paths together with all computational, combinatorial, or dynamical data attached to those paths. Then the following hold:*

- (a) **Classical Recovery:** *If $\mathcal{G}$ is stripped of all computational or combinatorial decorations, it reduces (up to equivalence) to the classical ∞ -groupoid of M , recovering the standard Homotopy Hypothesis: “Homotopy types are equivalent to ∞ -groupoids.”*
- (b) **Enrichment:** *The enriched groupoid $\mathcal{G}$ encodes not only the homotopy type of M , but also the complete combinatorial, computational, or dynamical structure of the system—so that two decorated spaces are equivalent if and only if their enriched groupoids are equivalent (as symmetric monoidal ∞ -categories, or in an appropriate enriched sense).*
- (c) **Failure of Classical Correspondence:** *There exist decorated geometric manifolds M and N such that $M \simeq N$ as topological spaces (i.e., weak homotopy equivalence holds), but their enriched fundamental groupoids $\mathcal{G}_M$ and $\mathcal{G}_N$ are not equivalent in any enriched ∞ -categorical sense. More formally, there exists a category $\mathcal{ESM}^+$ of smooth or stratified manifolds with flow-preserving morphisms, torsion structure, and chirality data such that: No functor $F : \mathcal{ESM}^+ \rightarrow \infty\text{-Gpd}$ induces a homotopy equivalence that simultaneously preserves path components and the torsion-invariant structure, unless F is enriched with additional framing or coordinate data.*
- (d) **Computational Duality:** *For any such enriched groupoid, there exists a “Curry–Howard dictionary” between:*
 - (a) decorated paths as proofs, and
 - (b) decorated paths as programs,*so that the groupoid serves simultaneously as a classifier of homotopy types and as a substrate for geometric computation. So that: Decorated ∞ -groupoids $\simeq$ Decorated topological spaces*

Remark 4.7. It would be amiss not to mention Grothendieck’s Homotopy Hypothesis, which equates the world of homotopy types with that of ∞ -groupoids, classifying spaces up to continuous deformation via algebraic symmetry. However, in the regime of geometric computation, the fundamental groupoid now plays a dual role: it is both a classifier of paths and a dynamical engine for computation and flow. This suggests a possible enrichment of the hypothesis, in which ∞ -groupoids may encode intrinsic computational or dynamical structure—a new bridge between geometry, logic, and computation foreshadowed by Grothendieck’s original vision.

Remark 4.8 (On the Application of the EGHH vs. GHH). If d factors through $\Pi_1(M)$, then

$$\Pi_1^S(M) \simeq \Pi_1(M) \times \mathbf{BS}$$

and the geometric realization of the nerve yields a classical homotopy 1-type ($B\Pi_1(M) \times BS$ when S is a group). In the geometric (thin) case above, d typically does not factor through $\Pi_1(M)$, and the ordinary nerve-realization forgets the smooth holonomy/torsion data. To retain it, one must work in a smooth/Lie setting, motivating an extended homotopy hypothesis relating decorated ∞ -groupoids to decorated topological/smooth spaces.

To provide a rigorous setting for the maximal abelian regime, we explicitly construct a Lie groupoid.

Definition 4.10 (S-decorated Lie Groupoid). If $\mathcal{G} \rightrightarrows M$ is a Lie groupoid and S a Lie group, an *S-decorated Lie groupoid* is a smooth functor $d : \mathcal{G} \rightarrow \mathbf{BS}$, where $\mathbf{BS}$ is the one-object Lie groupoid with endomorphism group S .

Definition 4.11 (Underlying Lattice/graph Datum). Fix a finite connected graph (e.g. a finite GL patch)

$$\Gamma = (V, E),$$

where V is the set of sites and E the set of (undirected) adjacency links. Choose an orientation on each undirected edge, written as a subset $E^+ \subseteq \vec{E}$ of oriented edges, so that every undirected edge $\{i, j\} \in E$ appears exactly once as either $e : i \rightarrow j$ or $\bar{e} : j \rightarrow i$ in E^+ . Write $s(e), t(e) \in V$ for the source and target of an oriented edge e .

Definition 4.12 (Core Configuration Space in the QF Regime). The quasi-fluidic core state of the lattice is a *link phase assignment*

$$M_{\text{QF}} := U(1)^{E^+} \cong (S^1)^{|E^+|}.$$

An element $U \in M_{\text{QF}}$ is a family $U = (U_e)_{e \in E^+}$ with $U_e \in U(1)$. The value on the oppositely oriented edge is determined by inversion:

$$U_{\bar{e}} := U_e^{-1}.$$

Remark 4.9 (Chirality as Orientation Convention). In this interpretation, chirality is *not* a separate state coordinate. Instead, the handedness of the underlying helix fixes (or suggests) a preferred orientation for reading the link phase; reversing the convention replaces U_e by U_e^{-1} . Thus, the “chirality bit” is absorbed into the choice of orientation on each link and is invisible to gauge-invariant observables.

Definition 4.13 (Gauge Group and Action on Link Phases). Let the (site) gauge group be

$$G_{\text{QF}} := U(1)^V.$$

For $g = (g_i)_{i \in V} \in G_{\text{QF}}$ and $U = (U_e)_{e \in E^+} \in M_{\text{QF}}$, define the action $g \cdot U \in M_{\text{QF}}$ by the standard link transformation law

$$(g \cdot U)_e := g_{s(e)} U_e g_{t(e)}^{-1}, \quad e \in E^+.$$

Definition 4.14 (QF Core Lie Groupoid: the Gauge Action Groupoid). The quasi-fluidic core Lie groupoid is the action groupoid

$$\mathcal{G}_{\text{QF}} := G_{\text{QF}} \ltimes M_{\text{QF}} \rightrightarrows M_{\text{QF}}.$$

An arrow is a pair (g, U) with $g \in G_{\text{QF}}$ and $U \in M_{\text{QF}}$. The source and target maps are

$$s(g, U) = U, \quad t(g, U) = g \cdot U,$$

and composition is

$$(h, g \cdot U) \circ (g, U) = (hg, U), \quad h, g \in G_{\text{QF}}.$$

For finite Γ , this is a finite-dimensional Lie groupoid (indeed a Lie group action on a compact torus).

Remark 4.10 (Gauge Invariants live on the Quotient Stack). Physically meaningful quantities are gauge-invariant functions on M_{QF} (equivalently, functions on the quotient stack $[M_{\text{QF}}/G_{\text{QF}}]$). The groupoid $\mathcal{G}_{\text{QF}}$ encodes *gauge redundancy* (equivalence of descriptions). Time evolution (e.g. gradient flow of a local gauge-invariant energy built from plaquette terms) is additional structure: a flow on M_{QF} that is G_{QF} -equivariant and hence descends to the quotient. We treat this dynamical layer separately from the gauge groupoid. [23]

This work presents one possible interpretation of how a homotopy-theoretic framework for computation could be formulated. We offer it in the hope that it will be considered, refined, or challenged by the broader mathematical, computer science, and physics community.

4.7 Jet Bundles and Moduli Spaces on Chiral Manifolds

We note that the fiber bundle structure of a single twin helix manifold is trivial: as a bundle, it is globally diffeomorphic to a solid torus. This apparent simplicity is misleading for computation. Here, there is no topological twisting (no monodromy in the D^2 -fibers). Consequently, when geometric computation is introduced—via operations on chirality, orientation, and flow—the information lives not in the topological class of the bundle but in the dynamics and group actions we impose by gluing, deformations, and parameter updates.

As a motivating comparison, jet bundles on the product $S^1 \times D^2$ admit global coordinates, so derivative data and feedback laws can be managed globally. By contrast, in a nontrivial fibration (e.g. the Hopf fibration), monodromy produces path-dependent phases and constraints that act like ‘quantum-style’ control effects, often complicating and sometimes fundamentally constraining jet computation.

We use the trivialization $THM \cong S_\theta^1 \times D_{(r,\varphi)}^2$ to isolate two primitive invariants that will persist in more elaborate settings: (a) a continuous S^1 -valued *phase* accumulated along the θ -ring (“one turn”) and (b) a discrete *chirality* bit from the moduli of chiral embeddings. We deliberately avoid bundle/holonomy language here; the full $U(1)$ picture is introduced later.

Definition 4.15 (Quasi-holonomic Phase Functional on a Trivial Ring). Given a 2π -periodic angular rate $\Omega(r, \theta)$, define the phase of one turn at radius r by

$$\Phi(r) := \int_0^{2\pi} \Omega(r, \theta) d\theta \in \mathbb{R}/2\pi\mathbb{Z}, \quad U(r) := e^{i\Phi(r)} \in S^1.$$

For loops at the same radius, concatenation adds phases and multiplies U ’s: $U_{\gamma_2 \circ \gamma_1}(r) = U_{\gamma_2}(r) U_{\gamma_1}(r)$. If $\Omega \mapsto \Omega + \partial_\theta f$ with $f(r, \theta)$ 2π -periodic in θ , then $\Phi(r)$ (hence $U(r)$) is unchanged.

Proposition 4.3 (Radial Shear Identity). Set $A := \Omega(r, \theta) d\theta$ and $F := dA = \partial_r \Omega(r, \theta) dr \wedge d\theta$. For $r_1 < r_2$,

$$\log \frac{U(r_2)}{U(r_1)} = i \int_{[r_1, r_2] \times S^1} F \quad \text{in } i\mathbb{R}/2\pi i\mathbb{Z}.$$

In particular, if $\partial_r \Omega \equiv 0$ (no radial shear) then $U(r)$ is independent of r ; otherwise the difference of phases between two radii equals the integral of the 2-form F over the annulus.

Remark 4.11. Even with $c_1 = 0$ (topologically trivial), geometric control $\Omega(r, \theta)$ generates nontrivial S^1 gates $U(r)$ and a robust composition law. Later, we will recognize A as a $U(1)$ connection 1-form on the θ -ring and F as its curvature, so the identity above becomes the familiar holonomy–curvature relation.

Likewise, let us consider the moduli space of chiral curves. Since the moduli space of the full 4D THM is vast (requiring, in principle, a Hilbert space due to the infinity of possible cross-sectional 2-manifold joins), it is pragmatic to work with toy models and restricted cases.

Lemma 4.4 (Chiral Splitting for Regular Space Curves). Let $\mathcal{M}$ be the space of C^3 embeddings $\gamma : S^1 \rightarrow \mathbb{R}^3$ such that the curvature $\kappa_\gamma(s)$ and torsion $\tau_\gamma(s)$ of the Frenet–Serret frame exist and satisfy

$$\kappa_\gamma(s) > 0 \quad \text{and} \quad \tau_\gamma(s) \neq 0 \quad \text{for all } s \in S^1.$$

Equip $\mathcal{M}$ with the C^3 topology, and form the moduli space $\mathcal{M}/\sim$ where $\sim$ is generated by orientation-preserving Euclidean motions of $\mathbb{R}^3$ and orientation-preserving reparametrizations of S^1 . Then $\mathcal{M}/\sim$ has (at least) two connected components distinguished by the sign of torsion. In particular, a right-handed helix and a left-handed helix lie in different connected components.

Proof. Since $\tau_\gamma : S^1 \rightarrow \mathbb{R}$ is continuous and nowhere zero, its sign is constant on S^1 . Define the chirality invariant

$$\chi(\gamma) := \text{sign}(\tau_\gamma(s)) \in \{+1, -1\},$$

which is independent of s .

The torsion is invariant under orientation-preserving Euclidean motions, and orientation-preserving reparametrizations preserve the sign of τ as well; hence χ descends to a well-defined map on the moduli space $\chi : \mathcal{M}/\sim \rightarrow \{+1, -1\}$. Moreover, χ is locally constant (small C^3 perturbations cannot create a zero of τ without leaving $\mathcal{M}$), so the preimages $\chi^{-1}(+1)$ and $\chi^{-1}(-1)$ are disjoint open subsets whose union is all of $\mathcal{M}/\sim$. Therefore $\mathcal{M}/\sim$ is disconnected. $\square$

This is related to Lemma 1.1 but done in reverse. While the underlying objects vary smoothly within each regime, the presence of a discrete geometric invariant (here, chirality on the nondegenerate locus) induces a genuine decomposition of the corresponding moduli space into disconnected components. We may therefore regard the component label as an intrinsic “bit” carried by the geometry: it is stable under all permitted smooth deformations. It can change only via a controlled singular transition in which the invariant becomes undefined (or by extending the symmetry group of the model). In this precise sense, discrete information can arise from continuous substrates as a consequence of moduli bifurcation.

4.8 Lancret’s Ghost?

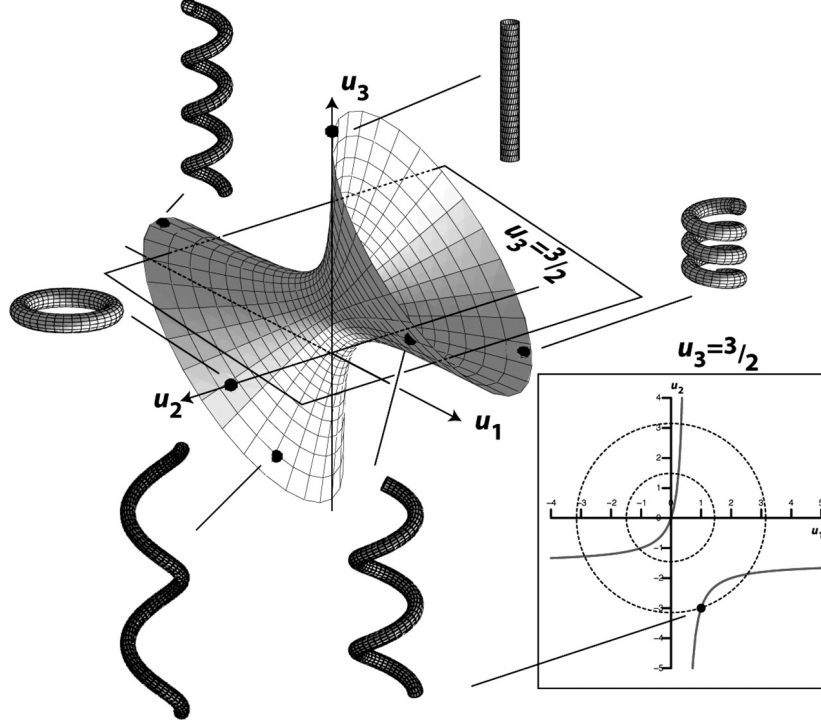


Figure 4: Homology, Helicity, and Hyperbolicity: a “geometric” Rosetta Stone. (Image Courtesy of Alain Goriely, “Helices” 2006)

Helices are mathematical objects rich in dynamical variables—as explored in this section—whose properties interact synergistically but without the constraints of maximal symmetry. This very lack of symmetry, unfortunately, appears to have led to helices being misunderstood as structured objects. A helix is a manifold, but it is also a vector. And as we have seen, maximal algebraic symmetry inevitably destroys geometric expressiveness—the very quality needed for a dynamical system of physical computation, where symbolic encoding of state is essential. In fact, any form of computation (even total recursive proof assistants) requires at least some modicum of expressiveness, because without it, one is left only with propositional logic. Helices can dance with ones and zeros, but a sphere will never lie.

So, despite their ubiquity in physical systems—from DNA strands to solitons to protein folding—helices are strikingly absent from modern algebraic geometry. This omission is *not* accidental: the standard helix is defined by *transcendental* functions (e.g., $\gamma(t) = (a \cos t, a \sin t, bt)$), placing it outside the algebraic category! Algebraic geometry, by its very design, privileges curves and surfaces defined by polynomials—algebraic varieties—over transcendental ones. However, this prioritization systematically filters out a broad class of naturally occurring, dynamically rich shapes: those that appear constantly in physics, chemistry, and biology.

In this sense, the helix is a kind of orphan in the modern mathematical landscape: too smooth for combinatorics, too transcendental for algebraic geometry, and too infinite for finite computation. Yet,

for a theory of geometric computation—especially one grounded in constructible physical objects—the helix rightfully takes center stage.

Indeed, recent investigations into the internal representations of large language models [24] indicate that gradient-based optimization naturally converges on a “generalized helix” topology for performing arithmetic operations. This suggests that the helical manifold is not merely a physical curiosity, but a universal computational primitive for systems requiring the simultaneous integration of cyclic (phase) and linear (magnitude) data, precisely the constraints present in gauge theory.

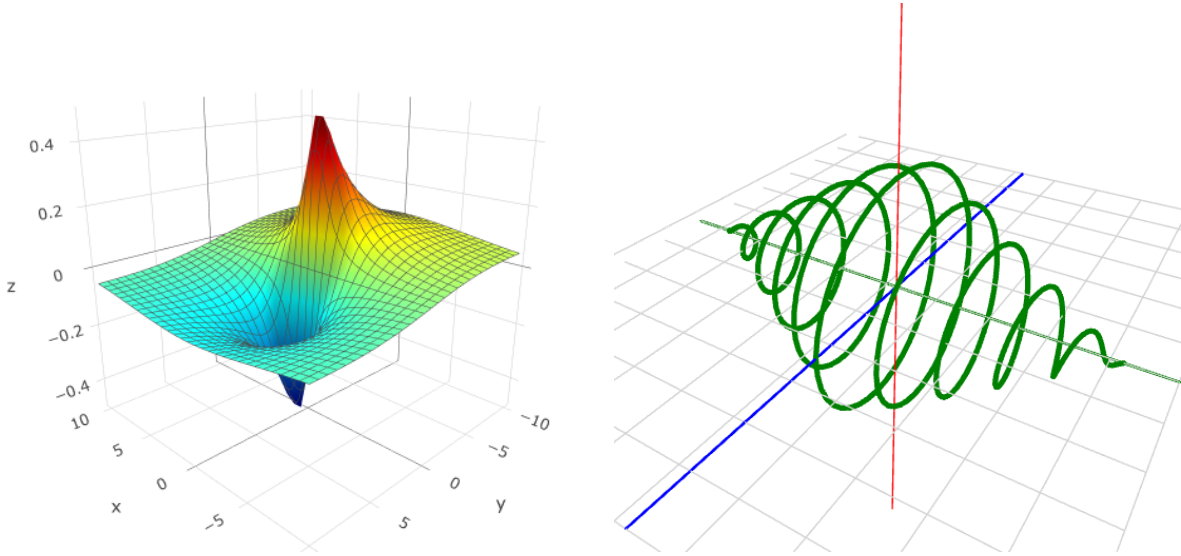
Currently, however, we lack a general form for the moduli spaces of helical manifolds that reveals their deeper structure. Lancret’s Theorem (1802) tells us that a space curve in $\mathbb{R}^3$ is a circular helix if and only if the ratio of its torsion τ to its curvature κ is constant along the curve, i.e., $\tau/\kappa = C$. **This centuries-old result may hide much deeper implications.** In our view, constructing and characterizing the moduli space of helical manifolds is the most urgent and challenging direction for future research. Such a moduli space is likely to be a highly complex, infinite-dimensional orbifold with wild local and global structure. The expressive power of helices is remarkable: they admit a countably infinite variety of cross-sections, as well as parameters for twisting, pitch, and chirality, yielding a vast configuration space. Mapping this moduli space in detail presents a challenge of the same order as the motivic integration of Calabi-Yau varieties—a Kontsevich-style formalism. Nevertheless, understanding its structure is crucial for advancing both the theory and applications of geometric computation.

While helices offer remarkable efficiency for encoding computational structure, it is essential to note that not all helices in nature are used for computation, nor do they dominate large-scale architectures. Helical forms excel at packing information, facilitating transmission, and exploiting symmetry at small scales (think, DNA and chiral molecules). However, at larger scales, nature often favors other symmetries—branching, fractal, planar, or modular structures—driven by stability, energetic, and functional constraints. Thus, the computational potential of helices arises not from their ubiquity but from their unique ability to localize and propagate information under appropriate conditions.

Remark 4.12 (Overlooked But Important Reading on Helicity). For a continuum-mechanics foundation on why helices are generic energy minimizers (and how curvature/torsion couple), see Goriely [25]; for a crisp, canonical derivation of when minimizers are helical with pitch/radius set by constraints, see Chouaieb et al. [26]. Developable-ribbon energetics and torsion-curvature locking—which motivate “channel” formation in cross-sections—are captured by Starostin-van der Heijden [27]. On the geometric side that is often missed, Adachi [28] provides a leading example of how (helical) Killing fields and foliations structure flows, aligning with our base-join transport picture. Ferrández Izquierdo’s recent monograph [29] will also provide a focused, modern treatment of helical geometries to unify these threads for newcomers.

For all we know? Perhaps the key to modern quantum theory lurks quietly in the diagrams of a century-old German treatise (Loria [30]) on transcendental curves—waiting for someone to notice that helices and hyperboloids speak a universal geometric tongue.

5 Geometric Computability Theory



(a) BPST instanton: a finite-action, localized topological solution to the Yang–Mills equations, exhibiting geometric twist and winding.

(b) Complex quantum wave: a superposition of oscillations, illustrating the intertwined, helical modes of computation.

Figure 5: Familiar examples of geometric configurations corresponding to Type 3 and Type 1 *THM*.

It is at this point that we can now formally turn our attention to the foundations of quantum physics—an excellent source for fertile cross-pollination of ideas with work from mathematicians like Kontsevich [31], Connes [32], and physicists like Witten [33] tackling questions related to quantization, fundamental force symmetries, and spin, to name but a few.

5.1 Curvature as Computation

After assembling *THM* in the abelian, quasi-fluidic band—where computation is well captured by finite local operations on geometric primitives—we now enter the *fully fluidic* regime. Regard the helix lattice as a bundle with connection. Let

$$\pi : P \rightarrow B$$

be a principal G -bundle over a computably presented base B (the working lattice region), with associated state bundle $E := P \times_G F$ whose fiber F encodes the finite-dimensional internal variables of a twin helix cell. A connection $A \in \Omega^1(B; \mathfrak{g})$ prescribes parallel transport, and its curvature

$$F = dA + A \wedge A \in \Omega^2(B; \mathfrak{g})$$

measures path-dependence. In this language, a *join* is no longer purely combinatorial: it is the composition of parallel transports along adjacent interfaces. When G is abelian (e.g. $U(1) \simeq SO(2)$), joins commute; once G is non-abelian, joins become order-sensitive and composition acquires memory.

Example 5.1 (Holonomy as Computational Memory). For a loop γ based at $x \in B$, the parallel transport

$$U_\gamma = \mathcal{P} \exp \left(\int_\gamma A \right) \in G$$

is the *holonomy*. The subgroup $\text{Hol}_x(A) \subseteq G$ generated by all such U_γ is the effective symmetry seen at x . In the abelian limit $U_{\gamma_1} U_{\gamma_2} = U_{\gamma_2} U_{\gamma_1}$ and only additive flux matters; in the non-abelian case, path ordering is essential and different execution paths may yield nonconjugate elements, even for homotopic loops when $F \neq 0$. The gauge-invariant readouts are the *conjugacy classes* $[U_\gamma]$ (or class functions such as $\text{tr}_\rho(U_\gamma)$). The following now hold:

- (a) **Registers as loops.** Minimal cycles in the lattice carry holonomies $\{U_{\gamma_k}\}$ serving as state registers; the state is the family $\{[U_{\gamma_k}]\}$.
- (b) **Updates as transport.** Driving fields update A and hence $\{U_{\gamma_k}\}$; gluing cells changes the loop basis.
- (c) **Curvature as storage.** F encodes what cannot be gauged away—persistent, path-dependent context (geometric memory).

This is the geometric analogue of Wilson loops: the observable is the holonomy, *not* the potential itself. In the quasi-fluidic abelian band, holonomy reduces to a scalar phase (orientation logic, integer twist counters). Entering the fluidic/gauge band promotes these to matrix phases (non-abelian Berry/Wilczek-Zee transport) [34], and join symmetry breaks via $[U_{\gamma_1}, U_{\gamma_2}] \neq e$.

From abelian phases to non-abelian transport. Each twin helix cell has a finite internal state space; as parameters vary over B these states form a vector bundle $E \rightarrow B$ (the *band bundle*). Adiabatic transport induces a connection on E . If $\text{rk } E = 1$ over a region, the induced connection is abelian (Berry phase), hence commuting joins. When degeneracy persists ($\text{rk } E \geq 2$ on an open set), the connection becomes non-abelian (Wilczek-Zee): parallel transports no longer commute and path ordering becomes a resource. Base anisotropy and helix-helix coupling are precisely the mechanisms that create protected degeneracy and thus promote $U(1)$ to $SU(2)$ -type dynamics.

Definition 5.1 (Join Commutator in the Interface Groupoid). Let $\mathcal{J}$ be the interface groupoid whose objects are boundary faces and whose morphisms are admissible joins (parallel transports along interfaces). For $J_1, J_2 \in \text{Hom}_{\mathcal{J}}(X, Y)$ with common source/target, set

$$[J_1, J_2] := J_1 J_2 J_1^{-1} J_2^{-1}.$$

Joins *commute* if $[J_1, J_2] = \text{id}$ in $\text{Hom}_{\mathcal{J}}(X, X)$.

Computation now proceeds by progressively internalizing curvature; symbolic joins become ordered transports, and the state is read off from conjugacy classes of holonomies. Expressivity exceeds purely symbolic models because time-ordering, loop creation/annihilation, and curvature redistribution generate behaviours not captured by classical finite-step semantics.¹² Physically, the twin helix fiber is the matter degree of freedom, and A is the control field; noncommutativity (path dependence) turns the join algebra into a genuine gauge computation.

Spin textures as fluidic defects. A *THM* defines an internal orthonormal 2-frame; choosing a unit normal gives a map $n : B \rightarrow S^2$. Its defects organize memory channels:

- **Vortex/phase defects** (*abelian*). Windings in $U(1)$ along loops (π_1) ,
- **Skyrmion-type textures** (*non-abelian*). Degree of n on embedded 2-spheres $(\pi_2(S^2) \cong \mathbb{Z})$,
- **Linked textures** (*Hopf-type*). Linking of preimages $n^{-1}(p)$ and $n^{-1}(q)$ for $p \neq q$ $(\pi_3(S^2) \cong \mathbb{Z})$ in 3D thickenings).

These classes appear in the holonomy spectrum: abelian vortices shift scalar phases; non-abelian textures produce non-conjugate holonomies around homologous loops, yielding distinct class-function readouts. The central split in summary:

- (a) **Symbolic/Quasi-fluidic.** G is effectively $U(1)$, joins commute, memory is a scalar phase or integer flux.
- (b) **Fluidic/Gauge.** G is non-abelian (from helix symmetry and interface groupoid), joins do not commute, memory lives in holonomy conjugacy classes; curvature stores context.

The base topology controls available memory channels: on contractible B with $F=0$ holonomy is trivial up to conjugacy; nontrivial F , defects, or handles produce noncommuting generators. In the terms of Lemma 2.4 and Table 7, flipping across from the quasi-fluidic abelian band to the fluidic/-gauge band trades raw orientation microstates for noncommuting holonomy, where the true expressive capacity of geometric computability is realized. Fully fluidic GCMs begin to resemble endless streams of information as increasingly chaotic computational phenomena appear.

¹²We refrain from machine-model claims. Under continuum idealizations, non-abelian gauge dynamics are known to encode phenomena beyond classical Turing models; our interest here is programmability via $(A, F, \text{Hol}(A))$.

Example 5.2 (Spectral Flow as a Computable Readout). Driving the system around a control cycle S^1 promotes B to $B \times S^1$. The net change in holonomy class across a period is governed by the integral of curvature over spanned 2-cycles (a Thouless-type pump):

$$\Delta(\text{class function on } U_\gamma) = \int_{\Sigma \subset B \times S^1} \text{tr}(F) \quad (\text{abelian}), \quad \Delta(\text{spec}(U_\gamma)) \text{ by } \int \text{tr}(F \wedge F) \quad (\text{non-abelian}).$$

Thus, periodic control implements a bordism in parameter space, and evaluation of class functions on Wilson loops gives an integer (or discrete spectrum) that is stable and algorithmically retrievable in our symbolic presentation.

5.2 The Algebra of Flow

Starting in the quasi-fluidic regime, interactions are not purely symbolic: they are geometric and dynamical—the information is encoded in the differential structures of both flow and form. In the fully fluidic regime, the cross-sectional geometry, as discussed in Section 2.6, reappears. Why the focus?

Two kinds of analog computation. Computation as we know it seems to split as such:

- (*Algebraic analog flow*) Rank-1 bundles with abelian holonomy ($U(1)$): states are continuous but updates commute; residual symmetry imposes selection rules; behavior is predictable.
- (*Geometric analog chaos*) Rank- ≥ 2 degeneracy with non-abelian holonomy ($SU(2)$ -type): path ordering matters; joins do not commute; internal mixing encodes control flow; behavior is history-sensitive. In Felleisen’s sense [8], these differ at the level of *core constructs*, not mere “data types.”

Symmetry selection rules (abelian kinematics). When a local annulus $A \simeq S^1 \times I$ around a helix cell carries a residual C_n symmetry, transport admits a harmonic decomposition $e^{im\theta}$ in which the connection couples only modes $m \equiv m' \pmod{n}$. Transport block-diagonalizes by angular-momentum class modulo n , and the induced holonomies on each block commute. In this quasi-fluidic band, joins commute and memory reduces to scalar phases or integer twist counters. Breaking C_n (or leaving a round annulus) removes these selection rules; see Appendix A.8 for the calculation.

Promoting $U(1) \rightarrow SU(2)$ by degeneracy. Helix-helix coupling plus base anisotropy can maintain a rank- ≥ 2 degenerate internal subspace across a region of B . Adiabatic transport on this subbundle is then *non-abelian* (Wilczek-Zee): path-ordered products of transports do not commute, and the state carried by a loop is a conjugacy class $[U_\gamma]$ rather than a single phase. Practically, anisotropy supplies off-diagonal couplings while coupling between the twin strands stabilizes the degeneracy that upgrades $U(1)$ to $SU(2)$ -type dynamics.

Definition 5.2 (Macro-expressiveness). For two geometric computation models A, B , write $A \succeq_{\text{macro}} B$ if there is a finite library of local transports/joins/defect insertions in A whose *gauge-invariant readouts* (class functions on holonomy) can *macro-express* every primitive of B with bounded overhead, preserving observational equivalence at the boundary. Abelian $U(1)$ lacks path ordering and conjugation gates; the non-abelian regime supplies them, hence the jump in macro-expressiveness.

Base topology and gauge-invariant readouts. On contractible patches with $F = 0$, holonomy is trivial up to conjugacy, so no persistent state is stored. Nontrivial memory channels arise from curvature, defects, or handles. Gauge transformations act by conjugation on holonomies; the physically meaningful readouts are *class functions* of U_γ (e.g. traces in fixed representations).

Computational tradeoffs: algebraic vs. geometric analog.

- **Memory carrier.** *Algebraic analog* ($U(1)$): a scalar phase per loop. *Geometric analog* ($SU(2)$): a conjugacy class per loop (equivalently, spectra or traces $\text{tr}(U_\gamma), \text{tr}(U_\gamma^2), \dots$).
- **Registers and gates.** Registers are chosen loops $\{\gamma_k\}$. In $U(1)$ gates add phases (commuting joins); in $SU(2)$ gates are ordered transports with nontrivial commutators $[U_{\gamma_i}, U_{\gamma_j}] \neq \text{id}$.

- **Control knobs.** (a) introduce base anisotropy beyond the first harmonic; (b) maintain internal degeneracy (rank ≥ 2 band bundle); (c) add defects/handles so $\pi_1(B)$ generates noncommuting holonomies; (d) redistribute curvature to reprogram stored context.
- **Verifiability.** $U(1)$: readout is a phase/integer. $SU(2)$: readout uses class functions (e.g. traces of Wilson loops) that remain algorithmically retrievable in our symbolic presentation.

Since the base geometry controls capacity, on simply-connected regions with $F = 0$, holonomy classes are trivial; introducing curvature, handles, or protected degeneracy creates noncommuting generators and promotes the system from the quasi-fluidic abelian band to the fluidic/gauge band. This is the bridge where join algebra becomes order-sensitive and holonomy conjugacy classes serve as programmable memory.

Now the internal geometry of the cross-section becomes a new axis of calculation. Beyond circles, one can admit distinct flow channels, branch lines, and defect cores within the same cross-section; these change the *isotypic decomposition* of transport and create/maintain degeneracies. Higher symmetry (round annulus) yields fewer angular sectors and abelian blocks; broken symmetry and multi-channel shapes create protected mixing and non-commuting gates. Thus cross-section complexity controls the number and kind of “channels” available for computation: not just *which* states, but *how* transport moves *within* a state.

Remark 5.1. Only the geometric analog can outstrip algebraic field-operation models under standard readouts: chaotic/PDE dynamics can exceed polynomial/field-arithmetic semantics as shown by Pour-El and Richards study of the wave equation [35]. In our constructive calculus (Section 5.5), such steps appear as *effects* with witnesses (no hidden oracles), preserving normalization and observational equivalence on gauge-invariant readouts.

5.3 Helicity and the Origin of Quantum Spin Networks

Although this paper did not originally aim to adjudicate unification programs, aspects of the geometric computation model resonate with the combinatorial, categorical, and background-independent *perspectives* emphasized in Loop Quantum Gravity.

As developed throughout this work, helices exhibit exceptionally rich geometric behaviour. The pre-quantum computational geometry of twin helices—where information resides in flows, holonomies, and deformations—can supply a substrate from which discrete boundary data (as in spin-foam state spaces) may emerge after coarse-graining. Rather than postulating quantum discreteness as primitive, it is reasonable to view quantum structure as an invariant/stable phase of the underlying geometric dynamics, much as solitons or topological defects arise from continuous systems. In this picture, coarse-graining the micro-geometry of helices can yield spin-network-like graphs (cf. Section 2.5); the effective pure-state manifold of a qubit is $\mathbb{CP}^1 \cong S^2$ (the Bloch sphere), while discreteness enters at the level of projective readout rather than the underlying state space.

Our viewpoint also aligns with strands of work beyond LQG: non-Markovian/correspondence ideas in quantum foundations (e.g. Barandes [36]), relativistic quantum information (Fuentes-Schuller [37]), the positive geometry program in scattering amplitudes (Arkani-Hamed et al. [38]), and deterministic explorations (’t Hooft [39]).¹³ In our reading, discrete quantum geometry then appears as a phase transition within a more general continuous geometric logic—without discarding the manifold substrate.

Example 5.3 (The Quantum Tetrahedron). Fix a working region B and a principal G -bundle $P \rightarrow B$ with connection A , as in the fluidic discussion. Choose a cellular decomposition dual to the helix lattice (plaquettes/handles adapted to the THM_4 pattern), and let Γ be the resulting 4-valent graph:

- **Edges** $e \in E(\Gamma)$. Homology classes of minimal loops threading twin helix channels. The nonabelian holonomy U_e around e has conjugacy class $[U_e]$; its spectrum (eigenphases) is the gauge-invariant “register value.”
- **Edge labels.** Are extracted by coarse-graining class functions of U_e . For $G = SU(2)$, the spectrum organizes into unitary irreps; we record the isotypic content by an $SU(2)$ spin $j_e \in \frac{1}{2}\mathbb{N}$. Operationally, j_e is the minimal label whose character χ_{j_e} matches the measured class functions $\{\text{tr}(U_e^k)\}$ on e .

¹³Regarding superdeterminism: GCT rejects this. Measurement settings are treated as exogenous boundary data; at the ontic level, this induces measurement dependence via a global, time-symmetric constraint, while linear/gauge symmetries preserve operational free choice and no-signalling.

- **Vertices** $v \in V(\Gamma)$. These arise where four channels meet (the V_4 cell). Gauge covariance imposes the Gauss law $\prod_{e \ni v} U_e \simeq \text{id}$ (up to conjugation); in representation terms this singles out an $SU(2)$ intertwiner $\iota_v \in \text{hom}(\bigotimes_{e \rightarrow v} V_{j_e}, \bigotimes_{e \leftarrow v} V_{j_e})$.

Thus each coarse cell yields a labeled pair $(\Gamma, \{j_e\}, \{\iota_v\})$, i.e. a spin network state in the standard sense. No discreteness hypothesis was used: it is the *stability* of the holonomy conjugacy classes under small deformations (and the V_4 join rule) that makes the labels robust.

Choice of group: Z_4 vs V_4 ? The two independent binary symmetries of a twin helix cell (left/right, up/down) generate $V_4 \simeq Z_2 \times Z_2$. Classifying surface modes or boundary excitations by Z_4 collapses these axes; phrasing them as V_4 cleanly factors the parity data that seed the four-valent node. In practice, this matches the base 2-geobit state space and yields a sharper route from micro-geometry to the 4-valent “quantum tetrahedron” intertwiners.

Spin foams as computational bordisms. Helix updates (gluing, defect motion, curvature redistribution) generate a 2-complex $\mathcal{F}$ whose faces track the evolution of edge holonomies and whose vertices encode local join moves. Coarse-graining sends such an update to a labeled 2-complex (a spin foam) whose faces carry j -labels and whose edges carry intertwiners, precisely the data of LQG transition amplitudes. In our setting, amplitudes arise from products of local class functions evaluated on Wilson loops, making the construction functorial in computable bordisms. The “foam” is therefore an emergent bookkeeping of gauge computation, not a fundamental discretization of spacetime.

Spin networks correctly capture the *gauge-invariant* content (conjugacy classes and intertwiners) of nonabelian holonomy on coarse loops and junctions; spin foams capture its elementary updates. In the helix picture, these are *phase data of transport* through THM_4 cells, stabilized by join rules and curvature, rather than postulated “atoms of space.” Background independence is recovered at the coarse level because all readouts are class functions (insensitive to coordinate choices), even though the micro-dynamics live on a smooth 4-manifold.

Proposition 5.1 (Coarse-graining to Spin Networks). *There is a canonical coarse-graining functor*

$$\mathcal{C} : \frac{\text{helix configs on } (B, A)}{\text{computable gauge bordism}} \longrightarrow \frac{\text{spin networks } (\Gamma, \{j_e\}, \{\iota_v\})}{\text{graph isomorphism}}$$

sending minimal loop holonomies to edge labels and join constraints to intertwiners. Moreover, $\mathcal{C}$ is compatible with the composition of updates: a computable bordism of helix configs maps to a spin foam between the corresponding spin networks.

Proof sketch. Gauge-invariant readouts are class functions of holonomy; spectral data of U_e define j_e via the $SU(2)$ character table. Local gauge constraints at V_4 junctions select the invariant subspace, i.e. an intertwiner. Elementary updates act by conjugation and fusion/splitting, which descend to foam faces/edges. Functoriality follows from the locality of joins and the multiplicativity of class functions along concatenated loops. $\square$

Baez’s and Barrett’s Quantum Tetrahedra inside THM_4 At a single V_4 cell, four edge labels $\{j_1, \dots, j_4\}$ and an intertwiner ι define a quantum tetrahedron. In the helix model, these labels come from the spectra of the four minimal loop holonomies, while the join law enforces ι . The usual area/angle relations are thus coarse summaries of underlying nonabelian transport around the cell.

Physical evidence of note for GCT/LQG. Recent and highly invaluable work by Bliokh et al. [40] has demonstrated that the surface Maxwell modes at media interfaces are governed by a topological invariant valued in Z_4 . Their approach, rooted in phase winding, represents a significant advance in our understanding of topological invariants in photonic systems. However, in my view, this classification does not fully account for the independent binary symmetries present in the physical configuration—specifically, the left/right and up/down combinations encoded in a THM_4 construction. While the choice of Z_4 as an invariant is logical within their framework, a reformulation in terms of the Klein four-group V_4 more accurately reflects the underlying symmetry structure. The viewpoint directly

parallels the base 2-geobit state space and LQG tetrahedra, providing both a more granular symmetry classification and a richer foundation for both physical insight and computational applications.

Summary. Spin networks/foams are not competitors to geometric computation; they *are* its coarse invariants. The helix micro-geometry supplies the nonabelian holonomy that the LQG combinatorics were designed to encode. In this sense, LQG’s discrete data are recovered as a stable phase of a continuous, programmable gauge dynamics, and need not be elevated to fundamental kinematics.

5.4 On the Expressive Power of Physical Computation

Before stating our final claim, we collect a few canonical gauge-theoretic solutions whose geometry already exhibits the helical/twisting features emphasized in this work. The point is not to assert equivalence, but to note recurring *mechanisms*—periodicity, compactification twists, unstable saddles, and chiral transport—by which the connection A accumulates phase or matrix holonomy along preferred directions.

Calorons. Periodic instantons on $\mathbb{R}^3 \times S^1$, whose Euclidean time compactification enforces a helical/phase structure along the circle [41].

Twisted solutions. Gauge fields on compactified or cylindrical backgrounds with nontrivial boundary conditions; the twist enforces a globally enforced phase (or matrix) winding [42].

Sphalerons. Unstable saddle points in electroweak theory; they sit atop energy barriers and encode large, topological reconfigurations (holonomy changes) along interpolation paths [43].

Witten’s twisted constructions. Lower-dimensional/topologically twisted reductions where the effective degrees of freedom are organized by holonomy classes and cohomological data [44].

Non-abelian plane waves. Chiral, order-sensitive transport in non-abelian gauge theory; parallel transport along wavefronts produces noncommuting holonomies [45].

Taken together, these exemplars highlight three recurring themes: (a) *periodic/compactified directions* that act as helical axes (calorons, twisted sectors), (b) *unstable saddles* that mediate large holonomy moves (sphalerons), and (c) *intrinsically chiral, non-abelian transport* (plane waves). In what follows, we leverage exactly these mechanisms—now encoded by holonomy class functions on chosen loops—to motivate the usage of geometric computation without committing yet to any specific mass-gap conclusion.

While this discussion does not claim to exhaust the intricate connections between geometric computation and modern field theory, it is instructive—cf. Fig. 5a—that gluing two helical manifolds along a common boundary produces patching data directly analogous to the clutching that underlies BPST instantons. We now enter the final of the four computational regimes, *emergent gauge*.

Each unit helix manifold H_A, H_B is modeled by open sets $U_A, U_B \subset M$ carrying local trivializations of a principal G -bundle $\pi : P \rightarrow M$. Gluing the helices along their common base is encoded by a transition map

$$g_{AB} : U_A \cap U_B \longrightarrow G,$$

which acts on the fibers; on the overlap, the base identification is the identity. In gauge theory, g_{AB} specifies how local data patches are related; geometrically, it records how internal frames are rotated when crossing the interface. Deformations of the gluing (and of g_{AB}) change the induced holonomies on linking loops and thus realize discrete computational updates. In this sense, a unit helix behaves solitonically, and computation in this regime naturally traffics in the same patching/moduli structures that organize the instanton sector of Yang–Mills theory. This positions geometric computability as a concrete starting point for a *computational Yang–Mills* framework: a GCM presented as a bundle with connection whose updates are local in the bundle charts and functorial under gluing.

$$\begin{array}{ccc} \text{Local helix chart } (U_A) & \xrightarrow{g_{AB}} & \text{Local helix chart } (U_B) \\ \downarrow & & \downarrow \\ \text{Trivialization } P|_{U_A} & & \text{Trivialization } P|_{U_B} \end{array}$$

The diagram summarizes the configuration space of gluings: once a connection is chosen and equations (e.g. ASD) are imposed, one arrives at a moduli space modulo gauge—the familiar organizing object for instantons.

Example 5.4 (Minimal Bundle–Connection Data). Let $P \rightarrow M$ be a principal G –bundle with local 1–forms $A_A \in \Omega^1(U_A, \mathfrak{g})$ and $A_B \in \Omega^1(U_B, \mathfrak{g})$ related on overlaps by

$$A_B = g_{AB}^{-1} A_A g_{AB} + g_{AB}^{-1} dg_{AB}, \quad F_B = g_{AB}^{-1} F_A g_{AB},$$

with $F_A = dA_A + A_A \wedge A_A$. Viewed as a time–bordism on $M \times I$ (temporal gauge), the BPST sector corresponds to $G = SU(2)$ and anti–self–dual curvature,

$$F + *F = 0.$$

On a compact 4–manifold $\widehat{M}$ (a compactification of $M \times I$), the instanton number is

$$k = \frac{1}{8\pi^2} \int_{\widehat{M}} \text{tr}(F \wedge F) \in \mathbb{Z},$$

and the boundary Chern–Simons functional changes by $\Delta \text{CS} = \int_{\widehat{M}} \text{tr}(F \wedge F) = 8\pi^2 k$ under standard normalizations.

In the abelian/quasi–fluidic limit $G = U(1)$, a loop $\gamma \subset M$ carries the computational phase

$$d_\tau([\gamma]) = \text{Hol}_\gamma(A) = \exp\left(i \int_\gamma A\right), \quad d_\chi \in C_2 \text{ (chirality),}$$

and these labels transform covariantly under a change of patching g_{AB} . Gluing two helices along $U_A \cap U_B$ modifies the transition map ($g_{AB} \mapsto h g_{AB}$) and hence the induced holonomies on linking loops—implementing a discrete update at the interface. In the $SU(2)$ sector, a computational step can be idealized as traversing an ASD configuration on $M \times I$; the induced change in the boundary Chern–Simons functional is governed by k , providing a topological gate in the emergent–gauge regime.

Hence, in geometric computation, abelian solutions correspond to the stable storage and propagation of data—periodic, commutative, easily reversible. Non–abelian solutions correspond to the transformation of data: their flow actively encodes logic gates, information processing, and state transitions that cannot be reduced to simple, commutative propagation. The highest regime—emergent gauge—allows for history–dependent, topological computation, where even the process of transitioning between states (as in sphalerons—see Manton [46]) becomes part of the logic. The current hypothesis in this paper, that the geometry of a Yang–Mills field is, in all likelihood, some form of analog computation composed of helices, is outlined in 9.¹⁴

Table 9: Geometric computational regimes and the equations of motion

Abelian Regimes		Non–Abelian Regimes	
Symbolic	Quasi–Fluidic	Fluidic	Gauge
Discrete logic states, stable storage	Periodic or cyclic state transitions, memory elements	Data dynamically transformed by non–commutative flows	Topological transitions, history–dependent computation
Classical EM fields, digital bits	Calorons, periodic instantons, “clock” solutions	Non–abelian plane waves (Coleman), helical/twisted solutions (Witten)	Sphalerons, instantons, tunneling between vacua

Example 5.5 (The Computational Consequences of Non–Abelian Plane Waves). As a motivating example of one solution cited above, consider the appearance of non–abelian plane waves as described by legendary Harvard physicist Sidney Coleman:

$$L = -\frac{1}{4} \text{tr}(F_{\mu\nu} F^{\mu\nu}) = \frac{1}{2} \sum_{a=1}^n [(E^a)^2 - (B^a)^2] =: \frac{1}{2} (E^2 - B^2).$$

This is discussed by Kastrop [48], where he notes (with some exasperation) that plane–wave solutions with equal electric and magnetic energy densities satisfy $E^2 = B^2$ and hence $L = 0$. In the non–abelian

¹⁴As to why the Universe chooses to *express itself* this way? The author kindly asks that questions be directed elsewhere. For starters, see Penrose [47].

case, such waves carry internal structure: along the propagation direction, the gauge potential undergoes nontrivial parallel transport, yielding non-vanishing holonomy in the gauge group. Equivalently, the field configuration traces a nontrivial path in the group manifold even though the local scalar invariant $E^2 - B^2$ vanishes.

Within a geometric–computational viewpoint, this slice $L = 0$ may be regarded as a *switching hypersurface* in configuration space: crossing it changes the group–theoretic state realized by the parallel transport while preserving the light-like character of the wave. The family of such plane waves thus furnishes a controlled alphabet of *signal trajectories*: each solution implements a definite group action (holonomy) along its world-volume, so compositions of waves correspond to compositions of group elements, with path ordering in the non-abelian case. In this sense, classical Yang–Mills solutions encode a concrete logic of state transitions, with the group providing the algebra and the spacetime evolution providing the flow.

Remark 5.2 (Euclideanization and Clocking). While the physical Yang–Mills mass gap is posed on Minkowski spacetime $\mathbb{R}^{1,3}$, our constructive model is formulated on a *Euclidean* 4-manifold. Concretely, we work after a Wick rotation $t \mapsto -i\tau$. Hence, the background metric is Riemannian, and the fourth coordinate can be used to encode an extended, chirality-resolving structure (e.g., Type 2 *THM*) without committing to a Lorentzian time foliation.

Operationally, temporal evolution is replaced by global pulses/flow propagation that provide an internal clocking of evaluations. Lorentzian observables are recovered by analytic continuation: correlation functions are computed in the Euclidean model and (where reflection positivity holds) reconstructed to Minkowski amplitudes via the usual Osterwalder–Schrader logic. In our calculus, “Wick rotation” is a controlled effect: Euclidean evaluation (for existence/compactness) and Lorentzian reconstruction (for scattering/causality) are separate, typed phases of the same program. This retains the computational advantages of a 4D Euclidean setting while maintaining a principled path back to $\mathbb{R}^{1,3}$ for physical predictions.

5.5 Constructive Computational Gauge Theory

We can and should frame quantum gravity, and more generally gauge theory, as a problem of expressiveness versus verifiability. If we allow “all histories” (arbitrary geometries, topology change, gauge redundancy, unbounded recursion in the construction of spacetimes), amplitudes become ill-defined and intractable. If we clamp down too hard, we lose physically relevant states and dynamics. **Functional programming offers a blueprint for balancing these extremes.** Our proposal is a constructive computational gauge theory that strikes a principled middle ground: a typed, linear, total, effect-controlled calculus of geometries. Concretely, boundary data (3-geometries with gauge labels) are the types; spacetime regions (4-dimensional histories/cobordisms) are the terms; and gluing is composition. This gives a compositional semantics familiar from TQFTs but designed to scale beyond the purely verifiable/topological setting (i.e., Chern-Simons).

Table 10: Programming concepts and quantum gravity analogues

Programming Concept	Quantum Gravity Analogue
Types	Boundary states (3-geometries with gauge data)
Terms / Programs	4-geometries (cobordisms, histories)
Composition	Gluing of spacetime regions
Linear types	Conservation laws, unitarity (no-cloning of boundary data)
Totality	Termination of the “geometry evaluator” (finite amplitudes)
Effects & handlers	Coarse-graining and renormalization
Dependent types	Gauge and diffeomorphism constraints

Readers are asked to consider the correspondences in Table 10. Three design choices enforce computability and physics: linearity, totality, and effects. Linearity tracks boundary degrees of freedom as conserved resources (no cloning/erasure), so unitarity and charge conservation are built into the typing discipline rather than imposed post hoc. Totality means the “geometry evaluator” (our state-sum/variational executor) always normalizes: amplitudes exist and are finite in the core fragment.

The phenomena that usually force uncontrolled manipulations, such as coarse-graining, stochastic mixing, and renormalization, are modeled explicitly as algebraic effects with handlers. In this way, renormalization becomes a controlled transformation of programs rather than an ad hoc subtraction. Dependent types encode gauge and diffeomorphism constraints at the level of well-typedness, so invariances propagate mechanically through compositions.

Within this calculus, amplitudes are evaluations, symmetries live in the types, and RG/coarse-graining are effect handlers. The proposed helical primitives provide the concrete generators of histories: smooth, orientable flows that carry discrete topological labels (orientation/chirality) alongside continuous geometry. This marries the “continuous versus discrete” tension: spectra and curvature are continuous objects; quantum numbers arise as stable, counted winding data. Practically, the workflow is: specify typed boundary data; assemble regions from helical primitives; compose; evaluate; and, where needed, apply effect handlers that implement scale changes with proofs of soundness.

The payoff is a language that is *expressive* enough to describe nontrivial gauge dynamics and background independence, yet *restricted* enough to prove normalization, locality/compositionality, and anomaly-freeness in the core. Extensions (matter content, topology change, nonperturbative sectors) are added modularly as new effects or controlled type extensions, preserving verification theorems as we widen scope. In short, constructive computational gauge theory provides a semantics where we can calculate, compose, and certify. For the foundational work on constructive quantum field theory, see Baez, Segal, and Zhou [49]. Our approach here is in this spirit, but computational. Likewise, see Adlam [50] for recent philosophical work on constructive spacetime axiomatics.

On QFT, and methods of meaning. Per the EVT (2.6), we take the view that QFT as a single continuum probability measure in 4D is likely too expressive to be globally verifiable without running into tractability bounds. Primitive probabilities on infinite-dimensional functional spaces entail inheriting measure-theoretic pathologies. Renormalization already reveals QFT as a form of self-editing computation: its effective laws depend on scale, with counterterms and field redefinitions acting as code transforms; the interpreter (RG) rewrites the program (the Lagrangian). In such a macro-expressive system, the EVT predicts the failure of global algorithmic verification. Our remedy is not to deny QFT’s empirical success, but to replatform it. While QFT can and will continue *language* for scientists to create *shared systems of meaning* for the phenomena they encounter in their day-to-day work (e.g., condensed matter, quantum information), challenges such as QG require exploration of new computational approaches.

From external lattices to an intrinsic, lattice gauge calculus. Lattice gauge theory discretizes a continuum on an external grid: links carry group elements, plaquettes carry curvature, and one hopes to recover physics in a subtle continuum limit. But computability is not baked in; it is delegated to numerics.

By contrast, GCM lattices are of a different kind. The “lattice” is the 1-skeleton generated by minimal holonomy loops of the helix geometry itself; nodes are join interfaces (Gauss constraints), edges are Wilson-loop registers, faces carry curvature flux. No continuum limit is imposed from outside: the cell structure is induced by the manifold’s own group-theoretic and topological constraints. Spectral gaps and forbidden transitions then arise from intrinsic geometry rather than from a choice of mesh.

Operationally, this translates to:

- **State space.** A presheaf of local moduli (connections/holonomy classes on cells) over a helix-adapted cover; *sheafification* enforces gluing laws (Gauss, Bianchi) and yields global states.
- **Evaluation.** Amplitudes are functorial evaluations of class functions on Wilson loops (edge registers) subject to node intertwiners; normalization is a theorem of the typed core.
- **Scale change.** Refinement/coarse-graining is an explicit algebraic effect that changes the cover and updates moduli by certified handlers; proofs of invariance live in the type/effect system.

In this setting, the “sum over histories” is replaced by *sheafification + evaluation*: one computes global sections (or their invariants) of the sheafified moduli rather than integrating over an externally supplied configuration measure.¹⁵

¹⁵“It’s nice to have a framework where you can only have one Lagrangian, but it would be even nicer to have some reason that this particular algebra would inevitably appear as an answer to some question.” —E. Witten [51]

Remark 5.3 (Why Not Classical $\mathbb{R}$?). In the fluidic/gauge band, one is tempted to model directly over the classical continuum. However, unrestricted real computation (cf. Blum–Shub–Smale A.5) admits hyper–Turing power and brittle pathologies (exact tests, branch–on–sign over nonrobust inputs, oracle–like primitives). This undermines the two pillars we require: *executability* (every step is effectful but total) and *verifiability* (small perturbations do not change the decision class). Our constructive stance preserves smooth geometry while preventing hidden oracles.

Geometric reals? We separate *semantics* (smooth reasoning) from *execution* (effective approximation):

- (a) **Computable reals** $\mathbb{R}_{\text{comp}}$. externally, numerical data (couplings, boundary values, readouts) live as computable reals given by fast Cauchy names (or, when possible, algebraic numbers with exact arithmetic). Equality is replaced by interval refinement; separation proofs guard branching. This is the target of all executable algorithms.
- (b) **Synthetic/smooth reals.** $\mathbb{R}_{\text{syn}}$ internally, differential reasoning uses the smooth reals of a well–adapted SDG model (Kock–Lawvere axiom, nilpotent $d\varepsilon = 0$), so vector calculus and curvature remain first–class without appeal to pointwise limits.¹⁶
- (c) **A bridge.** A semantic–to–execution functor sends internal smooth expressions (curvatures, class functions on holonomy) to external names in $\mathbb{R}_{\text{comp}}$, with *proof obligations* discharged by our effect handlers (for limits/coarse–graining).

The aim is not to deny the continuum but to choose a base field that respects computability. Parameters are measured to finite precision; algorithms require effective representations. Thus we use rationals/algebraics/ $\mathbb{R}_{\text{comp}}$ for execution, while $\mathbb{R}_{\text{syn}}$ supplies calculus internally. This pragmatic split aligns with the observation (cf. Szudzik [52]) that physical models do not require arbitrary reals to be *expressed* or *simulated*, even if smooth structure is invaluable for *reasoning*.

Design axioms: equality, limits, and effects. We enforce three constraints in the core calculus:

- **No exact branching on undecidable predicates.** Tests are certified by separation bounds (or deferred as obligations).
- **Limits are effects.** Completion, closure, and renormalization are algebraic effects with handlers that carry witnesses of convergence/stability.
- **Gauge invariants only.** Readouts are class functions of holonomy (e.g. traces), ensuring stability under small perturbations and gauge changes.

Remark 5.4 (Disclaimers on Classical QFT). Our setting concerns analog geometrodynamics on a noncompact configuration space. “Gaps” (mixing rates, spectral edges of transfer generators) are typically defined by nested sup / inf over unbounded domains. In general, these constants need not be uniformly computable from the defining data. We therefore (a) state results in terms of *semi–decidable* or *effectively approximable* quantities, or (b) confine genuinely noncomputable constants to effectful interfaces outside the core (they cannot influence totality/normalization proofs). Likewise, this is a computability warning label for our own continuum models; we cannot use our model to solve the Clay Yang–Mills problem [53], which requires a fully rigorous quantum theory.

On finite information and intuitionist physics. We concur with the operational point that experimental data carry finite information, and that uncountable precision should not be a primitive of algorithms, as brought up by Prof. Gisin [54]. Where we differ is methodological: we retain a smooth internal semantics ($\mathbb{R}_{\text{syn}}$) to keep geometry local and functorial, while ensuring that *every* externally visible quantity is realized in $\mathbb{R}_{\text{comp}}$ with certified error control. Thus, the continuum remains a modeling tool; computation proceeds on effectively presentable numbers.

Choosing a constructive base allows us to maintain the familiar analysis when needed, while ensuring that every step of the formalism is executable. Our contention here is that the universe operates on smooth, transcendental flows; however, topology and quantization necessitate coarse–graining, which

¹⁶Any well–adapted model suffices; no commitment to a specific topos is needed, but an implementation is provided in A.7 for convenience.

projects these flows onto the discrete structures humans use. It is hoped that constructive proof presented here on the existence of GCMs may begin to help settle longstanding issues. *Both* are real: the continuum substrate *and* the discrete emergent layer! What we call “physical computation” is a geometric process that sits precisely at this boundary, with its behavior governed by the EVT Trilemma. In particular, nothing in the approach depends on sub-Planckian claims; we avoid making uncountable structure a primitive of the dynamics. A high-level road map toward the Yang-Mills mass gap in computational gauge theory is outlined below:

- (a) **Define Computational Semantics for Gauge Theory** — Define a typed, linear, total calculus whose types encode boundary data (3-geometries with gauge labels) and whose terms encode histories (4-dimensional “programs” composed by gluing).
- (b) **Build Computable Geometric Bundles** — Give the calculus a denotational semantics as a computable fiber-bundle (connection, parallel transport) formulated in a constructive/synthetic differential setting over a computable base field. This preserves the differential structure while ensuring that all primitives are evaluated effectively.
- (c) **Assess the Scaling Limit to Continuum Yang-Mills** — Prove that families of programs approximate continuum Yang-Mills in a controlled scaling limit. Concretely, a “differential computational jet calculus” provides error bounds showing convergence of correlation functions/Wilson loops to those of continuum Yang-Mills as the program lattice is refined (see A.2. for details)
- (d) **Prove the Gap and its Stability** — Establish a spectral gap (uniform lower bound on excitation energies) for the constructive model using the calculus’ linear/compactness properties; then show this lower bound survives the scaling limit, yielding a continuum mass gap. Here, the constructive setting replaces global non-compact searches with local-to-global, type-theoretic invariants.

All physically relevant predictions are recoverable with parameters and observables taken from an effective number system (e.g., computable reals). This keeps the continuum as a limit, not as a prerequisite, and lets “proof by construction” coexist with the analytic tools physicists use.

Rather than pursuing unification for its own sake, it is more likely that progress in mathematical physics can be achieved by developing powerful new tools for concrete, foundational problems such as the Yang-Mills mass gap. GCT, by providing a new lens for both discrete and analog field theories, has the potential to make previously intractable problems accessible to rigorous analysis. In doing so, it may render the broader unification of physical theories less a distant goal and more a natural outcome of deep mathematical understanding.

This approach reframes the Yang-Mills mass gap problem as a question of information-theoretic structure. Can the dynamics of a gauge field theory, defined over the continuum and acted on by a compact simple Lie group, enforce a nontrivial discrete gap in its computational state space? This translation from an energy gap to an information gap offers a new perspective on both the foundations of gauge theory and the limits of computability in physical systems. **In essence, to attempt to solve the mass gap, we cannot just try to compute the solutions to the Yang Mills equations, but rather we may have to compute *with* the solutions to the equations themselves.**

Atiyah’s warning and the work to come. Needless to say, the final step of the outline would involve a complex derivation process. In the upper regimes, one must appeal to cohomological, topological, or even quantum invariants to encode information, and the entire process of “computation” itself becomes vastly more abstract. The actual group needed to “power” such a computational process would likely be none other than $\mathbf{Diff}(\mathbf{M})$ itself. Recall the chart of n -logics from 7. This is the final entry at the top. Its geometric expressiveness—“Extreme,” its algebraic closure—“Maximal.” Under such a maximal symmetry group, preserving invariants becomes increasingly tricky. We can gain immense geometric power, but only at a high algebraic cost.

The landscape of twistor theory. While our constructions emphasize categorical and geometric composition, it would be amiss not to note that *twistor theory*, pioneered by Penrose and collaborators [55], gives a striking *holomorphic* encoding of four-dimensional conformal geometry and massless field equations. Concretely, the Penrose transform and the Ward correspondence identify cohomological data on projective twistor space with solutions of linear massless equations and, in the non-abelian case, (anti-)self-dual Yang-Mills on spacetime.

Reader’s compass. For our purposes, “twistor phase” simply means a regime where inherently noncommuting spacetime/gauge data admit a holomorphic reparametrization that turns analysis into cohomology. The symmetry group at play is the full conformal group $SU(2, 2) \cong \text{Spin}(2, 4)$; it is *not* an abelian corner. Thus the twistor limit is best read as a complex-analytic chart in which verification improves because equations become integrable/holomorphic—even though the underlying gauge content remains non-abelian. Nothing in this viewpoint licenses literal real-valued “infinite computation”: finite-information bounds (e.g. Bekenstein–Bousso [56]) still apply. The twistor phase should be read as a *cohomological* linearization window, not as physical hypercomputation.

Finally, our framework is naturally compatible with holomorphic tools that also appear in parts of the Geometric Langlands story (cf. Kapustin–Witten) [57] and with twistor approaches to scattering amplitudes; see, e.g., the mathematical discussion noted by Woit [58]. We do not claim a direct realization of Langlands duality; rather, we point to a common structural mechanism—holomorphic/complex-analytic repackaging of non-abelian field data—that GCT will interface with in future work.

On Calabi–Yau manifolds and stable loci. Our perspective is not string-theoretic, but it sits squarely within the geometric-analysis paradigm inaugurated by Calabi: seek canonical metrics by analytic deformation and flow, then read structure from the resulting moduli. On a compact Kähler manifold with $c_1 = 0$, Yau’s theorem guarantees a unique Ricci-flat Kähler metric in each Kähler class. This furnishes a canonical, *integrable* background with powerful linear tools (Hodge theory, period maps, variation of Hodge structure) and controlled deformations.

Reader’s compass. In our setting, Calabi–Yau (CY) structures are *stable bookkeeping loci*: places in the global moduli where curvature decouples enough that the effective local algebra is as close to commutative as the dimension allows. The harmonic representative of each cohomology class provides a linear “ledger” for fluxes and charges, and the Hodge decomposition/intersection form supply rigid accounting for admissible operations. For threefolds, the $(1, 1)$ and $(2, 1)$ sectors (Kähler vs. complex moduli) control, respectively, capacities and morphisms available to the substrate; at such loci, symbolic projections are maximally checkable because key questions reduce to cohomological linear algebra.

Technically, this is the state space where:

- (a) the effective coefficient algebra is (up to mild twists) E_∞ in abelian and maximally-abelian quasi-fluidic regimes, so excision holds and parallel composition is transparent;¹⁷
- (b) twisted and equivariant K -theory encode stable resource classes (channels, phases) and their integral constraints via periods, while variations of complex structure are recorded by period integrals and Picard–Fuchs equations;
- (c) moving away from a CY locus (e.g. conifold degenerations or large-complex-structure limits) introduces monodromy/Picard–Lefschetz transformations, which reintroduce noncommuting transport: checks become globally harder even as the space of behaviors enlarges.

Hence, CY manifolds are not a generic case; they are *integrable gate points*—symmetry-enhanced regions that abelianize the accounting and make verification accessible. They play an essential role in the lower (abelian) and quasi-fluidic phases as calibration loci and phase boundaries: near a CY stratum, the dynamics are governed by linear/cohomological data; moving away activates non-abelian holonomy and the twistor calculus native to the non-commutative regimes.¹⁸

On Seiberg–Witten and Atiyah–Singer. In the continuum, anomaly cancellation and spectral flow are governed by index theory: the analytic index of Dirac-type operators equals a topological index (Atiyah–Singer). In our framework, the corresponding Ward/typing identities can be seen as constructive index constraints: zero-mode counts and anomaly coefficients are enforced at the level of types and persist under gluing. Thus, index-theoretic stability is reflected as compositional invariants of our evaluator rather than as a posteriori analytic facts.

Likewise, the Seiberg–Witten equations exemplify how gauge-theoretic PDEs on 4-manifolds can yield compact moduli spaces with energy bounds and computable invariants. Our Euclidean computational formulation seeks the same structural features: finite-energy solution classes generated by

¹⁷ E_∞ here just means “homotopy-commutative”; readers may safely read it as “fully abelian up to coherent homotopy.”

¹⁸For readers fluent in string theory: one may regard a geobit as a deterministic analogue of a worldsheet defect. This viewpoint suggests a geometric mechanism reminiscent of the weak gravity conjecture [59]; we do not attempt a derivation here.

helical programs, compactness under refinement, and invariants derived from holonomy/typing. We emphasize that we do not claim new SW invariants; instead, we adopt the SW template with compactness and abelianized control as the target shape for our constructive certificates (e.g., gap lower bounds and well-behaved moduli).

Table 11: Classification of computational regimes by data type and algebraic structure

Symmetry Regime	Algebraic Structures	Data Types
Discrete / symbolic (finite cyclic C_n)	Finite rings/Boolean algebras	Bits, n -its
Analog abelian ($U(1)$)	Real/complex fields, Lie algebra $\mathfrak{u}(1)$	Circle phases, amplitudes
Analog non-abelian ($(SU(2), SU(3))$; gauge group actions)	Principle bundles, moduli spaces, ∞ -groupoids; (co)homology operations	Global classes (characteristic classes/Chern, Floer/Donaldson-type)

Disclaimer regarding entropic gravity and the Lorentzian. Lest the framework proposed be misinterpreted, it should be specified that geometric computation works at the pre-metric substrate and derives causal/optical structure via an effective constitutive tensor χ . In near-equilibrium regimes χ becomes metric-induced, recovering Lorentzian geometry and Schuller’s pre-metric electromagnetism [60] as an effective semantics. On larger scales, the thermodynamic equation of state reproduces Jacobson’s derivation of Einstein’s equations. So Schuller’s and Jacobson’s perspectives [61] appear as complementary limits of a single constructive framework: EM fixes the light cone via χ ; coarse-graining of the underlying computation promotes χ to $\chi[g]$ and yields GR. No contradiction is implied—only a layering of descriptions. Still, operational constraints require investigating the nature of the strong nuclear force before addressing black hole thermodynamics.

Likewise, we do not quantize gravity in the traditional sense. We do not treat the metric as an additional Yang–Mills field and impose a continuum path integral over geometries. Quantum features (probabilities, interference) arise from structured coarse-graining of deterministic geometric programs, while GR appears as the large-scale limit of sound gluing rules. Here, gravity is not “quantized” as a field; it is the constraint architecture of the computation, with quantum behavior emerging from how we compress and observe it.

5.6 On the Classification of Geometric Field Theories

We can now classify *geometric field theories* by specifying three layers of structure and composing the canonical maps between them:

$$\begin{array}{c}
\text{decorated state/transition groupoid} \\
\downarrow \text{higher rep.} \\
\text{fully dualisable object in a symmetric monoidal } (\infty, n)\text{-category} \\
\downarrow \text{Cobordism Hyp.} \\
\text{fully extended } (\infty, n)\text{-TFT}
\end{array}$$

Existence–uniqueness (Baez–Dolan [49]/Lurie [62]) identifies the last arrow once a fully dualisable object is supplied; the first arrow is where GCT lives, turning helix–decorated paths and bordisms into higher representations with explicit operational semantics.

Operational layer (constructive evaluator). As in Section 5.5, *types* are boundary data (3-geometries with gauge labels), *terms* are 4-dimensional histories (cobordisms), and *composition* is gluing. Linearity tracks conserved resources; totality guarantees normalization of evaluations in the core; coarse-graining/renormalization are treated as algebraic *effects with handlers*. For every regime below, the interface is the same: registers are Wilson loops/surfaces, readouts are gauge-invariant class functions of holonomy (e.g. traces), and evaluation is symmetric monoidal and local.

Regime Ladder Overview.

(a) **Symbolic.** *Input symmetry/labels:* decorated fundamental groupoid with finite abelian labels $(V_4, (C_2)^3)$. *Higher rep target:* finite-group 2-representations (objects = twist sectors, 1-morphisms = path homotopies). *Dualisability:* finite semisimple $\Rightarrow$ left/right duals exist. *Predicted TFT:* 2D Dijkgraaf–Witten. *Ops:* commuting joins; characters χ_g as readouts.

(b) **Quasi-fluidic.** *Input:* $(C_2)^4$ with 1- and 2-form labels (higher-form flux). *Higher rep:* 3-representations / crossed-module representations. *Dualisability:* finite higher group $\Rightarrow$ rigid. *Predicted TFT:* 3D higher-form DW / Yetter quantum double. *Ops:* loops and surfaces as registers; higher Wilsons; coarse-graining as 2-sheafification.

(c) **Fully fluidic.** *Input:* symmetry of a THM cell $D_4 \ltimes (C_2)^4$ acting on a decorated Π_1 . *Higher rep:* 4-representations / 2-connections on principal 2-bundles (toy non-abelian sector). *Dualisability:* restrict to compact structure groups; finite-depth truncations for constructivity. *Predicted TFT:* 4D BF (topological) as rigorous core; YM-term added as an *effect* in the evaluator. *Ops:* noncommuting joins; class functions of holonomy as readouts; renormalization as a certified handler.

(d) **Emergent gauge.** *Input:* bundle of simple Lie data (e.g. $SU(2)$) on helix-adapted charts. *Higher rep:* bundles of fully dualisable objects in $\mathbf{Alg}_\infty$ (e.g. E_2 -algebras of Wilson lines). *Dualisability:* compactness/rigidity (fusion) or factorization criteria, depending on dimension. *Predicted TFT:* 3D Chern–Simons (fully extended); 4D topological sector (BF/CS-like); YM handled as an *effect*. *Ops:* same interface; node constraints become intertwiners in the representation category.

Lemma 5.2 (Evaluator Factors Through the Classified TFT). *Let $\mathcal{D}$ be a symmetric monoidal (∞, n) -category and $X \in \mathcal{D}$ a fully dualisable object obtained as a higher representation of the decorated groupoid/bordism data. The constructive evaluator of Section 5.5 (which is symmetric monoidal, local, and sends start/stop to coevaluation/evaluation) factors uniquely through the fully extended TFT Z_X classified by X .*

Proof sketch. By locality and monoidality, the evaluator is a symmetric monoidal functor on the bordism (∞, n) -category. Its values on units and caps/cups agree with the duality data of X by construction, hence it is equivalent to Z_X by the Cobordism Hypothesis. $\square$

Table 12: Constructive classification: regimes, dualisable inputs, and predicted TFTs.

Group Type	Fully dualisable object supplied	Predicted (n) -TFT	Comments	Dualisability guardrails
Abelian	Finite-group 2-representation of the decorated groupoid (V_4)	2D Dijkgraaf–Witten	Factors through finite sets $\rightarrow \mathbf{Vect}$; commuting joins	Finite semisimple category; duals automatic
Maximally abelian	Discrete 3-representation of $(C_2)^4$ with higher gauge 2-forms	3D DW / Yetter double	$\mathbb{Z}_2$ higher-form lattice models; surfaces as registers	Finite higher group; rigid monoidal structure
Non-abelian	Smooth 4-representation of $D_4 \ltimes (C_2)^4$ inside decorated Π_1	4D BF; Yang–Mills via effect	Noncommuting holonomies; renorm as certified handler	Compact structure groups; depth truncations
Simple Lie	Bundle of dualisable objects in $\mathbf{Alg}_\infty$ (e.g. E_2 of $SU(2)$ lines)	Chern–Simons / (∞, n) -TFT; YM via effect	Matches the Cobordism Hypothesis with decorated invariants	Compactness or factorization-algebra criteria

Guardrails in 4D: Topological Core vs. Metric Effects. In four dimensions, the Cobordism Hypothesis classifies *fully extended* topological field theories by *fully dualisable* objects in an appropriate symmetric monoidal (∞, n) -category. This classification is inherently *topological*: it is driven by local-to-global gluing, duals/evaluations, and functoriality under bordism composition, rather than by metric-dependent dynamics. [63]

Pure Yang–Mills is therefore not an acceptable *core* object for dualisability statements: the Yang–Mills action depends on a choice of metric through the Hodge operator $*$, and it carries propagating local degrees of freedom. To keep the 4D semantics fully compositional, we isolate a metric-free *BF topological core* and treat the metric-sensitive Yang–Mills term as an *effect* handled by the constructive evaluator. Concretely, one may regard Yang–Mills in first-order form as a deformation of BF theory:

$$S_{\text{BFYM}}(A, B) = \int_M \text{tr}(B \wedge F_A) + \frac{g^2}{2} \int_M \text{tr}(B \wedge *B),$$

where A is a connection with curvature F_A and B is an adjoint-valued 2-form. The BF term is topological (metric-free), while the second term is precisely where metric/scale enters via $*$. In the formal “topological limit” $g \rightarrow 0$, one recovers BF theory; integrating out B recovers the usual second-order Yang–Mills action.

This mirrors the Plebanski [64] paradigm: four-dimensional gravity can be presented as *constrained BF theory*, [65] where a BF-type action is supplemented by *simplicity constraints* (or, equivalently, constraint potentials) to recover metric geometry and propagating modes. In our setting, the analogy is operational:

- **Core (dualisable).** BF-type topological semantics, used for the Cobordism-Hypothesis classification layer.
- **Effects (non-topological).** Metric-dependent dynamics (YM terms, constraint potentials, coarse-graining/RG) implemented as controlled effect-handlers in the evaluator.
- **Soundness criterion.** Effect-handlers must preserve symmetric monoidality (gluing/parallel composition) and must respect boundary gauge-invariant observables (e.g. class functions of holonomy such as Wilson loop traces).

All dualisability statements in this paper pertain to the BF core; the physically relevant Yang–Mills sector is recovered by applying the declared metric effects in a way that is compositional on boundaries.

Asymmetry of Regimes. Recall again the chart of n -logics 7. The four regimes split into “inner” (Symbolic, Fluidic) and “outer” (Quasi-fluidic, Gauge) types as detected by the join commutator 5.1:

- (a) **Quasi-fluidic (outer)** — $[J_1, J_2] = e$ and the local symmetry on phases is abelian. A K -equivariant change of variables puts interfaces in block form, so *independence is a coordinate choice* via decoupling if need be.
- (b) **Symbolic (inner)** — $[J_1, J_2] = e$ by construction; helices are *intentionally glued* through a shared discrete update table.
- (c) **Fluidic (inner)** — generically $[J_1, J_2] \neq e$ couplings are path-dependent via non-abelian holonomy and helices are *dynamically glued*.
- (d) **Gauge (outer)** — typically $[J_1, J_2] \neq e$, but there is gauge redundancy; after gauge-fixing, field lines behave as independent modules at the representation level, so *independence is a gauge choice* as the dynamics remains non-commutative.

Gluing and the sectors of ω -logic. There is a curious “outer vs. inner” asymmetry in the chart of n -logics. In the symbolic and fluidic bands, helices are defined as glued; in the quasi-fluidic and gauge bands, this need not hold. Under a C_2^4 embedding inside QF, joins commute yet are not always necessary (decoupling by normal form), whereas in the gauge band, joins play a different role altogether (gauge transformations). Thus, the observed breakdown of compulsory gluing toward the extreme ends of ω -logic signifies that gluing is not primitive there: it is coordinate- or gauge-dependent structure, rather than a topological invariant of the computation.

Question. *But why does nature bother with gluing helices?* A single helix’s axial $U(1)$ phase renders a gauge; with a twin, the relative phase $\Delta\theta \in S^1$ is gauge-invariant and robust. This yields an

abelian “phase logic” realizing the $C_2 \times C_2$ actions used in the symbolic regime. Likewise, gluing balances torque and stabilizes the framing, thereby making the interfaces implementable. The most commonly cited continuous symmetry of the helix (i.e., rotation about the screw axis) is not amenable to computation.

For a complete overview of the Computational Cobordism Hypothesis, see [A.7](#).

5.7 Holonomy Type Theory

It should be noted that two canonical join patterns for twin helices lead to distinct gauge–geometric behaviors.

Definition 5.3 (Twin Helical Manifold). A *base-joined* configuration (our *THM*) consists of two embedded helical surfaces $\Sigma_{\pm} \subset M$ whose boundary circles are identified along a smooth annulus $A \simeq S^1 \times I$, with orientations matched so that the resulting subset $\Sigma_+ \cup_A \Sigma_-$ is a smooth submanifold (no cone angle or cusp). A connection $A \in \Omega^1(M; \mathfrak{g})$ restricts smoothly across A ; its curvature F is bounded (no measure concentration). These are the objects used for deterministic logic flow, as in the **Schrödinger equation**.

Definition 5.4 (Apex Twin Helix). An *apex-joined* configuration is obtained by collapsing a boundary circle to a point, producing a conical join $C(S^1)$ (a stratified space with a point singularity). Equivalently, remove the apex and view the result as $M \setminus \{p\}$ equipped with a connection whose curvature may carry a localized charge at p (e.g. a Dirac monopole in the abelian case). Such joins model sources/sinks or defect cores as in optical vortices, but most prominently **Dirac cones**.

The operational distinction is holonomic:

Proposition 5.3 (Base vs. Apex Holonomy). *Let A be a connection on M .*

- (a) (Base-joined) *If F is smooth near the join annulus and B is simply connected there, then for sufficiently small loops γ linking the join, the holonomy class $[\text{Hol}_{\gamma}(A)]$ is trivial; memory arises only from the distributed curvature, not from a singular core.*
- (b) (Apex-joined) *If the join collapses to p and A has curvature with total flux through a small linking 2-sphere S_p^2 equal to a quantized charge, then small loops γ encircling p acquire nontrivial holonomy classes. In $U(1)$,*

$$\text{Hol}_{\gamma}(A) = \exp\left(i \int_{S_p^2} F\right) = e^{2\pi i k} \quad (k \in \mathbb{Z}),$$

and in the nonabelian case the conjugacy class of $\text{Hol}_{\gamma}(A)$ is determined by the localized topological data (e.g. degree of a normalized frame map, or the G -bundle class on S_p^2).

Thus, “smooth base joins” carry *transport* without intrinsic singular charge, whereas “apex joins” *store* charge as a defect that every small linking loop must detect. Both fit naturally in our holonomy-as-memory semantics; they simply instantiate different boundary/defect conditions.

Feature	Twin Helical Manifolds	The Apex Twin Helix
Join point	Smooth base join	Singular apex join
Topology	Globally smooth manifold	Manifold with point singularity
Flow	Bidirectional transport along an annulus	Radial (source/sink) around a defect core
Chirality	Controlled by local geometry	Emerges from defect topology
Singularities	Avoided (non-singular)	Essential (e.g. Dirac monopoles)
Field role	Symbolic/geometric logic (ordered joins)	Physical spinor dynamics
Purpose	Analog computational flow	Topological charge carriers

Resolution by blow-up. An apex singularity can be *resolved* by the real blow-up of M at p , replacing p with the unit normal S_p^2 . The connection extends over the exceptional sphere with a (possibly nontrivial) bundle; its Chern/characteristic class on S_p^2 is exactly the stored charge. In this sense, apex-joined and base-joined pictures are two presentations of the same data: defects as boundary classes vs. smooth transport with an extra S^2 boundary component carrying flux.

The “duality” is a clean dichotomy of join conditions: *smooth* joins emphasize programmable transport (deterministic logic), while *singular* joins emphasize conserved charges (defect logic). Hence,

helicity plays two separate roles in physics. Holonomy conjugacy classes and curvature flux rigorously control both.

Likewise, as illustrated in Fig. 5b, a complex plane wave

$$\Psi(x, t) = Ae^{i(kx - \omega t)}, \quad \Re \Psi = A \cos(kx - \omega t), \quad \Im \Psi = A \sin(kx - \omega t),$$

traces a Type 1 *THM* in space: the real/imaginary components are $\pi/2$ out of phase and form two counterwinding strands whose joints occur at equal-phase points. In Maxwell’s $U(1)$ sector (massless, no self-interaction), the physical photon inherits exactly this geometry: a pair of phase-locked, coaxial, counterwinding helices—the “helical orb.”

Question. *But why the difference? Why do photons appear as Type 1 while Yang–Mills instantons (Fig. 5a) present as Type 3 dual-cone configurations?*

Definition 5.5 (Helical Normal Forms). We say a localized geometric computational configuration on the twin helix substrate is:

- **Type 1 (Abelian Wave Normal Form).** A coaxial pair of counterwinding helices with a common propagation axis and phase-locked transverse frame; joints are equal-phase loci.
- **Type 3 (Dual-Cone Event Normal Form).** A compact, oriented cobordism in spacetime whose incoming/outgoing faces carry oppositely oriented helical frames, visualized as two opposed cones joined along a finite 3-volume.

Proposition 5.4 (Symmetry–Energetics–Boundary Selection Principle). *Given a gauge sector G and boundary data:*

- Abelian, propagating waves** (e.g. photons in $U(1)$) *minimize the geometric action by pure transverse frame rotation along an open worldline, yielding the Type 1 normal form.*
- Nonabelian, (anti)self-dual finite-action events** (instantons in $SU(2)/SU(3)$) *minimize by a compact, topologically charged 4D configuration interpolating between vacua with different $k \propto \int \text{tr}(F \wedge F)$. The oriented in/out faces enforce opposite frame orientations, producing the Type 3 dual-cone normal form.*

Proof sketch. (1) In the abelian case, holonomy abelianizes to a single phase; with no self-interaction or topological charge, the energetic minimizer is a traveling wave: equal-amplitude quadrature components assemble into a coaxial, phase-locked twin helix (Type 1). There are no junctions, cones, or domain walls to pin the frame. (2) For nonabelian (anti)self-dual solutions, the relevant invariant is:

$$k = \frac{1}{8\pi^2} \int \text{tr}(F \wedge F)$$

so finite action requires compact support in Euclidean spacetime and yields a vacuum-to-vacuum *cobordism*. The in/out orientations of the action density impose opposite normals on the two temporal faces of the event, and the lifted spin/transverse frame rotates accordingly, giving the dual, opposed cones (Type 3). These are *events* (tunneling) rather than asymptotic quanta. $\square$

Sector	Object	Dominant functional	Boundary class	Helical form
$U(1)$	photon	wave (no self-int.)	open, asymptotic	Type 1
$SU(2)/SU(3)$	instanton	(anti)self-dual soliton	vacuum $\rightarrow$ vacuum ($k \neq 0$)	Type 3

Table 13: Selection of helical normal forms by symmetry, energetics, and boundary data.

Summary. Photons are asymptotic Type 1 helical waves in an abelian phase channel; instantons are nonabelian, topological *events*—Type 3 dual-cone cobordisms—not particles. The distinct pictures are fixed by (a) symmetry (abelian vs. nonabelian), (b) energetics (wave transport vs. self-dual soliton), and (c) boundary class (open propagation vs. vacuum-to-vacuum interpolation).

Remark 5.5 (Disclaimers Regarding the SM and GUTs). Though we do not intend to execute such a derivation within this work, it should be said outright that GCT proposes that the observed Standard Model factorization $U(1) \times SU(2) \oplus SU(3)$ reflects two distinct computational primitives of the substrate: the ω -logic (binary orientation + phase rotation) underlying the chiral electroweak sector, and χ -logic (ternary triality/twist) underlying color. This accounts for the different anomaly structures, center symmetries, and IR phases of the two sectors. We offer this as an organizing hypothesis; it does not forbid an emergent simple-group unification (e.g., $SU(5)$) at higher energy scales, but it does not require it at the micro level.

5.8 Psi-Geometric

In light of our results, how can the existence of GCMs *help inform us* about potential interpretations of quantum mechanics?

It is plausible that the quantum state is an *ontic geometric structure*, a twin helix bundle with connection and density, while the usual wavefunction ψ is a *coordinate description* of that structure; probabilities arise from equivariant transport of density, and measurement is bordism-induced branching within the same geometry. This interprets ψ -ontic in the sense of PBR [66] (up to gauge), with an epistemic role for ψ as a representation.

Kinematic ontology. Fix a spacetime region M (nonrelativistic: $M \subset \mathbb{R}^3$ with time parameter; relativistic: a globally hyperbolic spacetime). The *beables* are:

- (O1) A twin helix fiber bundle $\pi : \mathcal{H} \rightarrow M$ with fiber carrying two oriented strands (chirality data), equipped with a $U(1)$ connection A (phase transport) and a smooth positive density ρ on M (helix-flux density).
- (O2) Global topological data: linking/holonomy classes of sections of $\mathcal{H}$ (“join” structure), which encode entanglement and Aharonov–Bohm-type effects.

Choosing a local trivialization of $\mathcal{H}$ yields fields (R, S) with

$$\psi(x) = R(x)e^{\frac{i}{\hbar}S(x)}, \quad R = \sqrt{\rho}, \quad \mathbf{v} = \frac{1}{m}(\nabla S - q\mathbf{A}),$$

and (relativistically) a conserved current J^μ with $\nabla_\mu J^\mu = 0$ and $J^\mu \sim \rho v^\mu$.

Dynamical Ontology.

- (D1) *Madelung/Schrödinger form.* Writing $\psi = Re^{iS/\hbar}$ gives the continuity and Hamilton–Jacobi pair

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \quad \partial_t S + \frac{(\nabla S - q\mathbf{A})^2}{2m} + V + Q = 0, \quad Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 R}{R}.$$

- (D2) *Helix-geometric form.* Q is a curvature/torsion scalar of the amplitude sheet induced by π and A ; $\mathbf{v}$ is the horizontal lift direction set by the connection. In relativistic settings, we use Tomonaga–Schwinger evolution $\Psi[\Sigma]$ on Cauchy surfaces Σ and define J^μ covariantly from the helix connection (spinorial or scalar sector as appropriate).

A measurement is a *bordism surgery* in the coupled system–apparatus helix bundle: new, boundary/join conditions modify the allowed horizontal lifts. Dynamically, this selects an attracting sector (branch) of the global helix geometry. The post-measurement ψ is the restriction/re-expression of the same ontic object to the surviving branch; no stochastic primitive is added.

Probabilities and the Born rule. The density $\rho = |\psi|^2$ is *equivariant* under the guidance flow:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0.$$

Given typical initial ensembles (Liouville/mixing with respect to the helix dynamics), frequencies in repeated experiments converge to $\int_\Omega \rho dx$ on any detector region Ω . Thus, probabilities are *epistemic-dynamical* (ignorance over initial micro-geometry) while the underlying evolution remains deterministic.

Multi-particle states are single glued geometric objects: linked/braided helix fibers with shared holonomy. Bell nonlocality arises from this *shared geometry* (one global guidance field) rather than superluminal signals; operational no-signalling is preserved. Aharonov–Bohm shifts are holonomy of A on a multiply connected base; interference channels move with $\Delta\varphi = (q/\hbar)\Phi$ while local forces vanish.

We use a covariant current J^μ and surface-based Tomonaga–Schwinger dynamics to avoid a preferred frame. Worldlines are integral curves of $v^\mu = J^\mu / \sqrt{-J^\nu J_\nu}$; statistics on any spacelike Σ agree by current conservation.

PBR and the Status of ψ . Let Λ denote the ontic state space of GCT beables:

$$\Lambda = \{(\mathcal{H} \rightarrow M, A, \rho; \text{holonomy/linking})\} / \text{gauge}.$$

- (a) **ψ –ontic (structural).** If ψ_1 and ψ_2 are not related by a gauge transformation and global holonomy equivalence, then the induced pairs $(\rho_1, \nabla S_1 - q\mathbf{A})$ and $(\rho_2, \nabla S_2 - q\mathbf{A})$ define *disjoint* subsets of Λ . Hence, distinct quantum states correspond to distinct ontic geometries—compatible with PBR.
- (b) **ψ –epistemic (representational).** A particular ψ is a *coordinate* on the gauge-equivalence class $[\lambda] \in \Lambda$ (choice of section/trivialization and coarse-graining). Different observers may use different ψ -charts for the same $[\lambda]$ without altering beables.

A Summary of Postulates.

- P1. Ontology.** Physical systems are twin helix bundles $(\mathcal{H}, A, \rho)$ over spacetime with specified holonomy/linking data. These structures are real (ontic).
- P2. Dynamics.** Evolution is deterministic guidance of defects/flux along the horizontal distribution set by A ; in coordinates, it reproduces Schrödinger/Dirac (Madelung) with Q a geometric scalar.
- P3. Equivariance.** $\rho = |\psi|^2$ is transported by the guidance field, yielding Born statistics for typical ensembles.
- P4. Measurement.** Coupling to an apparatus induces bordism surgery that selects an attracting branch; “collapse” is the re-expression of the same ontic geometry on that branch.
- P5. Nonlocality.** Entanglement is shared holonomy/bordism in one global object; no-signalling holds.
- P6. Relativity.** The law is tensorial (via J^μ) and surface-local (Tomonaga–Schwinger), avoiding a preferred frame.

This gives a ψ –ontic, computable, and geometric ontology (no configuration-space fluid, a concrete origin for the “quantum potential” Q , the shape operator of the helix amplitude sheet), and a unified account of AB phases, interference, and entanglement via holonomy and join structure. What remains open and falsifiable is (a) a complete relativistic foliation-independence proof for worldlines in interacting QFT; (b) a first-principles determination of abelian coupling (fine-structure constant) from helix micro-geometry, and (c) loop-level matching of abelian β -function from helix excitations.

Summary. In GCT, ψ is *ontic as geometry and epistemic as description*. The “real” thing is the helix bundle with its connection, density, and holonomy; ψ is a chart on that object. Probabilities arise from deterministic equivariant flow, and measurement is geometric branching, giving a PBR-compatible, computable interpretation of quantum mechanics.

Closing Remarks on THM_4 . We do not identify THM_4 with particular Standard Model particle families or with continuous gauge groups. At the level of kinematics, the twin helix substrate furnishes two independent $\mathbb{Z}_2$ invariants, yielding a minimal V_4 structure. These flips are analogous to parity-like and time-reversal-like operations on helical data, in the sense that they reverse handedness or orientation of the underlying geometric flow. Still, we do not claim they implement the spacetime P or T symmetries of the Standard Model, which are known to be violated in the electroweak sector.

That said, there is something suggestive about the Klein four-group; it is the minimal abelian structure where every non-identity is its own inverse. Historically, EPR highlighted a fundamental incompleteness under locality and separability; Bell [67] showed that the package is incompatible with quantum correlations. A deterministic, globally constrained model relaxes measurement independence

while preserving no-signalling, so Einstein, Podolsky, and Rosen's [68] impulse toward a deeper description is retained. But in the spirit of Bohm [69], the specific locality/separability question would receive, yet again, a negative answer.

Our universe may be built from the smallest groups, not just the richest.

A Appendix

A.1 “Hidden” Variables?

Question. *Proof of Yang–Mills instantons being composed of holonomic helices has so far not been confirmed. How then can we safely verify their existence?*

The issue is that the helical structure we posit resides in the *connection* A (parallel transport in the internal bundle). In contrast, direct observables are gauge-invariant functionals of the *curvature* $F = dA + A \wedge A$ or of holonomies (Wilson loops). A gauge transformation can erase any purely local feature of A ; what survives to observation are scalar densities such as $\text{tr } F_{\mu\nu} F^{\mu\nu}$ and $\text{tr } F \wedge F$, which appear as lumps, rings, or cone-like envelopes in lattice plots and experiments. The micro-handedness of a helical A thus projects to gauge-invariant *envelopes* rather than visible corkscrews.

Helical winding is a property of the *fiber* (the G internal directions) over spacetime. Detectors and standard lattice diagnostics live on the base manifold and access only fiber-invariant combinations. Projecting fiber twist onto base scalars discards orientation data but retains energy and topological charge, yielding the observed dual-cone profiles for instanton-like configurations. Real instruments and lattice operators average over space, time, and ensembles. Without a phase-locked reference to the internal connection, linear twist contributions cancel, leaving only quadratic or topological signals. Hence, the helical microstructure is washed out, while $\text{tr } F^2$ and $F \wedge F$ remain.

Many diagnostic solutions (instantons, calorons) are Euclidean saddles. The helix lives in the Euclidean/complexified connection; in real time, one measures its *consequences* (selection rules, rates, envelopes), not the helical motion itself. The appropriate observables are *nonlocal* and gauge sensitive:

- **Wilson-loop tomography.** Families of loops $\mathcal{W}(\gamma) = \text{tr } \mathcal{P} \exp \int_{\gamma} A$ detect non-abelian holonomy; loop-shape/systematic studies can reconstruct local twist consistent with $\text{tr } F^2$ envelopes.
- **Gauge-fixed diagnostics.** In a fixed gauge one may visualize the Chern–Simons 3-form K_{μ} to track handedness (useful for diagnosis, though not itself gauge invariant).
- **Analog platforms.** Phase-locked probes in $SU(2)$ optical media, cold atoms with synthetic gauge fields, or superconducting circuits access non-abelian holonomy directly, providing experimental surrogates for helical transport.

Summary. The helical substrate is “invisible” for the same reason the vector potential is: nature reveals only gauge-invariant aggregates locally and holonomy nonlocally. Helices manifest as the *geometry of transport* (path-ordered phases), while energy/topology plots display their *envelopes*. To expose the helices, one must either reconstruct in a fixed gauge or measure holonomy via nonlocal probes.

A.2 Higher Dimensional Algebra and Geometric Computational Field Theory

To our knowledge, no single algebraic framework presently captures computation in GCMs across the full symbolic-to-fluidic range. By this we mean a framework that simultaneously (a) composes systems under joins—both commuting and non-commuting, (b) handles local-to-global propagation through interfaces, and (c) couples continuous transport (flows/phases) with discrete readout. Lie groupoids (and stacky variants) currently provide the most serviceable language for local symmetry, defects, and gluing. However, a general categorical semantics that bridges a GCM’s geometry with its computability is still lacking.

As one passes from discrete to quasi-fluidic regimes, finite algebraic invariants rapidly lose discriminative power: abelian phase freedoms and continuous deformations overwhelm purely combinatorial structure. This suggests a hybrid treatment in which dynamical systems methods provide the evolution laws, while higher algebraic machinery provides the compositional spine. Accordingly, we use “algebra” here in a programmatic sense: the aim is a language that unifies discrete composition, continuous flow, and topological joining within a single, compositional setting.

Remark A.1. A viable semantics for GCMs must be simultaneously symbolic and geometric: it must (a) compose discrete states and interfaces (information-theoretic layer), (b) transport phases and flows continuously (geometric layer), (c) express gluing along boundaries and track joins, and (d) specify how update laws transform the underlying geometry (chirality/orientation/topological data).

The key obstacle is nonlocal constraint propagation across helices: interface compatibility is not decided on a single patch, and path-dependence (holonomy) persists under coarse discretization. Global

states are therefore not simple products of local states but solutions of descent/compatibility conditions along the interface groupoid (as in 4.2). Consequently, naive local linearization or discretization erases the very invariants that govern computation.

What is needed? A workable mathematical language for GCMs must support four capabilities and tie each to a concrete compositional device.

- (a) **Continuous transformation.** Represent smooth deformations (tube radius, phase, twist) and their flows.
- (b) **Local-to-global behavior.** Track how interface-local updates assemble to global states and constraints.
- (c) **Structural invariance and laws.** Express conservation and controlled change of chirality, orientation, and topological charges.
- (d) **Rich morphism structure.** Compose fragments along boundaries; allow deformations as higher morphisms.

Definition A.1 (Minimal Compositional Spine for GCMs). Let $\mathbb{G}$ be a symmetric monoidal double category of open GCMs:

- **Objects** (0-cells). Interfaces/boundaries carrying readout structure.
- **Horizontal** (1-cells). GCM fragments with specified interfaces (gluable bordisms equipped with interface rules $\mathcal{I}$).
- **2-cells.** K -equivariant deformations/flows between fragments (homotopies, pulses).
- **Monoidal product.** Disjoint union $\otimes$ (parallel composition); *horizontal composition*: gluing along interfaces.

Decorate $\mathbb{G}$ by a stack $\mathcal{S}$ of state groupoids over $\mathcal{E}SM^+$ that assigns to each open set the fiber $\Sigma = B \times T \times Z$ and enforces descent along the interface groupoid. Dynamics consists of (a) *pulses* as endo-2-cells (discrete steps), and (b) *flows* as internal vector fields on $\mathcal{S}$ (continuous evolution).

The above definition provides an outline of minimal workable semantics. In addition, the following areas of mathematics could be integrated:

Stacks and Descent Theory — can model local-to-global state by a stack $\mathcal{S}$ of groupoids (or higher groupoids) on the site $\mathcal{E}SM^+$: to each open U assign the state groupoid $\mathcal{S}(U)$ with K -equivariant structure (Def. 4.2); joins are encoded by descent along the interface groupoid. Nontrivial holonomy/-monodromy appears as nonvanishing obstruction classes in nonabelian Čech hypercohomology; in QF these reduce to abelian local systems (torus phases T and integer charges Z), while in the fluidic/-gauge regimes they become genuinely nonabelian. Thus, global configurations are *glued objects of $\mathcal{S}$* , not cartesian products of local states.

Stratified, Topological Field Theories — associate algebraic data to topological spaces. One can use a stratified, extended bordism category with defects (interfaces as 0-strata, GCM fragments as 1-strata, deformations/flows as higher strata) and a symmetric monoidal functor:

$$\mathcal{Z} : \text{Bord}_{d,\partial}^{\text{strat}} \longrightarrow \mathcal{C}$$

into a target higher category $\mathcal{C}$ (e.g., stacks, dg-categories). Functoriality under gluing expresses a “conservation of computational information” principle; factorization algebras can organize observables on $\mathcal{E}SM^+$. In the QF abelian subtheory, $\mathcal{Z}$ restricts to torus-valued local systems; allowing nonabelian holonomy recovers the fluidic/gauge behavior where $[J_1, J_2] \neq e$.

Synthetic Differential Geometry and Jets Bundles — would encode update laws and finite-order control by jet comonads J^k on fields (sections of bundles) and the Cartan/Spencer formalism; integrability and phase-slip events correspond to Spencer cohomology and failure of prolongation. Working internally to a differential cohesive ∞ -topos provides infinitesimals and differential modalities (e.g. $\int$, $\flat$, $\sharp$) so that vector fields, flows, and K -equivariant transport are first-class objects. QF corresponds to abelian transport on $T \times Z$ with thresholded readout on B ; fluidic/gauge add nonabelian connections and curvature.

Bordism Type Theory (*Sketch*) — is a dependent type theory internal to the differential cohesive setting with:

- *Types* as stacks of decorated manifolds (helical/stratified with K -equivariant state fibers Σ).
- *Terms* as global sections (configured geometries).
- *Id-types* as path objects (homotopies/deformations between configurations).
- Π - and Σ -types as homotopy right/left Kan extensions along fibrations modeling parameterized families and gluing.
- a sequent calculus whose cut/elimination rules match horizontal/monoidal composition in the double category of open GCMs.

A semantics functor interprets programs as morphisms in the compositional spine, with readout functors to cellular automata or torus rotations.

For abelian GCMs, ordinary bundle/connection language suffices. In the fluidic and gauge regimes, however, one must track not only fields and their holonomies along paths, but also *homotopies of gauge data*, junctions of line/surface operators, and coherent moves of defects under gluing. This is the natural domain of **higher gauge theory** [70]: objects encode boundary data, 1-morphisms encode bordisms with connections, and 2-morphisms encode gauge homotopies/defect moves; higher morphisms enforce the coherence needed for locality and composition. In short, higher categorical structure is not decorative—it is the minimal bookkeeping that makes non-abelian transport, defect fusion, and feedback under refinement precise.

In addition, our base symmetry $V_4 \simeq C_2 \times C_2$ embeds in $O(2)$. For constructions sensitive to spin/Pin data, we lift along $\text{Pin}(2) \rightarrow O(2)$; the preimage of V_4 is the quaternion group $Q_8 \subset \text{Pin}(2) \subset SU(2)$. Thus, orientation reversal and chirality are treated as genuine *equivariant* data, aligning with **Manolescu’s Pin(2)-equivariant Floer theory** [71]. Practically, the abelian/quasi-fluidic analysis uses the V_4 projection, while the fluidic/gauge analysis employs the $\text{Pin}(2)$ lift, ensuring compatibility with prospective Floer-homotopy refinements.¹⁹

Remark A.2. There are, of course, numerous avenues for future exploration, but for the time, this list must suffice. A full algebraic description of the actual behavior of a GCM likely demands a synthesis beyond the reach of any single mathematician—a genuinely collaborative effort. Groethendieck famously spoke of a rising sea in which complex problems could be gradually subsumed. However, when nature tries to frustrate humanity with phenomena such as quantum mechanics, sometimes no ocean is deep enough. To counter this, we may need to carve a new path, one that uses a river of flow.

A.3 Implications for Reversible Computing

GCT opens an avenue for *reversible computation*. This matters for physics because microscopic physical laws are, to an excellent approximation, information-preserving: classical Hamiltonian dynamics preserve phase-space volume (Liouville) and quantum dynamics evolve by unitary (or antiunitary) transformations. In both cases, the evolution can, in principle, be inverted. A computational model that respects this structure aligns directly with how physical systems propagate.

Let Conf be the space of configurations of a computing device. Then for:

- Finite/discrete models.** A step of reversible computation is a bijection $F : \text{Conf} \rightarrow \text{Conf}$. Composition gives a group $\text{Bij}(\text{Conf})$ of updates; gates such as Toffoli/Fredkin generate subgroups used in practice (see [72] for an overview).
- Classical continuum models.** Let (M, ω) be a symplectic phase space. An evolution $\{\phi_t\}_{t \in \mathbb{R}}$ is reversible when each ϕ_t is a *symplectomorphism* and $\phi_{t+s} = \phi_t \circ \phi_s$, $\phi_0 = \text{id}$, hence $\phi_{-t} = \phi_t^{-1}$. Hamiltonian flows are the canonical example.
- Quantum models.** On a Hilbert space $\mathcal{H}$, closed-system time evolution is implemented by a one-parameter unitary group $U_t = e^{-iHt}$, so $U_{-t} = U_t^\dagger$. More generally, any unitary (or antiunitary, for time-reversal) map is logically reversible; non-unitary channels (measurement, dissipation) are not.

¹⁹We withhold examination of Floers’ work and subsequent issues arising from it here not out of lack of importance, but because it requires an *entire paper unto itself* to do justice.

Reversible computation can be expressed as *isomorphisms* in a category of states and processes. If $\mathcal{C}$ is a category of configurations, a reversible step is an isomorphism $f : X \rightarrow Y$ with inverse $f^{-1} : Y \rightarrow X$ and $f \circ f^{-1} = \text{id}_Y$, $f^{-1} \circ f = \text{id}_X$. Collecting objects and isomorphisms yields a groupoid of configurations and reversible transitions. Equivalently, a reversible dynamics is an autoequivalence $F : \mathcal{C} \rightarrow \mathcal{C}$ with a quasi-inverse F^{-1} , i.e. $F \circ F^{-1} \cong \text{id}_{\mathcal{C}}$ and $F^{-1} \circ F \cong \text{id}_{\mathcal{C}}$. This captures “no information loss” at the structural level.

Example A.1 (Flows and Time-reversal). Let $\phi_t : M \rightarrow M$ denote a flow on a manifold M . If every ϕ_t is a diffeomorphism and $\phi_{-t} = \phi_t^{-1}$ for all t , the evolution is reversible. In classical mechanics this is strengthened by symplecticity $\phi_t^* \omega = \omega$, while mechanics conservation of phase-space volume and the existence of a well-posed inverse flow. In quantum theory, the analogous statement is $U_{-t} = U_t^\dagger$ for unitary U_t .

Reversibility is not merely an aesthetic constraint on computation; it mirrors the fundamental invertibility of closed-system physics.

- (a) **Information conservation** — implies reversible updates preserve *all* fine-grained information. Logical irreversibility (many-to-one maps) necessarily discards information and, when implemented physically, has thermodynamic cost.
- (b) **Microscopic reversibility vs macroscopic irreversibility** — means chaotic yet Hamiltonian systems are still reversible (invertible maps); apparent irreversibility arises from coarse-graining, noise, and coupling to environments. Computationally, introducing erasure, measurement, or stochastic resets corresponds to non-invertible maps and semigroup dynamics.
- (c) **Design principles** — are such that algorithms and architectures that factor into reversible cores (bijections/symplectomorphisms/unitaries) with controlled, explicit points of irreversibility (enable tighter accounting of energy and error, and interface cleanly with physics-based models).

Modeling computation by invertible transformations (bijections, symplectomorphisms, unitaries) places the theory squarely inside the same structural class as classical and quantum dynamics, with logical irreversibility reserved for explicit coarse-grainings or measurements.

A.4 Topological Kleene Field Theories and Computational Geometric Bordisms

Miranda, Peralta, and Gonzales [73] have recently introduced Topological Kleene Field Theory (TKFT), which aims to advance understanding of fluidic computation beyond classical Turing completeness.

At the heart of TKFT is the notion of a *smooth dynamical bordism*—a manifold connecting multiple disk boundaries, each embedded with a Cantor set of natural numbers, within which computation unfolds as a flow governed by partial recursive reaching functions. In this framework, computation is realized as trajectories in a Hilbert space, and the “pair of pants” topology elegantly encodes program branching, granting TKFT both expressive power and a high degree of abstraction.

However, the GCMs central to our theory do not naturally inhabit n -dimensional Hilbert spaces. This raises a conceptual question (echoing the cobordism hypothesis): how should we interpret “bordism” when the manifold itself is the computational substrate, rather than a medium mediating between external boundaries?

Definition A.2 (Computational Geometric Bordism). Fix a stack of computational structures $\mathcal{S}$ (e.g. principal G -bundles with connection, and decoration functors). An n -dimensional GCM is a pair (M, σ) with M an oriented smooth n -manifold (possibly with boundary) and $\sigma \in \mathcal{S}(M)$. A *computational geometric bordism* from (M_0, σ_0) to (M_1, σ_1) is a compact oriented smooth $(n+1)$ manifold W with boundary equipped with a collar-preserving identification

$$\partial W \cong \overline{M_0} \sqcup M_1,$$

together with a structure $\Sigma \in \mathcal{S}(W)$ whose restrictions along the collars agree with the endpoints: $\Sigma|_{M_0} = \sigma_0$ and $\Sigma|_{M_1} = \sigma_1$. Two such bordisms are equivalent if related by a boundary-fixing diffeomorphism and an isomorphism in $\mathcal{S}$.

Example A.2 (The Helix as a Geometric Bordism). Recall that a bordism between two closed n -manifolds M_0, M_1 is an $(n+1)$ -manifold W with boundary: $\partial W = M_0 \sqcup M_1$. If we embed the unit circle S^1 into $\mathbb{R}^2$ by the usual trigonometric map

$$(\theta, t) \mapsto (\cos(\theta + \lambda t), \sin(\theta + \lambda t), t), \quad (\theta, t) \in S^1 \times I.$$

and then thicken this embedding by the interval $I = [0, 1]$, we obtain

$$W = S^1 \times I \subset \mathbb{R}^3, \quad (\theta, t) \mapsto (\cos \theta, \sin \theta, t).$$

The boundary components $S^1 \times \{0\}$ and $S^1 \times \{1\}$ are two disjoint copies of the circle, and the cylinder W is a smooth $(1 + 1)$ -dimensional bordism connecting them.

A helix is a cylinder twisted by a chirality map: $(\theta, t) \mapsto (\cos \theta, \sin \theta, t + \lambda \theta)$, so its core curve winds through space while still furnishing an $(n + 1)$ -manifold whose boundary is the same pair of n -manifolds. Thus, a helix (or any smooth chiral flow) is a geometric computational bordism: the shape itself is the intermediary manifold realizing the cobordism relation.

In GCT, a more intrinsic and lightweight perspective is natural. Here, bordism is *internal* to the GCM: computation is realized as deformation or flow between different boundary states within the manifold itself. The “input” and “output” boundaries correspond to loci where flows implement morphisms. If TKFT aspires to capture the laws of fluidic computation in terms of a general-purpose “CPU” architecture—capable of universal computation via smooth dynamical flows—then GCT can be viewed as a more discrete, helical “GPU”: a specialized, fuzzy, and highly efficient processor for geometric tasks. We do not wish for persons to construe GCT as not a competitor to TKFT, but as a complementary framework with distinctive strengths and trade-offs. In principle, both approaches can be used in concert to address fundamental problems in topological physics from different vantage points.

A.5 Relationship to the Blum-Shub-Smale Model

The Blum–Shub–Smale (BSS) model [74] extends classical computability from $\mathbb{N}$ to $\mathbb{R}$ by postulating a register machine whose instructions are field operations and branches on real values. It yields a rich theory of decision/complexity over $\mathbb{R}$, but with an *algebraic* semantics predicated on unit-cost real arithmetic.

Our framework is inspired but orthogonal. We retain smooth reasoning internally (synthetic/smooth reals) and execute externally over effective number systems (computable reals/algebraics), while moving the *computational substrate* from algebra to geometry: We therefore use BSS only as *context*: it shows that enlarging the base field can change computability classes. In contrast, GCT changes the *operational layer*: computation is controlled evolution of geometric state, with gauge invariants as outputs and coarse-graining/renormalization modeled as algebraic effects (cf. Section 5.5). This preserves smooth calculus for reasoning, avoids hidden oracles in execution, and matches the holonomy-based semantics used throughout.

Definition A.3 (Geometric E_∞ -algebra). Let $\mathcal{C}_\infty$ denote the symmetric monoidal ∞ -category of smooth (derived) manifolds and smooth maps, equipped with the Cartesian product $\times$ as its tensor product. A GCM lattice is specified by

$$X \in \mathcal{C}_\infty, \quad \{\mu_n : X^{\times n} \longrightarrow X\}_{n \geq 0},$$

where the operations μ_n encode parallel composition (disjoint gluing) of n unit helix components. The family (μ_n) is required to satisfy the coherence relations of the E_∞ -operad in $\mathcal{C}_\infty$, making $(X, \mu_\bullet)$ a commutative E_∞ -algebra object is as follows:

- (a) The unit map $\mu_0 : * \rightarrow X$ corresponds to the empty (null) computation.
- (b) The binary product $\mu_2 : X \times X \rightarrow X$ realizes the geometric join of two independent lattice components.
- (c) Higher μ_n encodes n -way concurrent composition, commutativity holds up to coherent homotopy, giving the required flexibility for fluidic deformations.

In this language, a classical BSS machine is recovered by restricting to the full sub- ∞ -category of finite-dimensional real vector spaces with polynomial maps, whose objects form discrete E_∞ -algebras (strict commutative monoids). The GCM lattice generalizes this: it is still an E_∞ -monoid, but now geometry (i.e., smooth structure, group actions, bordisms) replaces uncountable real registers as the primitive data store.

$$\text{BSS (discrete commutative monoid)} \longrightarrow \text{GL (geometric } E_\infty\text{-algebra)}.$$

See Lurie [75] for the general theory of E_∞ -algebras in symmetric monoidal ∞ -categories. This synthesis of higher category theory and geometry moves us closer to a physical theory of computation.

A.6 On Casual Fermion Systems

Causal Fermion Systems (CFS), developed by Prof. Felix Finster and collaborators [76], is, from the GCT perspective, the current leading attempt, bar none, to rewrite the foundations of physics in a way that is *geometric-first* and *variational-first*, while treating standard quantum mechanics as a *limiting* (and in a sense, *effective*) description rather than an irreducible axiom. A treatment is mandatory; however, prior to the final sections of our journey, this proved difficult.

A CFS starts from an operator-theoretic object (a Hilbert space and a measure on a space of operators), and declares that space-time itself is not given a priori: it is reconstructed as the *support* of the measure. Dynamics is not postulated in the usual “Hamiltonian on a background manifold” way, but instead emerges from minimizing a nonlinear action functional (the *causal action*). In the appropriate continuum limit, one recovers familiar structures: Lorentzian geometry, classical gauge fields, and (in further approximations) the quantum-mechanical and QFT toolkits as effective formalisms.

The core data of a causal fermion system are often packaged as a triple

$$(\mathcal{H}, \mathcal{F}, \rho),$$

where:

- $\mathcal{H}$ is a Hilbert space of “physical” spinorial states (think: a space of Dirac-type solutions, but *not* necessarily tied to a pre-existing manifold in the fundamental formulation);
- $\mathcal{F}$ is a set of (typically finite-rank) self-adjoint operators on $\mathcal{H}$, with an imposed signature constraint (this is part of how “spin dimension” enters);
- ρ is a positive measure on $\mathcal{F}$ (the *universal measure*).

The conclusion is then:

$$M := \text{supp}(\rho)$$

is defined to be space-time. One then extracts causal relations, metric information, and spin structure from spectral data of operator products xy for $x, y \in M$. The dynamics are: vary ρ (subject to constraints) to minimize a positive action functional constructed from the eigenvalues of the products xy . The theory is therefore “equations of motion as an optimization principle on measures,” not “equations of motion as PDEs on a background manifold.”

In our regime language, CFS is best read as living predominantly in the gauge band, with an essential side-channel into the fully fluidic band. The most faithful placement is that CFS is a gauge theory of computation: non-commutative dynamics with a considerable quotient by redundancy, such that apparent modular independence is a gauge choice rather than a literal factorization. This is precisely the hallmark of the gauge regime in our classification: joins do not commute in general, but one regains a notion of “independence” after gauge-fixing; the state is stored in gauge-invariant aggregates (holonomies/curvature/spectral invariants) rather than in raw potentials.

CFS is often thought of as intimidating because it is broad (Finster is, by training, a mathematician first), but the overlap with GCT is actually clean at the conceptual level:

- (a) **Spacetime is derived, not assumed.** In GCT, we treat the manifold structure as constructed from computational geometry. CFS likewise treats spacetime as emergent (support of the universal measure) rather than a fixed stage.
- (b) **Physics is “action + constraints” at the fundamental layer.** In GCT, we repeatedly re-encounter the principle that what matters are invariants that are stable under the allowed deformations (gluing, transport, holonomy, charge). CFS makes this explicit by encoding dynamics as a variational principle on the measure itself.
- (c) **Gauge invariance is not a cosmetic symmetry: it is the computational quotient.** Our gauge regime is precisely the regime where the machine’s *actual* state is not the raw configuration, but the equivalence class under gauge transformations. CFS is philosophically the same move, except made at the operator level: local unitary freedom is built in, and only gauge-invariant spectral aggregates are physically meaningful.
- (d) **Quantum formalism as a readout/effective theory.** CFS is often read as “de-quantizing” quantum mechanics: the linear Hilbert-space formalism is not the fundamental law, but a limiting description arising from coarse-graining and continuum approximations. This resonates strongly

with the GCT posture that many “quantum” features are emergent bookkeeping for unstable geometric microstructures.

CFS Feature	GCT Interpretation
Universal measure ρ on operator space	Microstate distribution/ensemble over admissible geometric configurations (a “state-selection” layer)
Spacetime $M = \text{supp}(\rho)$	Working manifold (or effective base) as an emergent support of stable computational configurations
Spectral data of xy defines causal structure	Causality/communication constraints extracted from transport/holonomy invariants
Causal action principle (minimize action)	“Physics as compilation”: admissible computations selected by global consistency/extremal principles
Local unitary freedom (gauge)	Gauge regime: independence after gauge-fixing; observables are gauge-invariant aggregates
Continuum limit yields classical gauge fields + gravity	Gauge regime semantics: computation stored in curvature/holonomy classes; classical fields as macroscopic limit of microstructure

We need not adopt the full CFS formalism to benefit from it. What we all should take away is the *structural lesson*:

If one treats quantum theory as an effective linearization of a deeper geometric object, then the correct primitive is not “wavefunctions on spacetime” but rather a *geometric substrate* whose macroscopic shadows look like Hilbert-space quantum mechanics.

In this sense, CFS is not a competitor but a strong sanity-check: it shows that a long-running research program exists which is already comfortable with (a) emergent spacetime, (b) gauge-as-quotient, and (c) quantum mechanics as a derived language rather than a first principle. All three are compatible with the GCT regime taxonomy, and CFS can be read as a sophisticated inhabitant of the gauge band.

Summary. CFS is best understood here as an operator-level realization of the gauge regime: non-commutative dynamics with ample redundancy, where the physically meaningful information is the gauge-invariant residue. This makes it a natural “bridge theory” between our regime picture and conventional QFT language.

A.7 Towards the Computational Cobordism Hypothesis

The algebraic enrichment required by this framework demands categories that encode both geometric and computational structure. To this end, we introduce the category $\mathcal{ESM}^+$ (“Embedded Smooth Manifolds with Enrichments”) as the natural habitat for all GCMs.

Definition A.4 (Embedded Smooth Manifolds with Enrichments). Let **Man** denote the category of smooth manifolds and smooth maps. The *master category* $\mathcal{ESM}^+$ is a symmetric monoidal category whose:

- (a) **Objects.** Pairs (M, str) where M is a smooth (possibly stratified) manifold and str is a finite tuple of auxiliary geometric/computational structures over M (e.g. an orientation double cover, specified principal bundles and connections, distinguished vector fields/flows, discrete labels, etc.)
- (b) **Morphisms.** $(M, \text{str}) \rightarrow (N, \text{str}')$ are smooth maps $f : M \rightarrow N$ that preserve the declared structure (pull back covers/bundles, intertwine flows, respect discrete labels.)
- (c) **Monoidal product.** Disjoint union:

$$(M, \text{str}) \otimes (N, \text{str}') := (M \sqcup N, \text{str} \sqcup \text{str}'), \quad 1 := (\emptyset, \emptyset).$$

There is a faithful forgetful functor $U : \mathcal{ESM}^+ \rightarrow \mathbf{Man}$. We suppress set-theoretic size issues (all categories are essentially small).

Remark A.3. The choice $\otimes = \sqcup$ models parallel composition of independent systems; other monoidal products (e.g. fibered sums under boundary matching) can be introduced when needed, but are not required for the present development.

We separate the geometric substrate from its semantic evaluation target.

Definition A.5 (Differentially Cohesive Topos for GCMs). To reason natively about bundles, connections, curvature, and holonomy while retaining an executable semantics, we separate *geometry* from *execution* and connect them by a left-exact bridge. Let $\mathcal{E}_{\text{GCM}}$ be a (higher) differentially cohesive topos [77] in the sense of cohesive/hypoelliptic geometry (e.g. a smooth ∞ -topos of sheaves on Cartesian spaces). It comes equipped with the standard modalities

$$\Pi \dashv \flat \dashv \text{id} \dashv \sharp \quad \text{and} \quad \text{dR} : \mathcal{E}_{\text{GCM}} \rightarrow \mathcal{E}_{\text{GCM}}$$

so that one may speak internally of principal bundles, connections, differential characters, and holonomy. Decorated groupoids and bordisms land in $\mathcal{E}_{\text{GCM}}$ as objects carrying these structures.

Definition A.6 (Execution Ambient and Extraction). Let $\mathcal{D}_{\text{exec}}$ be a CPO-enriched cartesian closed category (a domains-style model for partiality, effects, and fixed points) whose numeric objects are effective (e.g. rationals, algebraics, or computable reals). A *left-exact, symmetric monoidal extraction*

$$\llbracket - \rrbracket : \mathcal{E}_{\text{GCM}} \longrightarrow \mathcal{D}_{\text{exec}}$$

preserves finite limits and the declared monoidal structure, and sends gauge-invariant functionals (e.g. class functions of holonomy, differential characters) to computable observables.

The overall pipeline is thus:

$$\text{decorated bordisms } \mathbf{Bord}_n^S \xrightarrow{\text{assignment}} \mathcal{E}_{\text{GCM}} \xrightarrow{\llbracket - \rrbracket} \mathcal{D}_{\text{exec}}$$

Remark A.4 (Modalities as Semantics). The cohesive modalities organize the readouts:

- Π (“shape”) extracts homotopy-level/topological content (e.g. conjugacy classes of holonomy).
- $\flat$ (“flat/discrete”) forgets curvature, yielding symbolic data for verification.
- dR produces differential-cohomological observables (differential characters).
- $\sharp$ (“codiscrete”) forgets topology when analytic control is required.

Operationally, “path integration is replaced by sheafification”: global states are computed as (homotopy) sheafified sections in $\mathcal{E}_{\text{GCM}}$, then evaluated by $\llbracket - \rrbracket$.

Remark A.5 (Fixed Points and Effects Without Oracles). Recursion, partiality, coarse-graining, and renormalization live in $\mathcal{D}_{\text{exec}}$ as CPO-based effects with handlers; fixed points exist by Scott–Kleene. Geometry remains smooth and local in $\mathcal{E}_{\text{GCM}}$. This separation keeps differential reasoning intact while ensuring all externally visible outputs are computable.

We now pass from the state/transition layer (decorated groupoids) to a control-flow layer where computations are realized as decorated bordisms.

Definition A.7 (Decorated Bordism Category). Fix a dimension $n \geq 0$, a decoration monoid/group $\mathcal{S}$, and a class of $(n+1)$ -dimensional compact smooth manifolds with boundary and $\mathcal{S}$ -decoration data compatible with composition (gluing). The *decorated n -bordism category* $\mathbf{Bord}_n^S$ has:

- Objects.** Closed (n) -manifolds with chosen $\mathcal{S}$ -decorations on their state/transition groupoids (e.g. $d : \Pi_1^*(\cdot) \rightarrow \mathbf{BS}$).
- Morphisms.** Compact n -manifolds W with boundary $\partial W \cong \overline{M}_0 \sqcup M_1$ carrying compatible $\mathcal{S}$ -decorations, composed by gluing along collars.
- Symmetric monoidal product.** Is disjoint union.

When a geometric substrate $(M, \text{str}) \in \mathcal{ESM}^+$ is fixed, we write $\mathbf{Bord}_n^S(M, \text{str})$ for the full subcategory of bordisms that embed in $M \times I$ and whose decorations agree with str on the boundary.

Conjecture A.1 (The Computational Cobordism Hypothesis (CCH)). *Let $\mathcal{ESM}^+$ be the ambient category, and let $\mathcal{E}_{\text{GCM}}$ be a topos. There is a classification principle such that symmetric monoidal functors*

$$Z : \text{Bord}_n^S(M, \text{str}) \longrightarrow \mathcal{E}_{\text{GCM}}$$

(geometric field theories) are in natural bijection with choices of fully dualizable decorated objects associated to (M, str) in a suitable symmetric monoidal $(\infty, 1)$ -completion of $\mathcal{ESM}^+$. Moreover, when $\mathcal{S}$ is trivial, and decorations factor through Π_1 , this reduces to the corresponding case of the (once-extended) Cobordism Hypothesis.

Three-line operational sketch.

- (a) **States/transitions.** Encodes local updates by an $\mathcal{S}$ -decorated (groupoid or action groupoid) $(\mathcal{G}, d)$ over boundary cuts.
- (b) **Control-flow.** Models a computation as a decorated bordism $W \in \text{Bord}_n^S(M, \text{str})$; composition = concatenation of phases.
- (c) **Evaluation.** A symmetric monoidal functor Z sends boundary data to objects of $\mathcal{E}_{\text{GCM}}$ and sends W to a morphism in $\mathcal{E}_{\text{GCM}}$, yielding denotational semantics for the computation.

Remark A.6 (Reduction to Classical Cases). If the decoration d factors through Π_1 and $\mathcal{S}$ is discrete/finite, then Z specializes to a finite-group TQFT. If $\mathcal{S}$ carries continuous abelian data (e.g. $S^1 \times C_3$), Z gives an abelian “gauge-like” semantics; non-abelian $\mathcal{S}$ interpolates to gauge-type models.

We now transition from the state/transition layer (decorated groupoids) to a control-flow layer, where computations are realized as decorated bordisms.

Example A.3 (Helical Bordism Computation). Let $M \cong S_\theta^1 \times D_{(r, \varphi)}^2$ carry a chirality label $\chi \in C_2$ and an orientation phase $\theta \in S^1$. Set

$$G = C_2 \times U(1) \times \mathbb{R},$$

acting on core states by

$$(\varepsilon, R_\alpha, t) \cdot (\chi, \theta, \phi, \dot{\phi}) = (\varepsilon\chi, \theta + \alpha, \Phi_t(\phi, \dot{\phi})),$$

where R_α is a rotation of the θ -coordinate and Φ_t is the time- t flow on internal variables.

Use the gauge-invariant monoidal target

$$\mathcal{S} := \text{Cl}(U(1)) \times C_2 \cong U(1) \times C_2, \quad (z, \eta) \cdot (z', \eta') := (zz', \eta \oplus \eta'),$$

where $\text{Cl}(U(1))$ denotes conjugacy classes (for $U(1)$ this is $U(1)$ itself). Fix a $U(1)$ connection A representing a chosen 1-cocycle in $H^1(M; U(1))$ and define the (symmetric monoidal) *decoration*

$$d : \Pi_1(M) \longrightarrow \mathcal{S}, \quad d([\gamma]) = (\text{Hol}_\gamma(A), \chi|_\gamma).$$

Let G act on $\mathcal{S}$ by

$$(\varepsilon, R_\alpha, t) \cdot (z, \eta) := (z, \varepsilon\eta),$$

so that d is G -equivariant: $d((\varepsilon, R_\alpha, t) \cdot [\gamma]) = (\varepsilon, R_\alpha, t) \cdot d([\gamma])$. Functoriality holds: $d([\gamma_1] \cdot [\gamma_2]) = d([\gamma_1]) \cdot d([\gamma_2])$, $d([\gamma]^{-1}) = d([\gamma])^{-1}$.

Consider the trivial bordism (cylinder) $W = S^1 \times I \subset M \times I$ with helical embedding

$$(\theta, t) \longmapsto (\cos(\theta + \lambda t), \sin(\theta + \lambda t), t), \quad (\theta, t) \in S^1 \times I,$$

whose boundary components are $S^1 \times \{0\}$ and $S^1 \times \{1\}$. Along a loop at fixed radius r , define a local angular rate $\Omega(r, \theta)$ and the one-turn phase gate

$$U(r) = \exp\left(i \int_0^{2\pi} \Omega(r, \theta) d\theta\right) \in U(1).$$

Then the decoration assigns

$$d([\gamma_r]) = (U(r), \chi).$$

Two loops with the same homotopy class but different radii $r_1 \neq r_2$ may yield $U(r_1) \neq U(r_2)$; hence the computational outcome depends on geometric control (not only on the bare π_1 class), while composition in $\mathcal{S}$ remains abelian and compatible with the groupoid structure and with the bordism functoriality of W .

A *geometric field theory* $Z : \text{Bord}_1^S(M, \text{str}) \rightarrow \mathcal{E}_{\text{GCM}}$ sends the boundary circle to an object $Z(S^1) \in \mathcal{E}_{\text{GCM}}$ and the decorated cylinder W to a morphism $Z(W)$ which, in this example, acts by multiplication by $U(r)$ on the appropriate component. When Ω is constant and χ is trivial, d factors through Π_1 and Z reduces to a discrete finite-group semantics.

A.8 Fourier Mode Calculations

Here, we examine kinematic selection effects in the idealized computational flow on the L^2 field of a trivial bundle. Let the cross-section be a disk D^2 in the xy -plane. The velocity field $\vec{v}(r, \theta)$ at each cross-section can be written in polar coordinates as

$$\vec{v}(r, \theta) = v_r(r, \theta) \hat{r} + v_\theta(r, \theta) \hat{\theta},$$

where v_r and v_θ are the radial and angular components, respectively.

Suppose the flow is smooth and incompressible, and consider the angular dependence of v_r and v_θ . Each component admits a Fourier expansion:

$$v_r(r, \theta) = \sum_{k=-\infty}^{\infty} a_k(r) e^{ik\theta}, \quad v_\theta(r, \theta) = \sum_{k=-\infty}^{\infty} b_k(r) e^{ik\theta}.$$

Imposing rotational symmetry of order n means requiring invariance under rotation by $2\pi/n$:

$$v_r(r, \theta + 2\pi/n) = v_r(r, \theta) \quad \forall \theta.$$

This forces all Fourier coefficients $a_k(r)$ and $b_k(r)$ with $k \notin n\mathbb{Z}$ to vanish. Thus, only harmonics with k divisible by n remain:

$$v_r(r, \theta) = \sum_{m=-\infty}^{\infty} a_{nm}(r) e^{inm\theta}.$$

In the absence of symmetry ($n = 1$), all $2K + 1$ modes with $|k| \leq K$ are permitted. Under n -fold symmetry, only those with $k = 0, \pm n, \pm 2n, \dots, \pm Mn$, with $M = \lfloor K/n \rfloor$, survive—yielding $2M + 1$ allowed modes.

We define the *relative expressiveness* as the fraction of permitted angular degrees of freedom:

$$E(n) := \frac{\text{Number of allowed modes for symmetry } n}{\text{Number for } n = 1} = \frac{2M + 1}{2K + 1}.$$

For example, with $n = 4$ and $K = 8$, we have $M = 2$ (i.e., $k = 0, \pm 4, \pm 8$), so $E(4) = 5/17 \approx 0.29$.

As $n \rightarrow \infty$, $M \rightarrow 0$ and $E(n) \rightarrow 0$; i.e., only the constant mode survives. The only flow invariant under arbitrary rotation is uniform, and all angular structure is lost. This directly parallels the earlier result that the only smooth closed 1-manifold with infinite rotational symmetry is the circle.

Thus, increasing symmetry necessarily reduces the space of physically expressible patterns. The tradeoff is exact, quantitative, and determined by the order of symmetry imposed on the system.

Remark A.7. If a field $F(r, \theta)$ on an annulus is invariant under all rotations $\theta \mapsto \theta + \phi$ ($U(1)$ symmetry), then in its Fourier expansion $F(r, \theta) = \sum_{m \in \mathbb{Z}} F_m(r) e^{im\theta}$ one has $F_m \equiv 0$ for $m \neq 0$. Equivalently, only the θ -independent ($m = 0$) mode survives; axisymmetric azimuthal components may remain, but no angular structure ($|m| \geq 1$) does.

A.9 Disclaimers

Regarding Fundamental Assumptions

The results presented herein **should under ABSOLUTELY NO CIRCUMSTANCES be interpreted as a challenge** to experimentally validated quantum field theories (such as Quantum Chromodynamics, Yang–Mills theory, or the Standard Model). Instead, they provide a rigorous geometric and deterministic framework offering new conceptual insights into symmetry, gauge structure, and hidden-variable formulations. This work aims to *enrich* our understanding of existing physics, not supplant or refute it!

Author's Note on Naming Conventions

Given that THM may turn out to be a fundamental building block of our universe/gauge theory, **I strongly feel that it should not bear anyone's name**, if this is the case. I understand why the naming of manifolds was done with Calabi-Yau, but in this case, my gut says doing the same is *probably* a very bad idea. I get that my wishes are unconventional, but I simply ask that they be respected, given the unusual circumstances.

That said, the focus of this paper is not personal legacy, but the advancement of geometric computability and its physical and pedagogical applications. Likewise, the author does not identify as a mathematician, nor is he a classically trained physicist; so he asks not to be labeled as either. The best term of description would be “computational scientist.”

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Conflicts of Interest

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Data Availability

There is no associated data with this manuscript.

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Likewise, the AI was tasked with designing the experiments above to detect the helices.

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